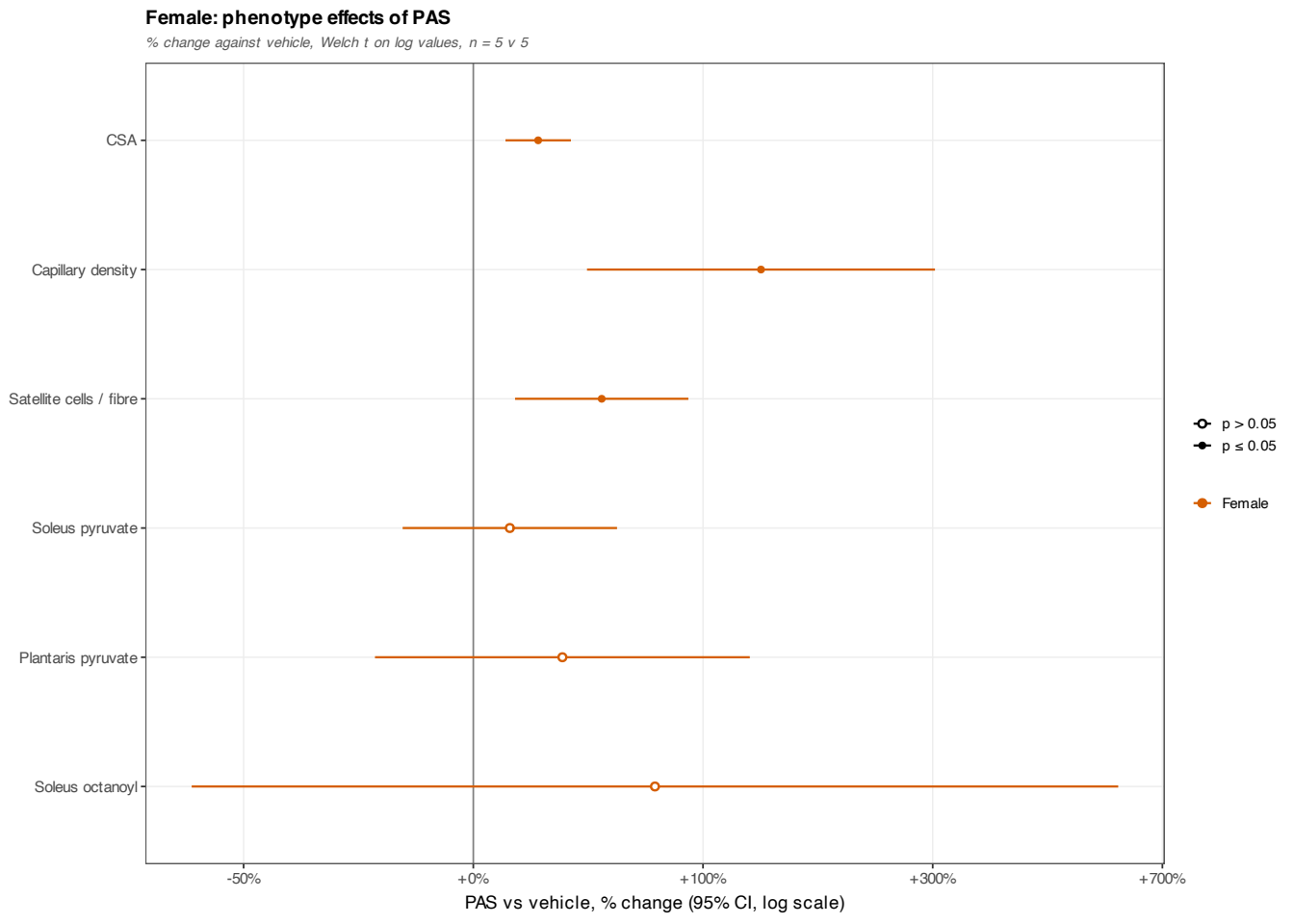
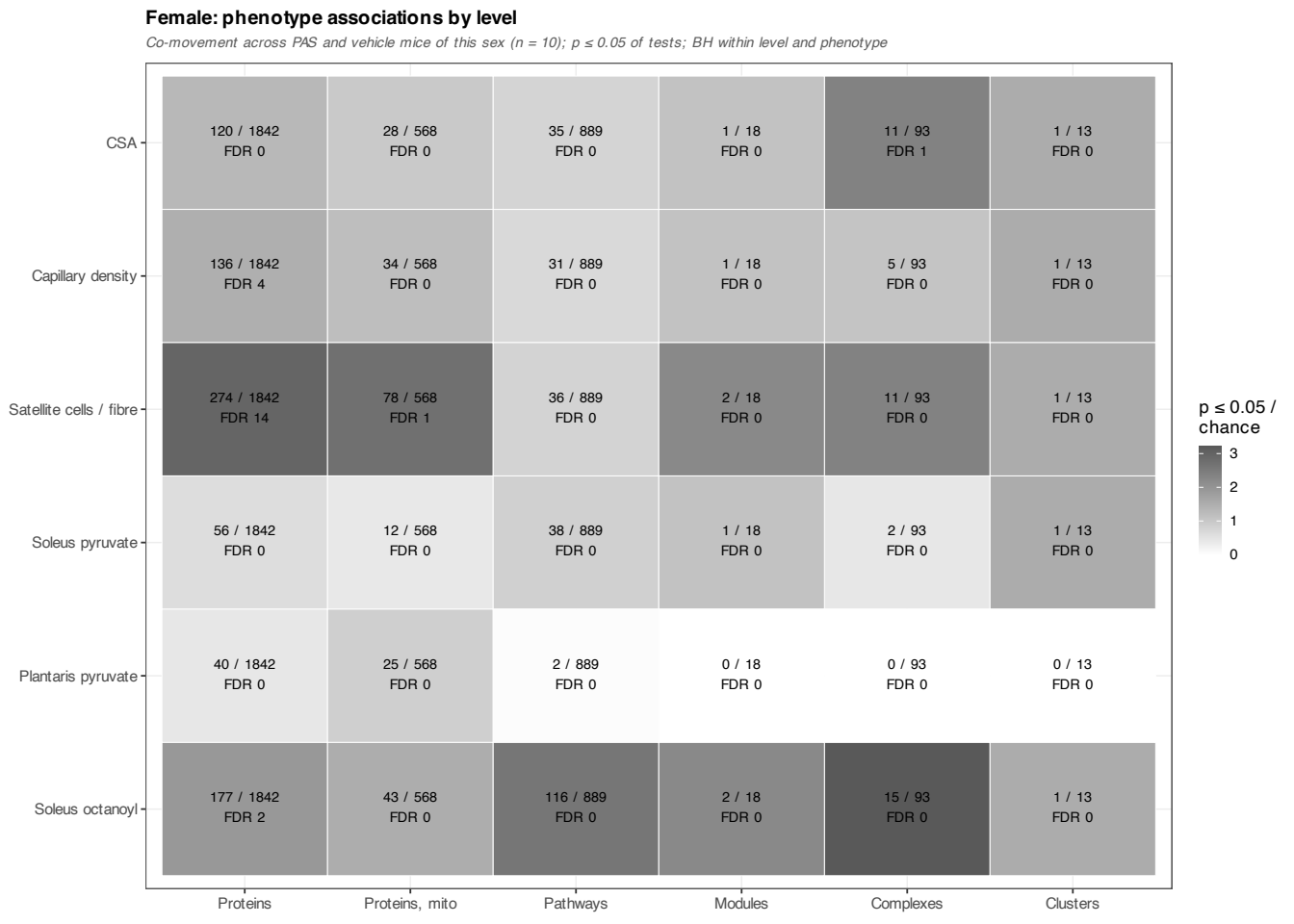
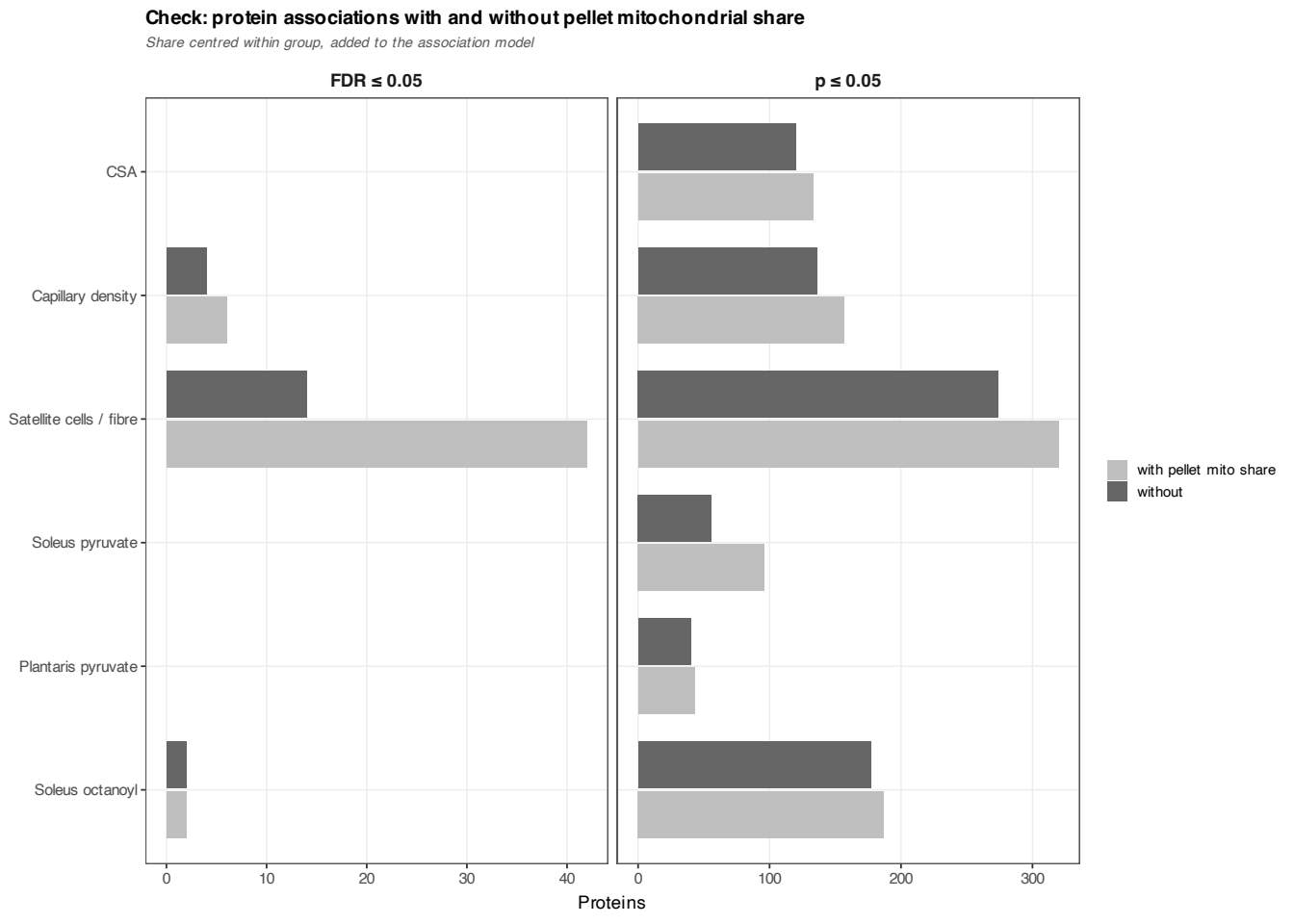
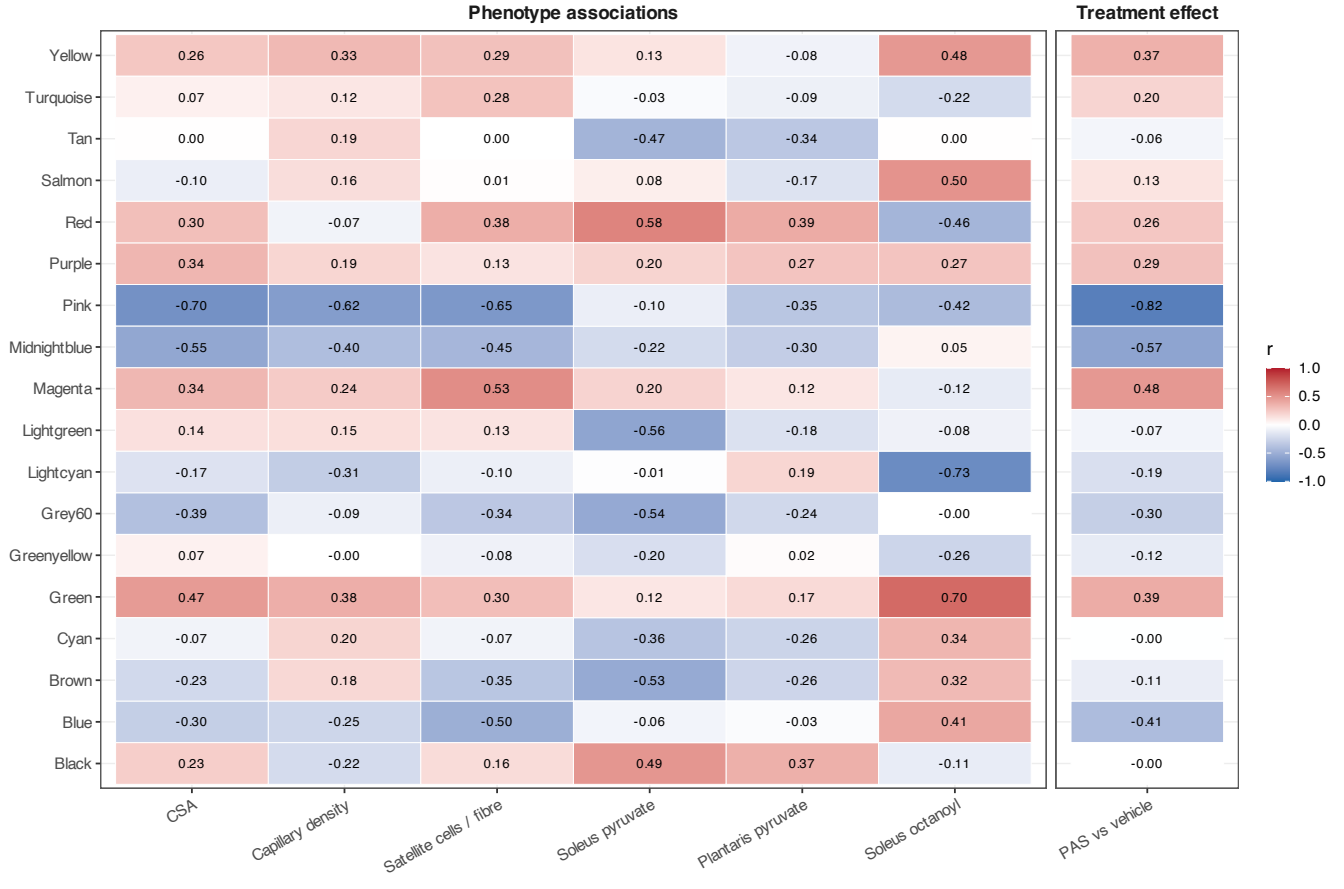

Female: phenotype effects of PAS
% change against vehicle, Welch t on log values, n = 5 v 5
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
p > 0.05
p ≤ 0.05
Female
-50%
+0%
+100%
+300%
+700%
PAS vs vehicle, % change (95% CI, log scale)

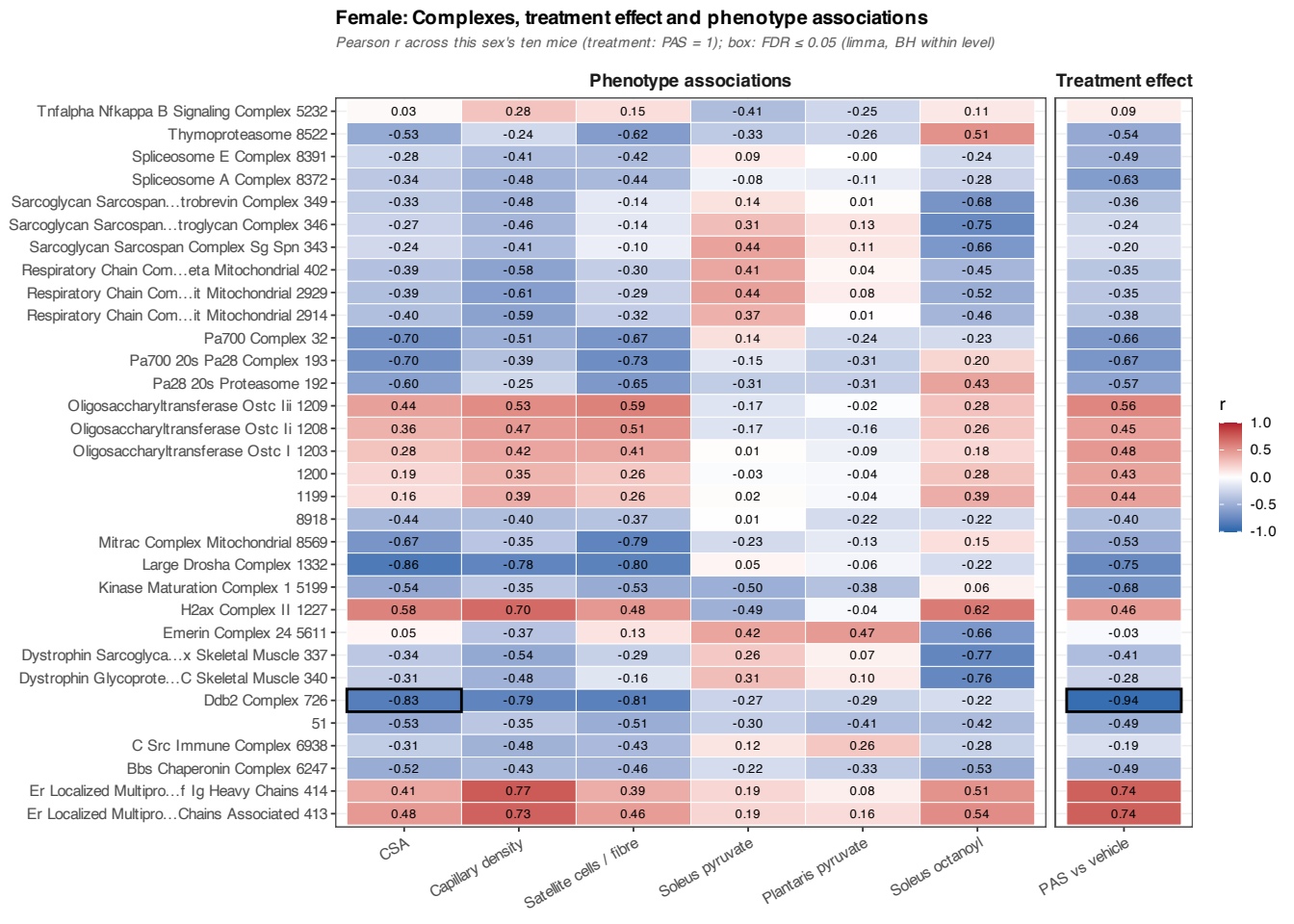

Female: phenotype associations by level
Co-movement across PAS and vehicle mice of this sex (n = 10); p ≤ 0.05 of tests; BH within level and phenotype
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
120 / 1842
FDR 0
28 / 568
FDR 0
35 / 889
FDR 0
1 / 18
FDR 0
11 / 93
FDR 1
1 / 13
FDR 0
136 / 1842
FDR 4
34 / 568
FDR 0
31 / 889
FDR 0
1 / 18
FDR 0
5 / 93
FDR 0
1 / 13
FDR 0
274 / 1842
FDR 14
78 / 568
FDR 1
36 / 889
FDR 0
2 / 18
FDR 0
11 / 93
FDR 0
1 / 13
FDR 0
56 / 1842
FDR 0
12 / 568
FDR 0
38 / 889
FDR 0
1 / 18
FDR 0
2 / 93
FDR 0
1 / 13
FDR 0
40 / 1842
FDR 0
25 / 568
FDR 0
2 / 889
FDR 0
0 / 18
FDR 0
0 / 93
FDR 0
0 / 13
FDR 0
177 / 1842
FDR 2
43 / 568
FDR 0
116 / 889
FDR 0
2 / 18
FDR 0
15 / 93
FDR 0
1 / 13
FDR 0
Proteins
Proteins, mito
Pathways
Modules
Complexes
Clusters
p ≤ 0.05 / chance
3
2
1
0

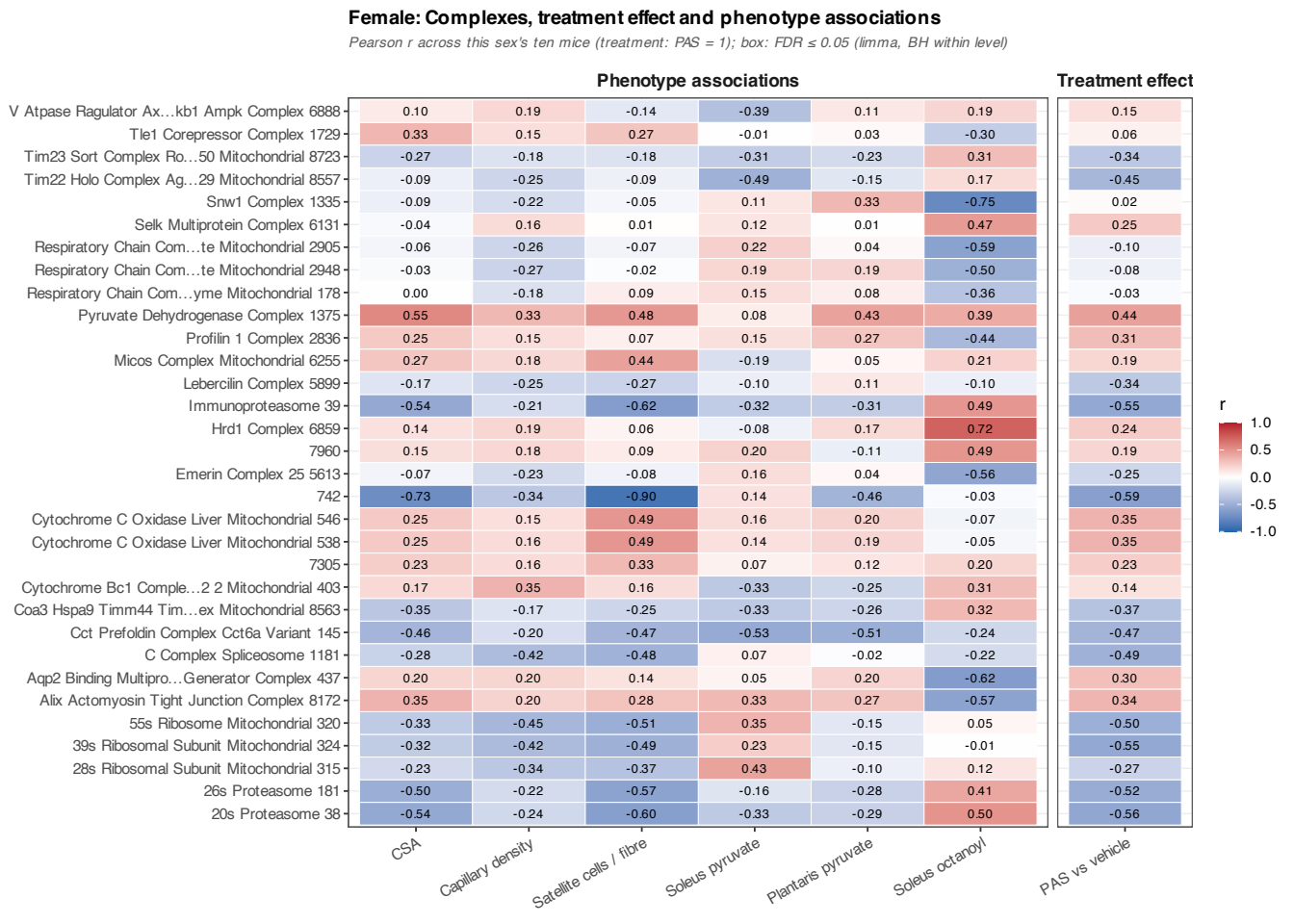

Check: protein associations with and without pellet mitochondrial share
Share centred within group, added to the association model
FDR ≤ 0.05
p ≤ 0.05
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
0
10
20
30
40
0
100
200
300
Proteins
with pellet mito share
without

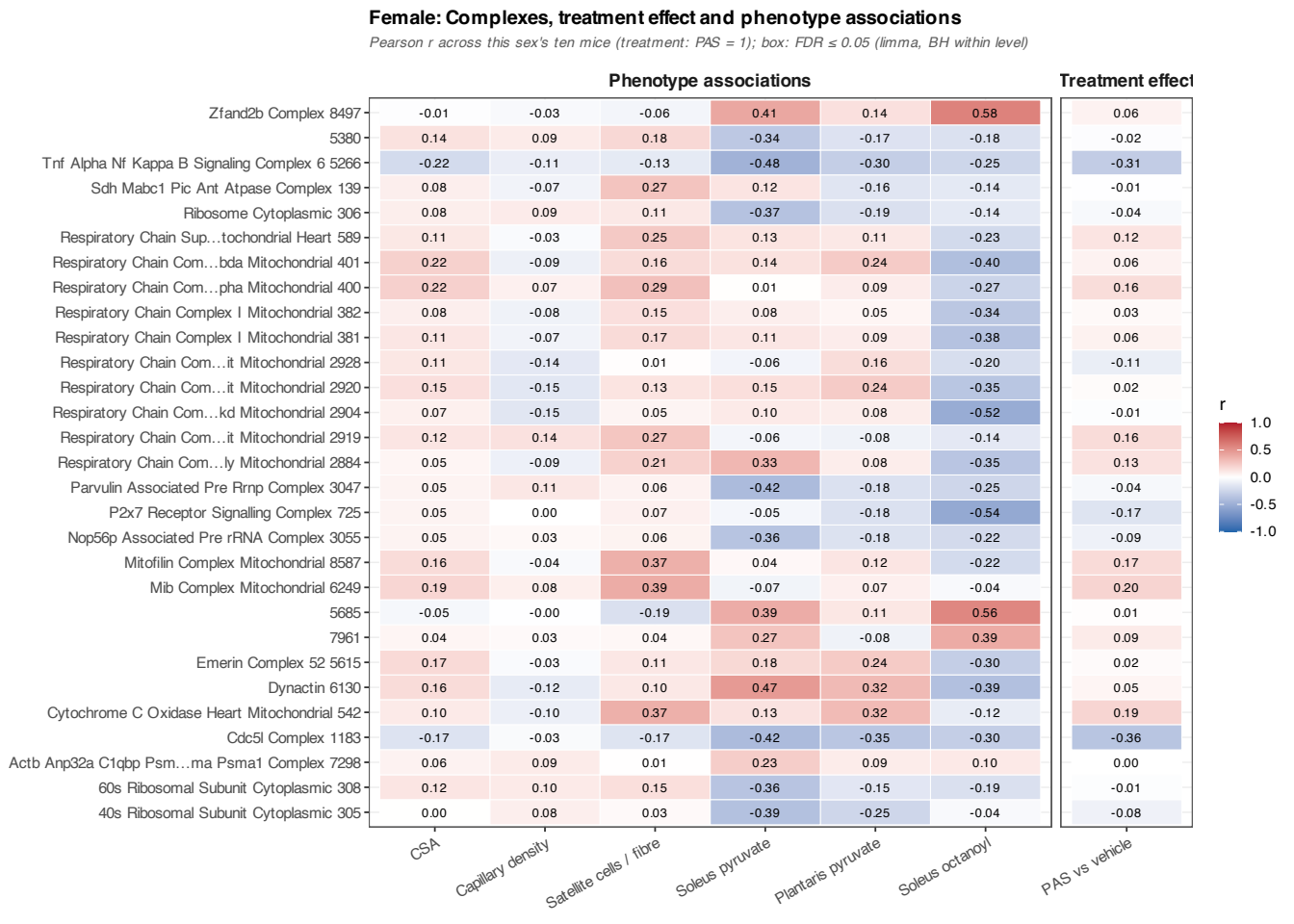
Female: Modules, treatment effect and phenotype associations

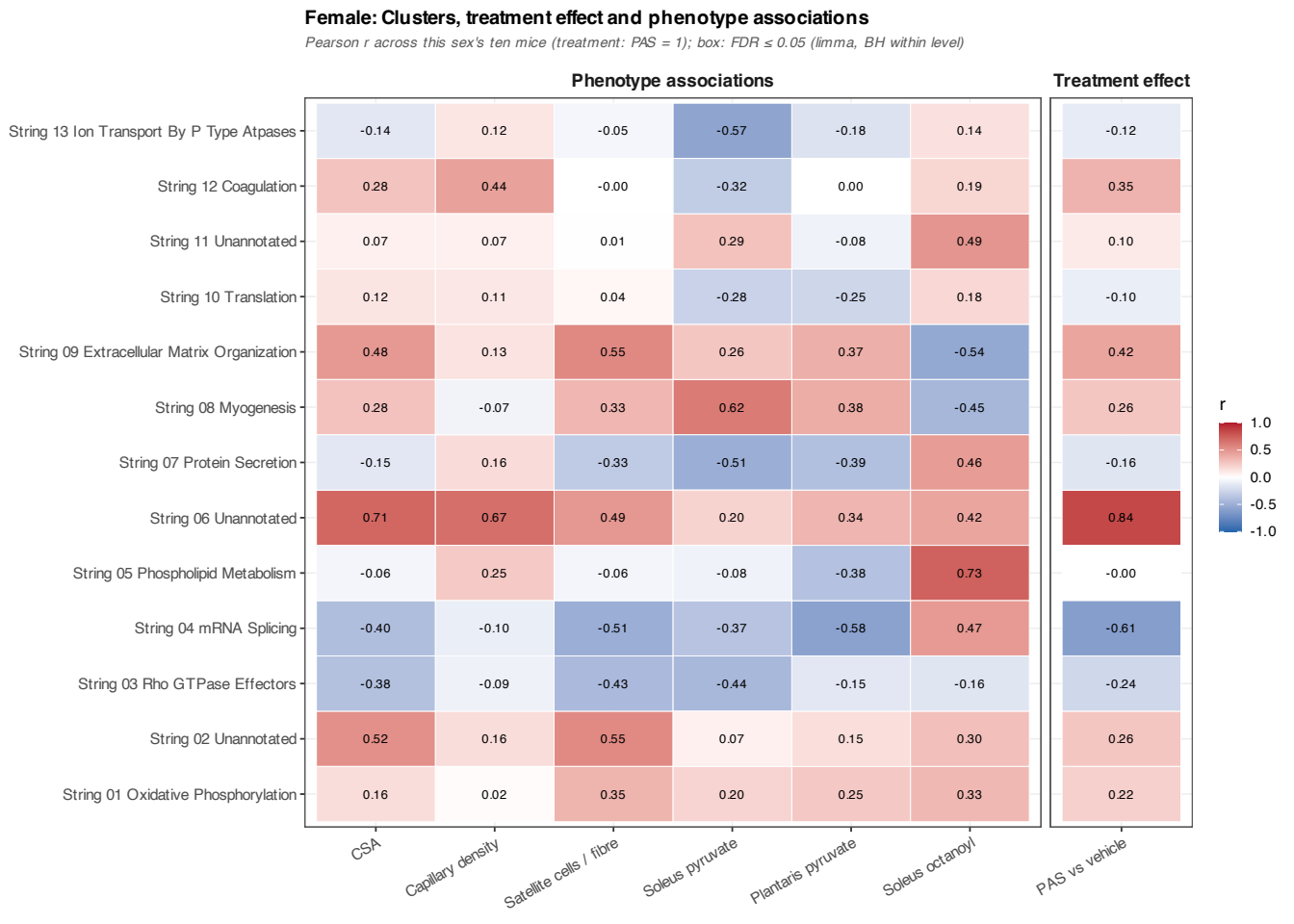
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)




Female: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Tnfalpha Nfkappa B Signaling Complex 5232 0.03 0.28 0.15 -0.41 -0.25 0.11 0.09
Thymoproteasome 8522 -0.53 -0.24 -0.62 -0.33 -0.26 0.51 -0.54
Spliceosome E Complex 8391 -0.28 -0.41 -0.42 0.09 -0.00 -0.24 -0.49
Spliceosome A Complex 8372 -0.34 -0.48 -0.44 -0.08 -0.11 -0.28 -0.63
Sarcoglycan Sarcospan…trobrevin Complex 349 -0.33 -0.48 -0.14 0.14 0.01 -0.68 -0.36
Sarcoglycan Sarcospan…troglycan Complex 346 -0.27 -0.46 -0.14 0.31 0.13 -0.75 -0.24
Sarcoglycan Sarcospan Complex Sg Spn 343 -0.24 -0.41 -0.10 0.44 0.11 -0.66 -0.20
Respiratory Chain Com…eta Mitochondrial 402 -0.39 -0.58 -0.30 0.41 0.04 -0.45 -0.35
Respiratory Chain Com…it Mitochondrial 2929 -0.39 -0.61 -0.29 0.44 0.08 -0.52 -0.35
Respiratory Chain Com…it Mitochondrial 2914 -0.40 -0.59 -0.32 0.37 0.01 -0.46 -0.38
Pa700 Complex 32 -0.70 -0.51 -0.67 0.14 -0.24 -0.23 -0.66
Pa700 20s Pa28 Complex 193 -0.70 -0.39 -0.73 -0.15 -0.31 0.20 -0.67
Pa28 20s Proteasome 192 -0.60 -0.25 -0.65 -0.31 -0.31 0.43 -0.57
Oligosaccharyltransferase Ostc Iii 1209 0.44 0.53 0.59 -0.17 -0.02 0.28 0.56
Oligosaccharyltransferase Ostc Ii 1208 0.36 0.47 0.51 -0.17 -0.16 0.26 0.45
Oligosaccharyltransferase Ostc I 1203 0.28 0.42 0.41 0.01 -0.09 0.18 0.48
1200 0.19 0.35 0.26 -0.03 -0.04 0.28 0.43
1199 0.16 0.39 0.26 0.02 -0.04 0.39 0.44
8918 -0.44 -0.40 -0.37 0.01 -0.22 -0.22 -0.40
Mitrac Complex Mitochondrial 8569 -0.67 -0.35 -0.79 -0.23 -0.13 0.15 -0.53
Large Drosha Complex 1332 -0.86 -0.78 -0.80 0.05 -0.06 -0.22 -0.75
Kinase Maturation Complex 1 5199 -0.54 -0.35 -0.53 -0.50 -0.38 0.06 -0.68
H2ax Complex II 1227 0.58 0.70 0.48 -0.49 -0.04 0.62 0.46
Emerin Complex 24 5611 0.05 -0.37 0.13 0.42 0.47 -0.66 -0.03
Dystrophin Sarcoglyca…x Skeletal Muscle 337 -0.34 -0.54 -0.29 0.26 0.07 -0.77 -0.41
Dystrophin Glycoprote…C Skeletal Muscle 340 -0.31 -0.48 -0.16 0.31 0.10 -0.76 -0.28
Ddb2 Complex 726 -0.83 -0.79 -0.81 -0.27 -0.29 -0.22 -0.94
51 -0.53 -0.35 -0.51 -0.30 -0.41 -0.42 -0.49
C Src Immune Complex 6938 -0.31 -0.48 -0.43 0.12 0.26 -0.28 -0.19
Bbs Chaperonin Complex 6247 -0.52 -0.43 -0.46 -0.22 -0.33 -0.53 -0.49
Er Localized Multipro…f Ig Heavy Chains 414 0.41 0.77 0.39 0.19 0.08 0.51 0.74
Er Localized Multipro…Chains Associated 413 0.48 0.73 0.46 0.19 0.16 0.54 0.74
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

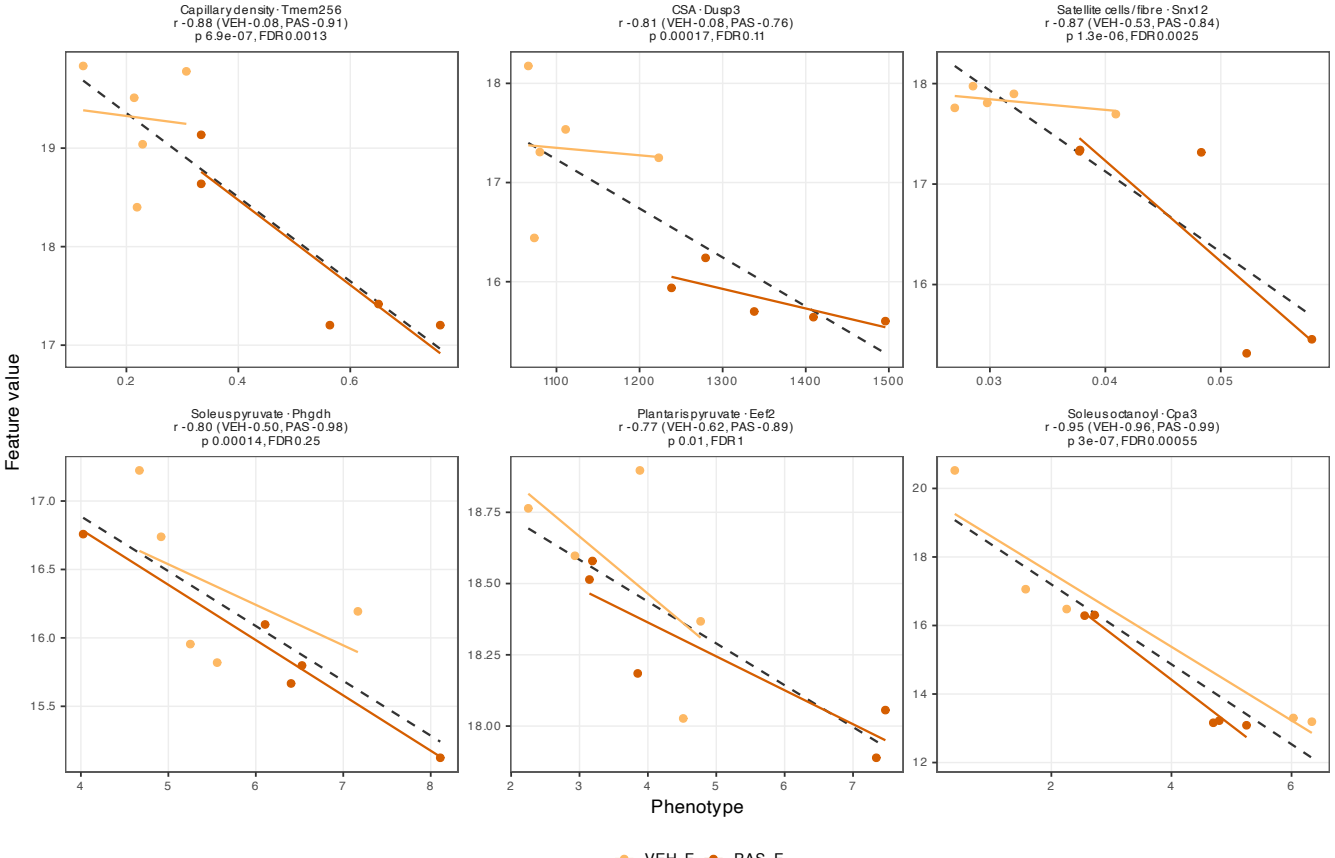

Female: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
V Atpase Ragulator Ax…kb1 Ampk Complex 6888
Tle1 Corepressor Complex 1729
Tim23 Sort Complex Ro…50 Mitochondrial 8723
Tim22 Holo Complex Ag…29 Mitochondrial 8557
Snw1 Complex 1335
Selk Multiprotein Complex 6131
Respiratory Chain Com…te Mitochondrial 2905
Respiratory Chain Com…te Mitochondrial 2948
Respiratory Chain Com…yme Mitochondrial 178
Pyruvate Dehydrogenase Complex 1375
Profilin 1 Complex 2836
Micos Complex Mitochondrial 6255
Lebercilin Complex 5899
Immunoproteasome 39
Hrd1 Complex 6859
7960
Emerin Complex 25 5613
742
Cytochrome C Oxidase Liver Mitochondrial 546
Cytochrome C Oxidase Liver Mitochondrial 538
7305
Cytochrome Bc1 Comple…2 2 Mitochondrial 403
Coa3 Hspa9 Timm44 Tim…ex Mitochondrial 8563
Cct Prefoldin Complex Cct6a Variant 145
C Complex Spliceosome 1181
Aqp2 Binding Multipro…Generator Complex 437
Alix Actomyosin Tight Junction Complex 8172
55s Ribosome Mitochondrial 320
39s Ribosomal Subunit Mitochondrial 324
28s Ribosomal Subunit Mitochondrial 315
26s Proteasome 181
20s Proteasome 38
0.10 0.19 -0.14 -0.39 0.11 0.19 0.15
0.33 0.15 0.27 -0.01 0.03 -0.30 0.06
-0.27 -0.18 -0.18 -0.31 -0.23 0.31 -0.34
-0.09 -0.25 -0.09 -0.49 -0.15 0.17 -0.45
-0.09 -0.22 -0.05 0.11 0.33 -0.75 0.02
-0.04 0.16 0.01 0.12 0.01 0.47 0.25
-0.06 -0.26 -0.07 0.22 0.04 -0.59 -0.10
-0.03 -0.27 -0.02 0.19 0.19 -0.50 -0.08
0.00 -0.18 0.09 0.15 0.08 -0.36 -0.03
0.55 0.33 0.48 0.08 0.43 0.39 0.44
0.25 0.15 0.07 0.15 0.27 -0.44 0.31
0.27 0.18 0.44 -0.19 0.05 0.21 0.19
-0.17 -0.25 -0.27 -0.10 0.11 -0.10 -0.34
-0.54 -0.21 -0.62 -0.32 -0.31 0.49 -0.55
0.14 0.19 0.06 -0.08 0.17 0.72 0.24
0.15 0.18 0.09 0.20 -0.11 0.49 0.19
-0.07 -0.23 -0.08 0.16 0.04 -0.56 -0.25
-0.73 -0.34 -0.90 0.14 -0.46 -0.03 -0.59
0.25 0.15 0.49 0.16 0.20 -0.07 0.35
0.25 0.16 0.49 0.14 0.19 -0.05 0.35
0.23 0.16 0.33 0.07 0.12 0.20 0.23
0.17 0.35 0.16 -0.33 -0.25 0.31 0.14
-0.35 -0.17 -0.25 -0.33 -0.26 0.32 -0.37
-0.46 -0.20 -0.47 -0.53 -0.51 -0.24 -0.47
-0.28 -0.42 -0.48 0.07 -0.02 -0.22 -0.49
0.20 0.20 0.14 0.05 0.20 -0.62 0.30
0.35 0.20 0.28 0.33 0.27 -0.57 0.34
-0.33 -0.45 -0.51 0.35 -0.15 0.05 -0.50
-0.32 -0.42 -0.49 0.23 -0.15 -0.01 -0.55
-0.23 -0.34 -0.37 0.43 -0.10 0.12 -0.27
-0.50 -0.22 -0.57 -0.16 -0.28 0.41 -0.52
-0.54 -0.24 -0.60 -0.33 -0.29 0.50 -0.56
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Female: Complexes, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
Zfand2b Complex 8497 -0.01 -0.03 -0.06 0.41 0.14 0.58 0.06
5380 0.14 0.09 0.18 -0.34 -0.17 -0.18 -0.02
Tnf Alpha Nf Kappa B Signaling Complex 6 5266 -0.22 -0.11 -0.13 -0.48 -0.30 -0.25 -0.31
Sdh Mabc1 Pic Ant Atpase Complex 139 0.08 -0.07 0.27 0.12 -0.16 -0.14 -0.01
Ribosome Cytoplasmic 306 0.08 0.09 0.11 -0.37 -0.19 -0.14 -0.04
Respiratory Chain Sup…tochondrial Heart 589 0.11 -0.03 0.25 0.13 0.11 -0.23 0.12
Respiratory Chain Com…bda Mitochondrial 401 0.22 -0.09 0.16 0.14 0.24 -0.40 0.06
Respiratory Chain Com…pha Mitochondrial 400 0.22 0.07 0.29 0.01 0.09 -0.27 0.16
Respiratory Chain Complex I Mitochondrial 382 0.08 -0.08 0.15 0.08 0.05 -0.34 0.03
Respiratory Chain Complex I Mitochondrial 381 0.11 -0.07 0.17 0.11 0.09 -0.38 0.06
Respiratory Chain Com…it Mitochondrial 2928 0.11 -0.14 0.01 -0.06 0.16 -0.20 -0.11
Respiratory Chain Com…it Mitochondrial 2920 0.15 -0.15 0.13 0.15 0.24 -0.35 0.02
Respiratory Chain Com…kd Mitochondrial 2904 0.07 -0.15 0.05 0.10 0.08 -0.52 -0.01
Respiratory Chain Com…it Mitochondrial 2919 0.12 0.14 0.27 -0.06 -0.08 -0.14 0.16
Respiratory Chain Com…ly Mitochondrial 2884 0.05 -0.09 0.21 0.33 0.08 -0.35 0.13
Parvulin Associated Pre Rrnp Complex 3047 0.05 0.11 0.06 -0.42 -0.18 -0.25 -0.04
P2x7 Receptor Signalling Complex 725 0.05 0.00 0.07 -0.05 -0.18 -0.54 -0.17
Nop56p Associated Pre rRNA Complex 3055 0.05 0.03 0.06 -0.36 -0.18 -0.22 -0.09
Mitofilin Complex Mitochondrial 8587 0.16 -0.04 0.37 0.04 0.12 -0.22 0.17
Mib Complex Mitochondrial 6249 0.19 0.08 0.39 -0.07 0.07 -0.04 0.20
5685 -0.05 -0.00 -0.19 0.39 0.11 0.56 0.01
7961 0.04 0.03 0.04 0.27 -0.08 0.39 0.09
Emerin Complex 52 5615 0.17 -0.03 0.11 0.18 0.24 -0.30 0.02
Dynactin 6130 0.16 -0.12 0.10 0.47 0.32 -0.39 0.05
Cytochrome C Oxidase Heart Mitochondrial 542 0.10 -0.10 0.37 0.13 0.32 -0.12 0.19
Cdc5l Complex 1183 -0.17 -0.03 -0.17 -0.42 -0.35 -0.30 -0.36
Actb Anp32a C1qbp Psm…ma Psma1 Complex 7298 0.06 0.09 0.01 0.23 0.09 0.10 0.00
60s Ribosomal Subunit Cytoplasmic 308 0.12 0.10 0.15 -0.36 -0.15 -0.19 -0.01
40s Ribosomal Subunit Cytoplasmic 305 0.00 0.08 0.03 -0.39 -0.25 -0.04 -0.08
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0


Female: Clusters, treatment effect and phenotype associations
Pearson r across this sex's ten mice (treatment: PAS = 1); box: FDR ≤ 0.05 (limma, BH within level)
Phenotype associations
Treatment effect
String 13 Ion Transport By P Type Atpases
String 12 Coagulation
String 11 Unannotated
String 10 Translation
String 09 Extracellular Matrix Organization
String 08 Myogenesis
String 07 Protein Secretion
String 06 Unannotated
String 05 Phospholipid Metabolism
String 04 mRNA Splicing
String 03 Rho GTPase Effectors
String 02 Unannotated
String 01 Oxidative Phosphorylation
-0.14 0.12 -0.05 -0.57 -0.18 0.14 -0.12
0.28 0.44 -0.00 -0.32 0.00 0.19 0.35
0.07 0.07 0.01 0.29 -0.08 0.49 0.10
0.12 0.11 0.04 -0.28 -0.25 0.18 -0.10
0.48 0.13 0.55 0.26 0.37 -0.54 0.42
0.28 -0.07 0.33 0.62 0.38 -0.45 0.26
-0.15 0.16 -0.33 -0.51 -0.39 0.46 -0.16
0.71 0.67 0.49 0.20 0.34 0.42 0.84
-0.06 0.25 -0.06 -0.08 -0.38 0.73 -0.00
-0.40 -0.10 -0.51 -0.37 -0.58 0.47 -0.61
-0.38 -0.09 -0.43 -0.44 -0.15 -0.16 -0.24
0.52 0.16 0.55 0.07 0.15 0.30 0.26
0.16 0.02 0.35 0.20 0.25 0.33 0.22
CSA
Capillary density
Satellite cells / fibre
Soleus pyruvate
Plantaris pyruvate
Soleus octanoyl
PAS vs vehicle
r
1.0
0.5
0.0
-0.5
-1.0

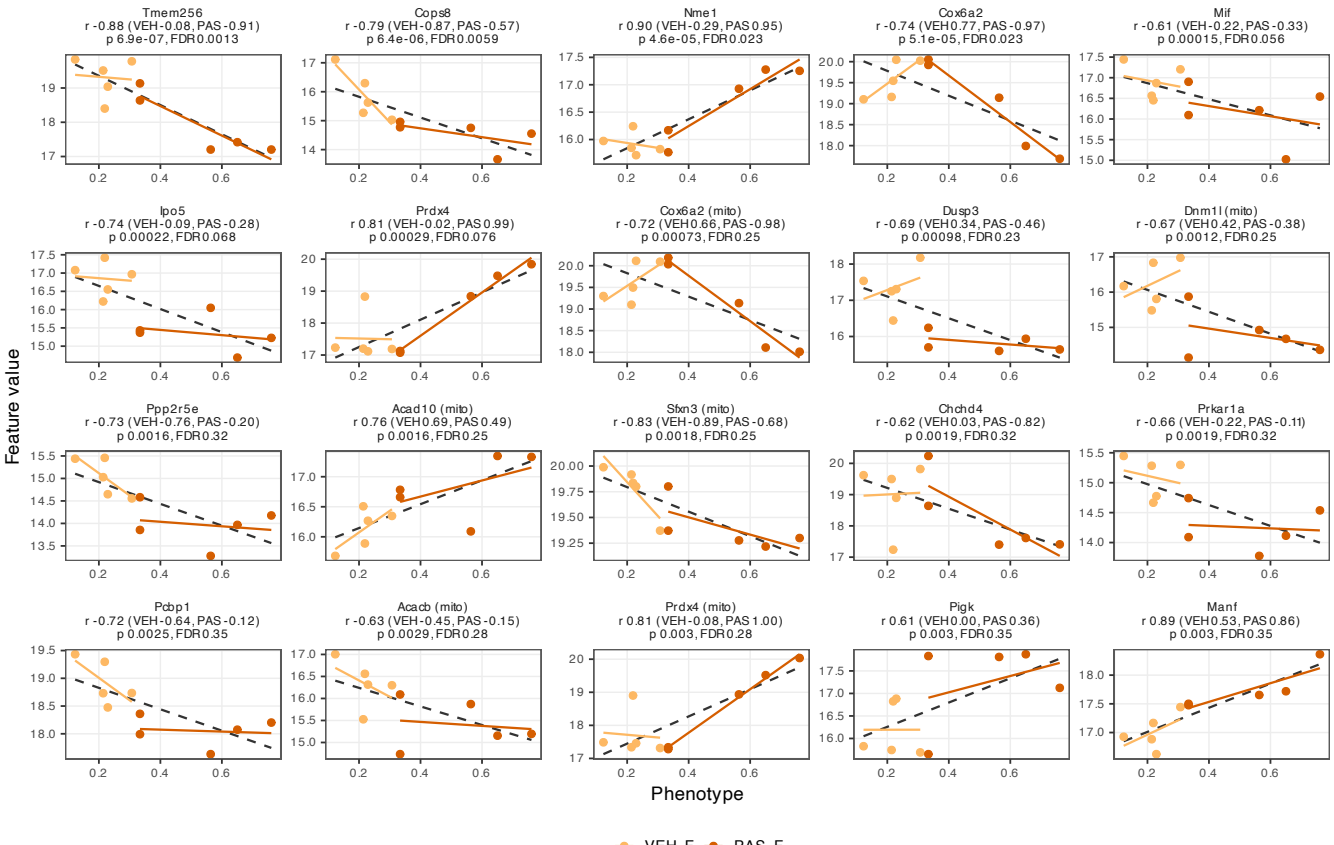
Female: strongest robust protein per phenotype

Smallest association p that keeps its sign and p ≤ 0.05 with any one mouse left out



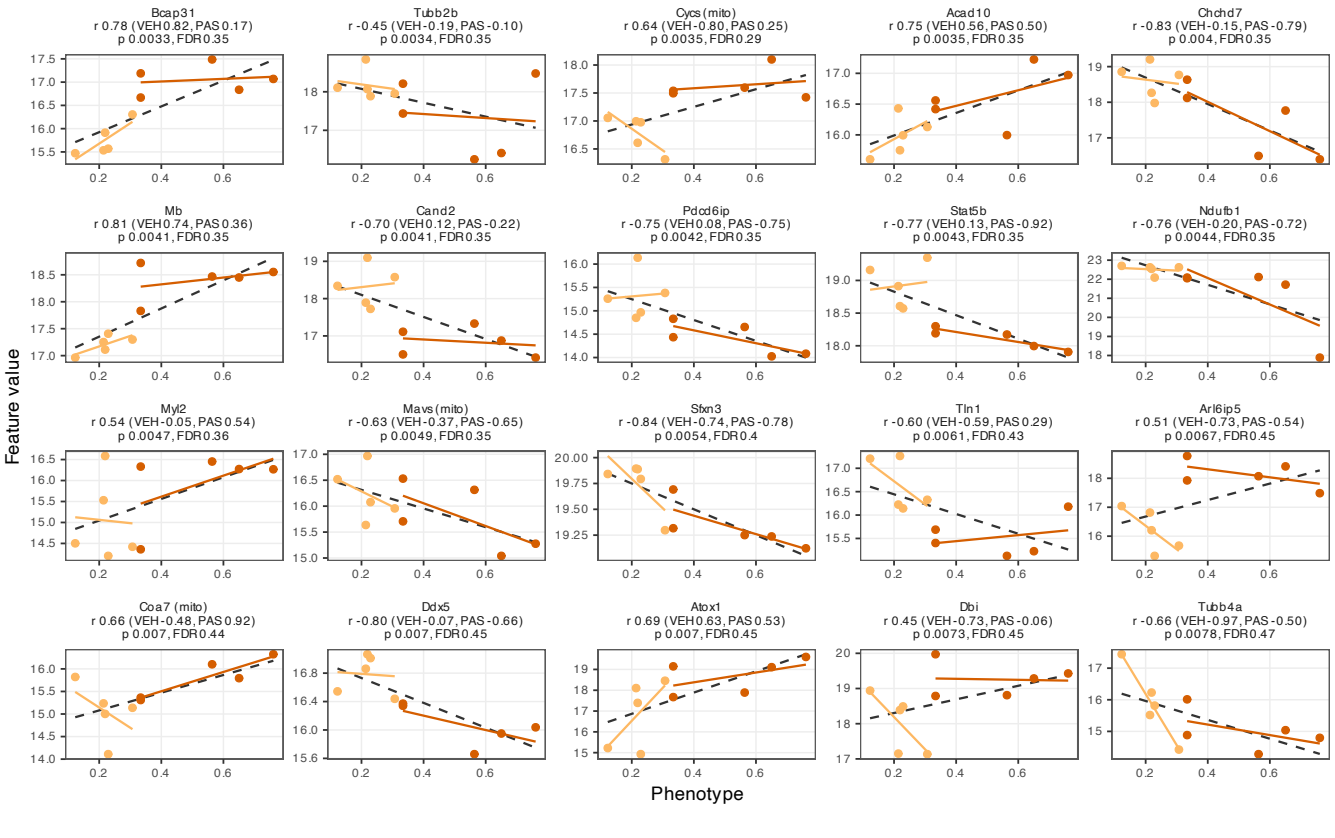
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



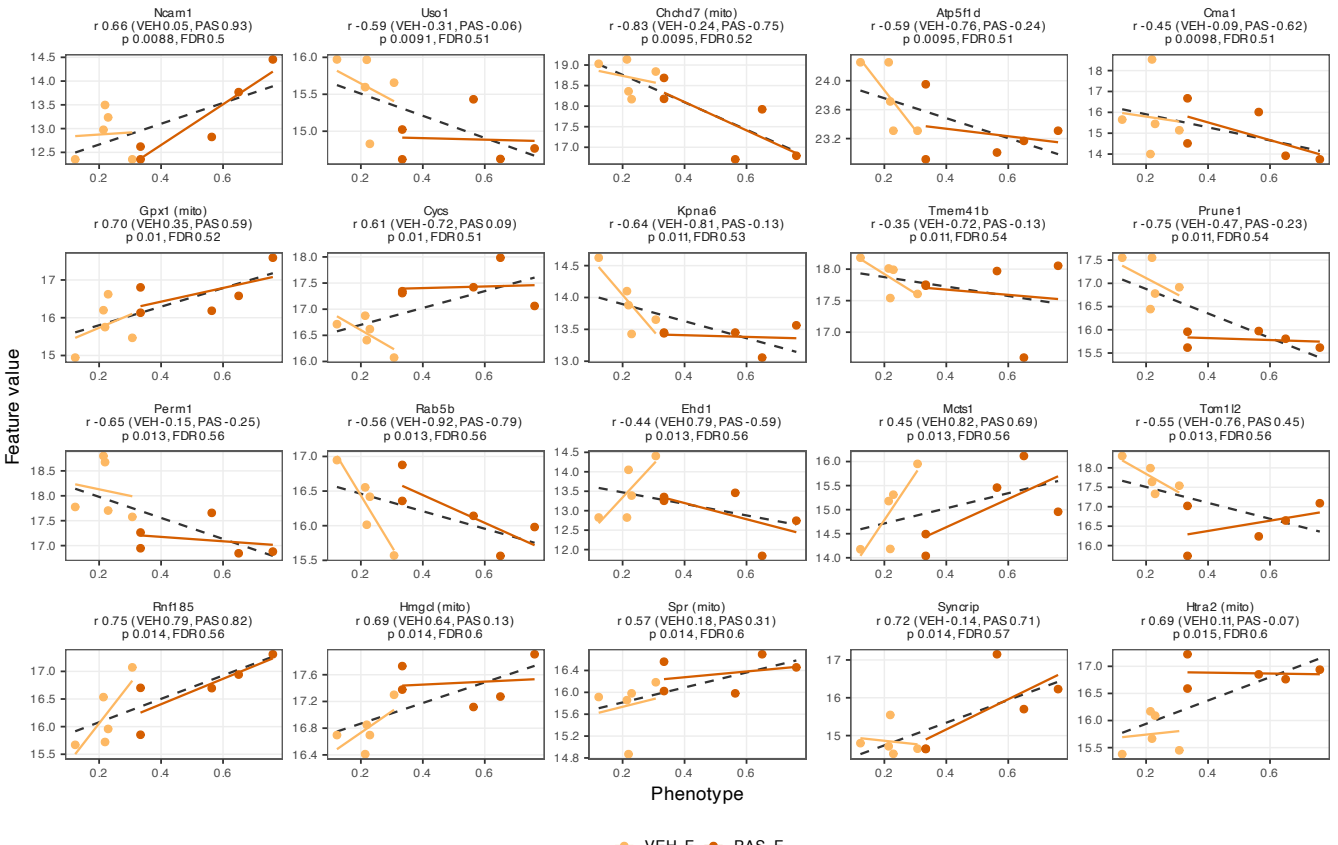
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



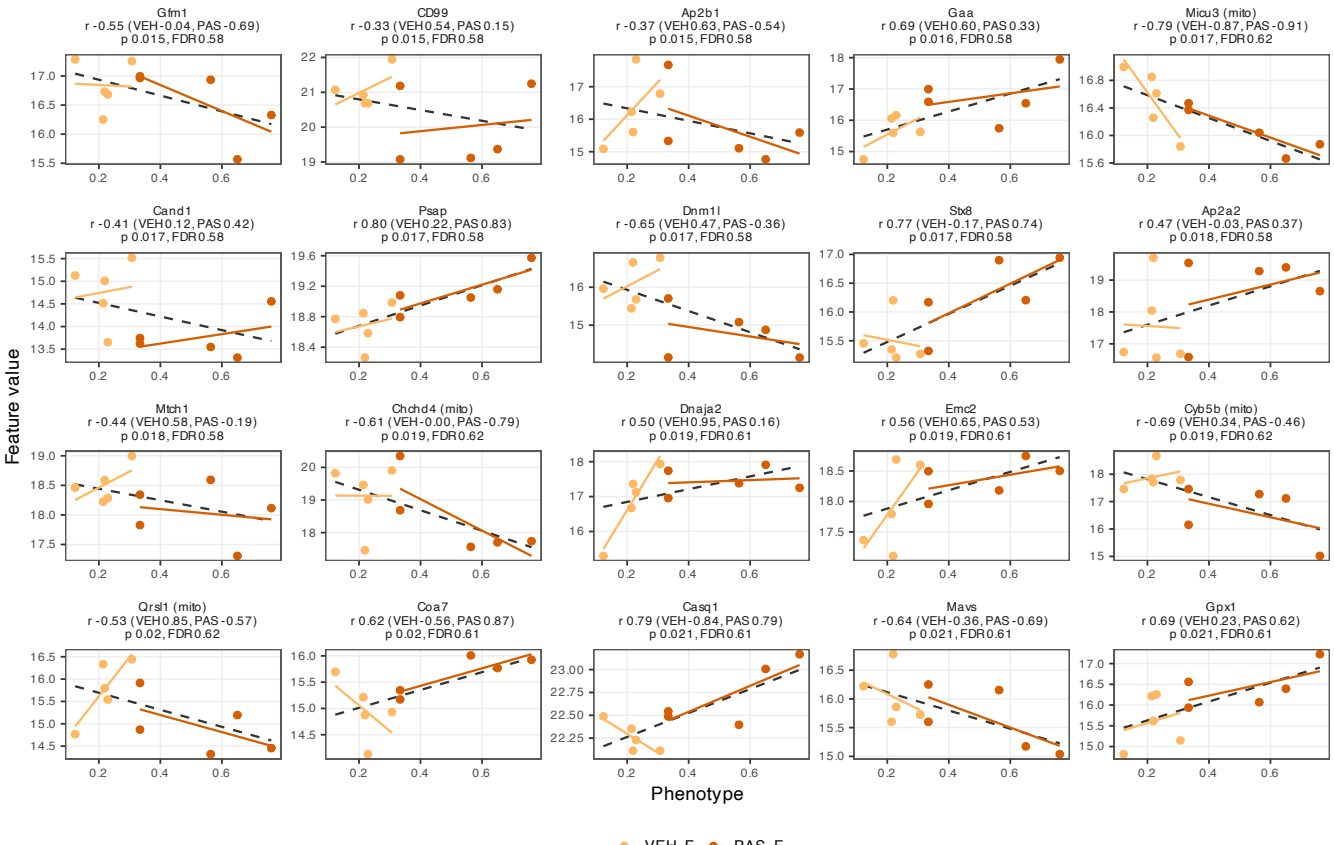
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



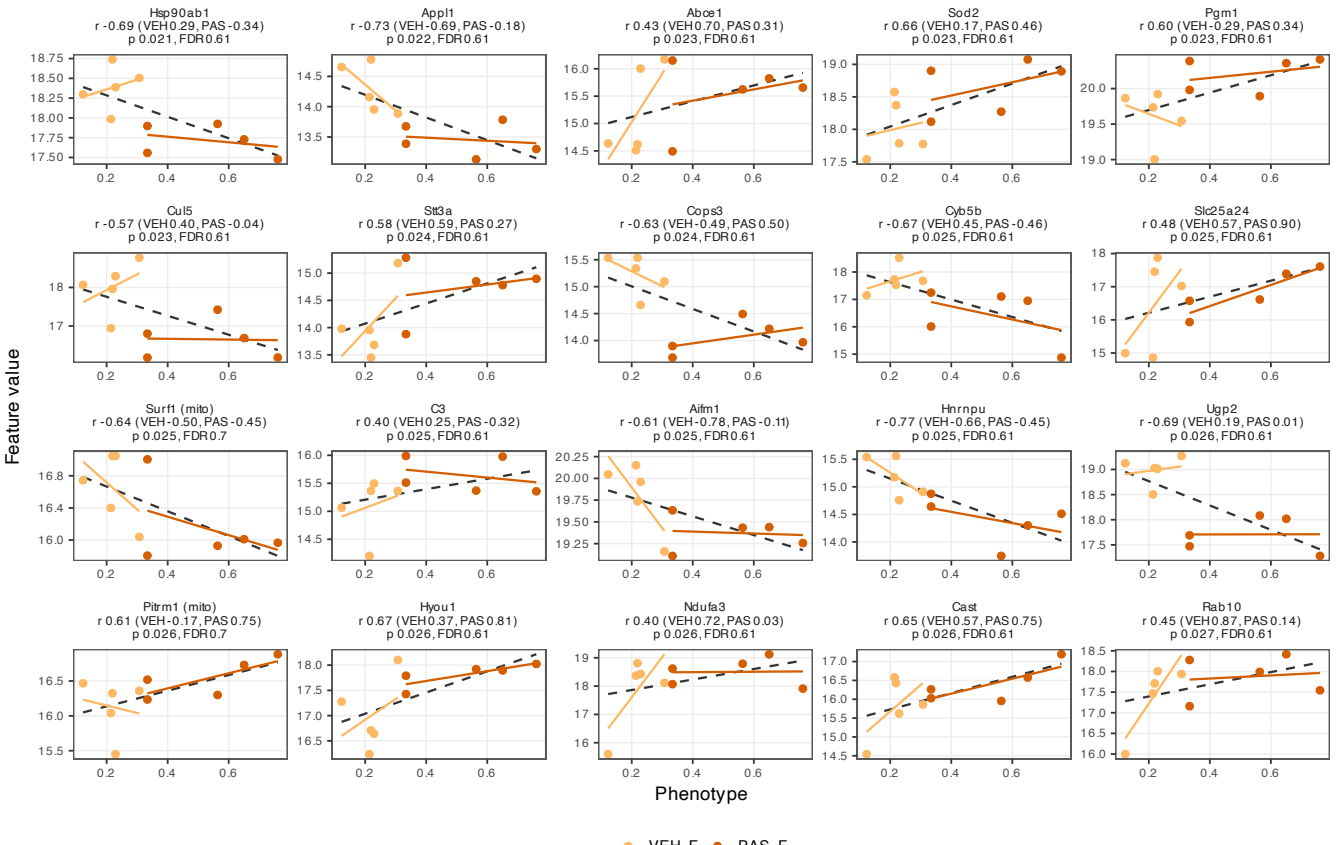
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



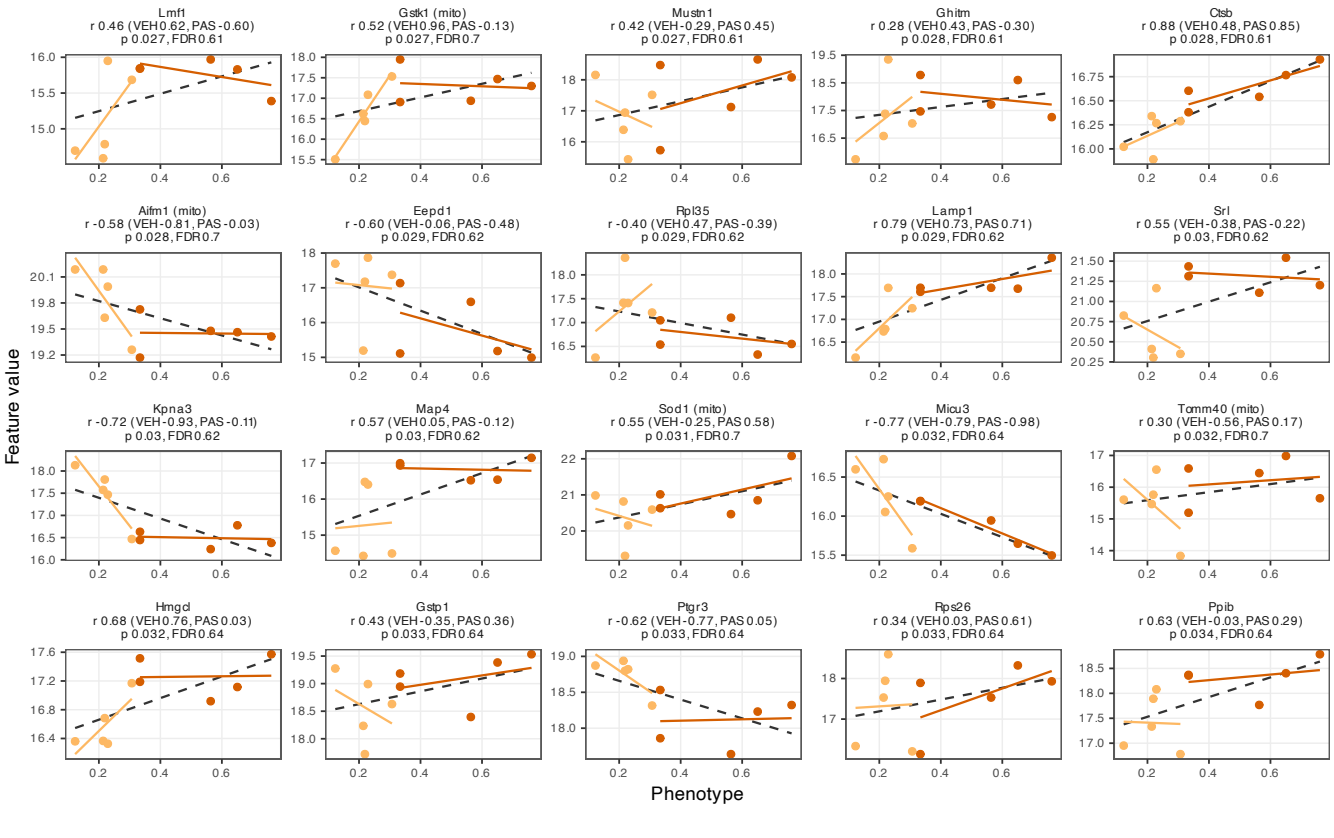
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



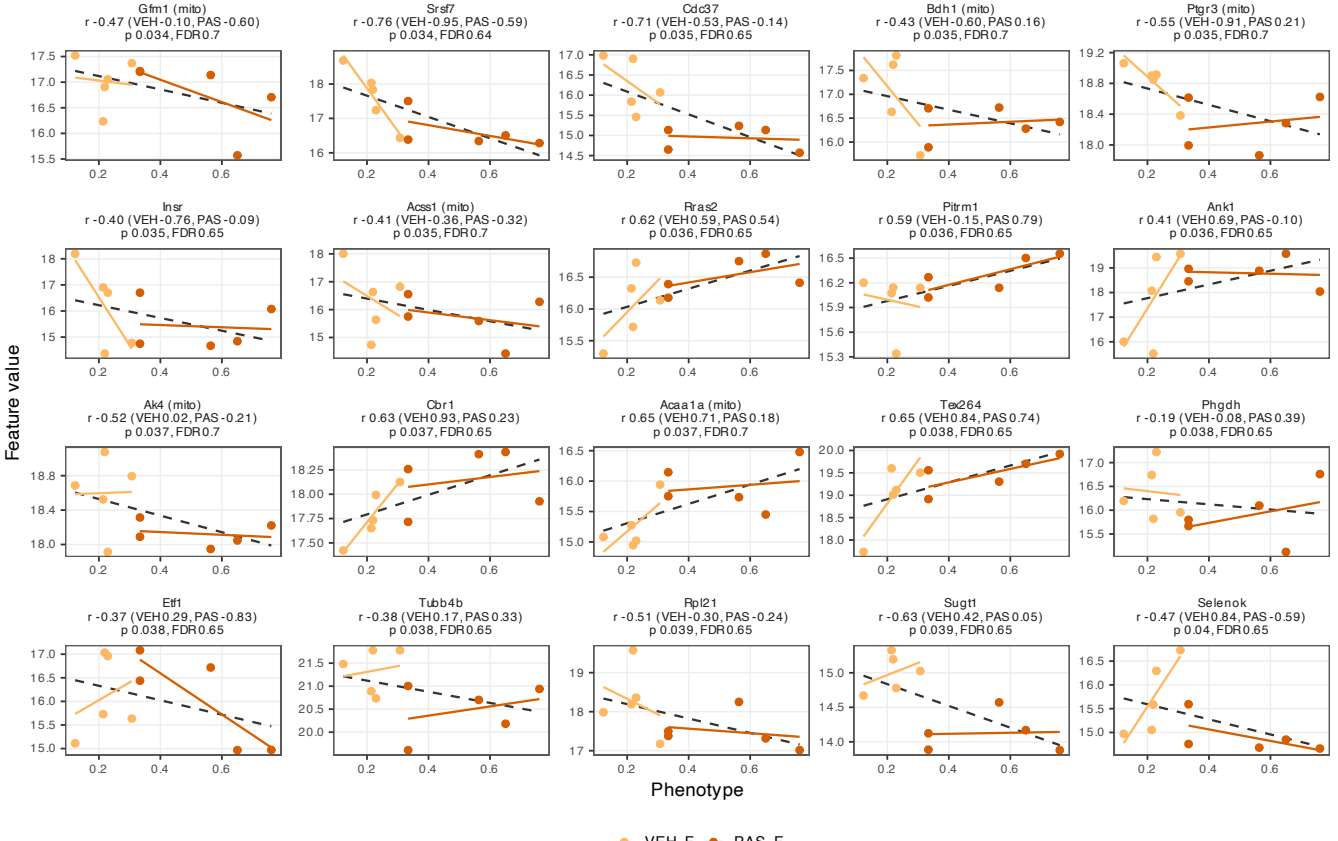
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



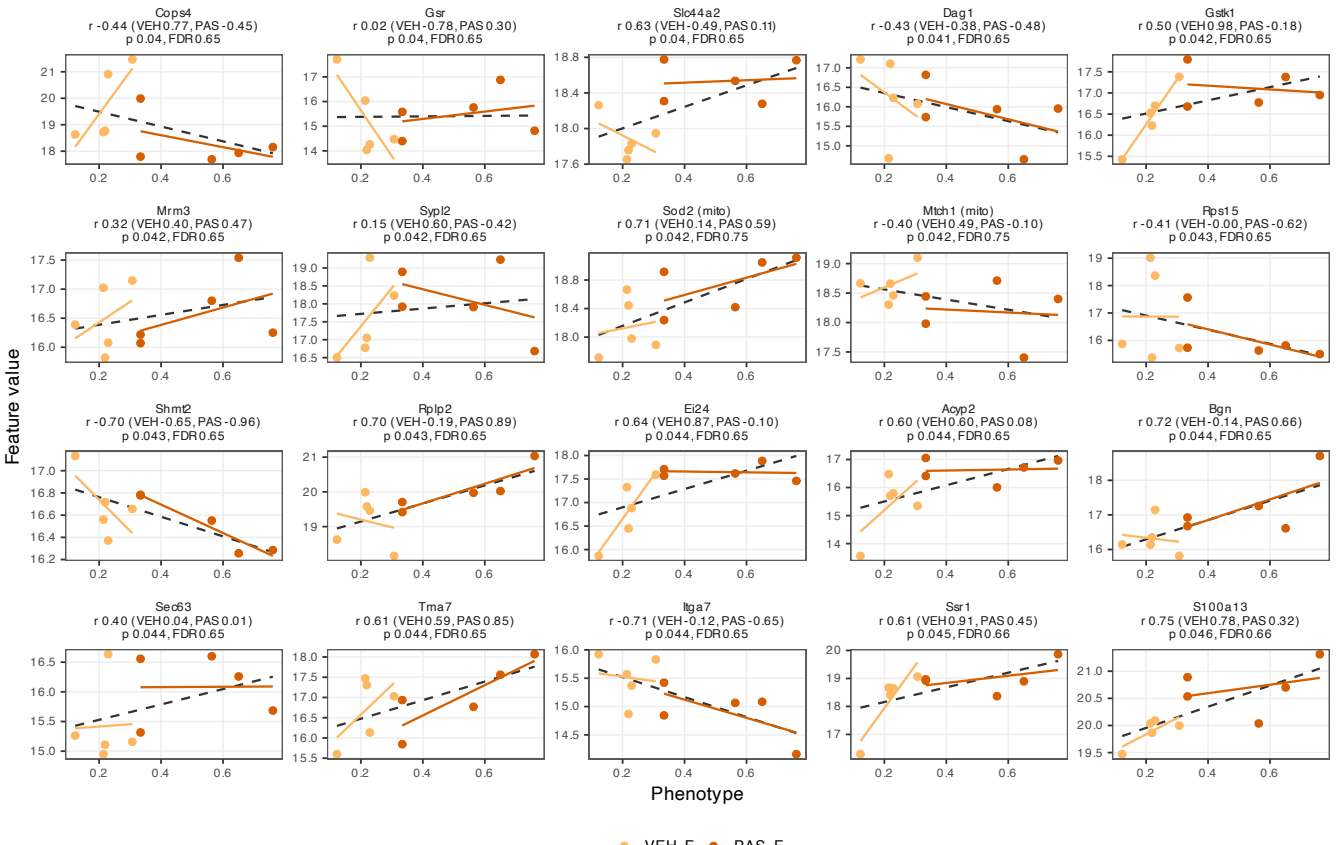
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



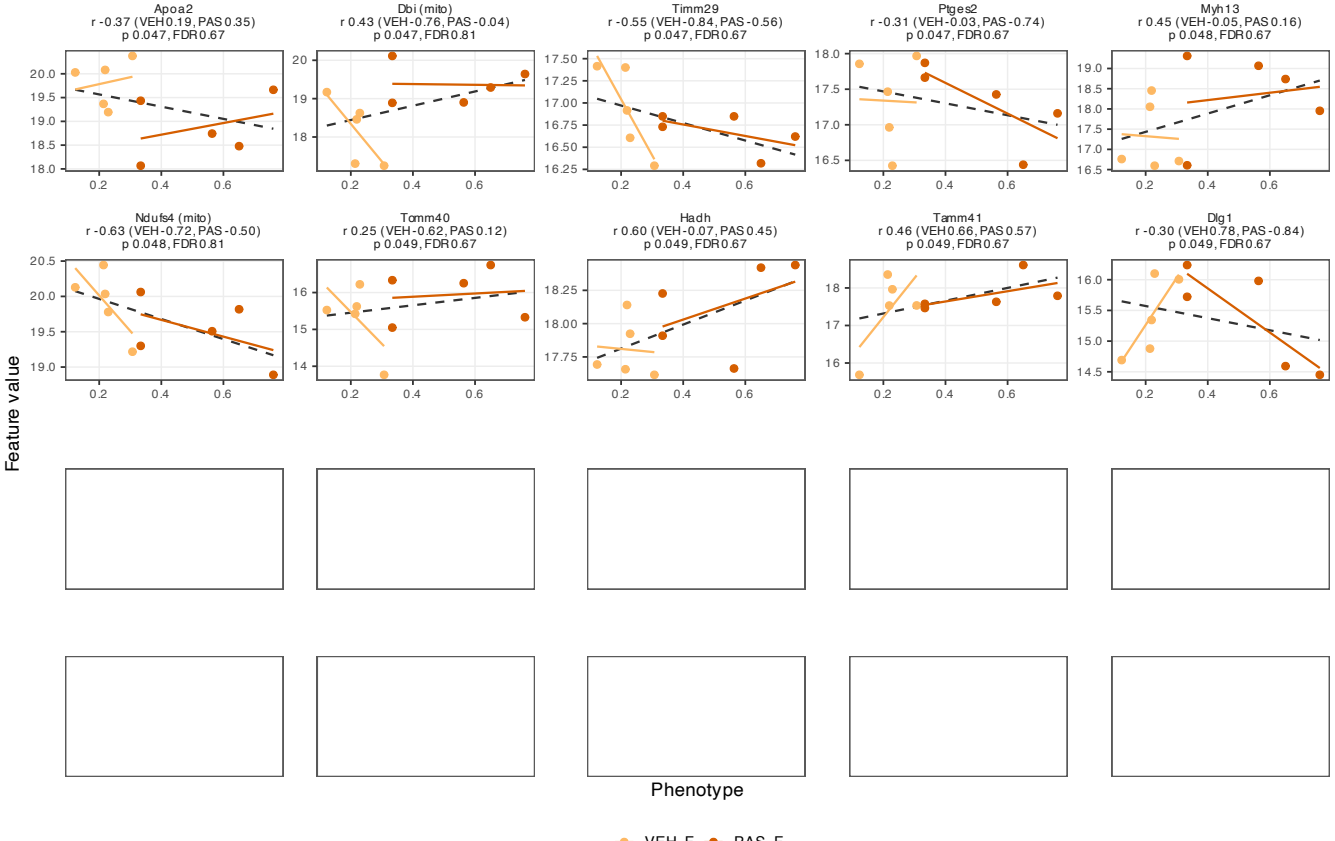
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



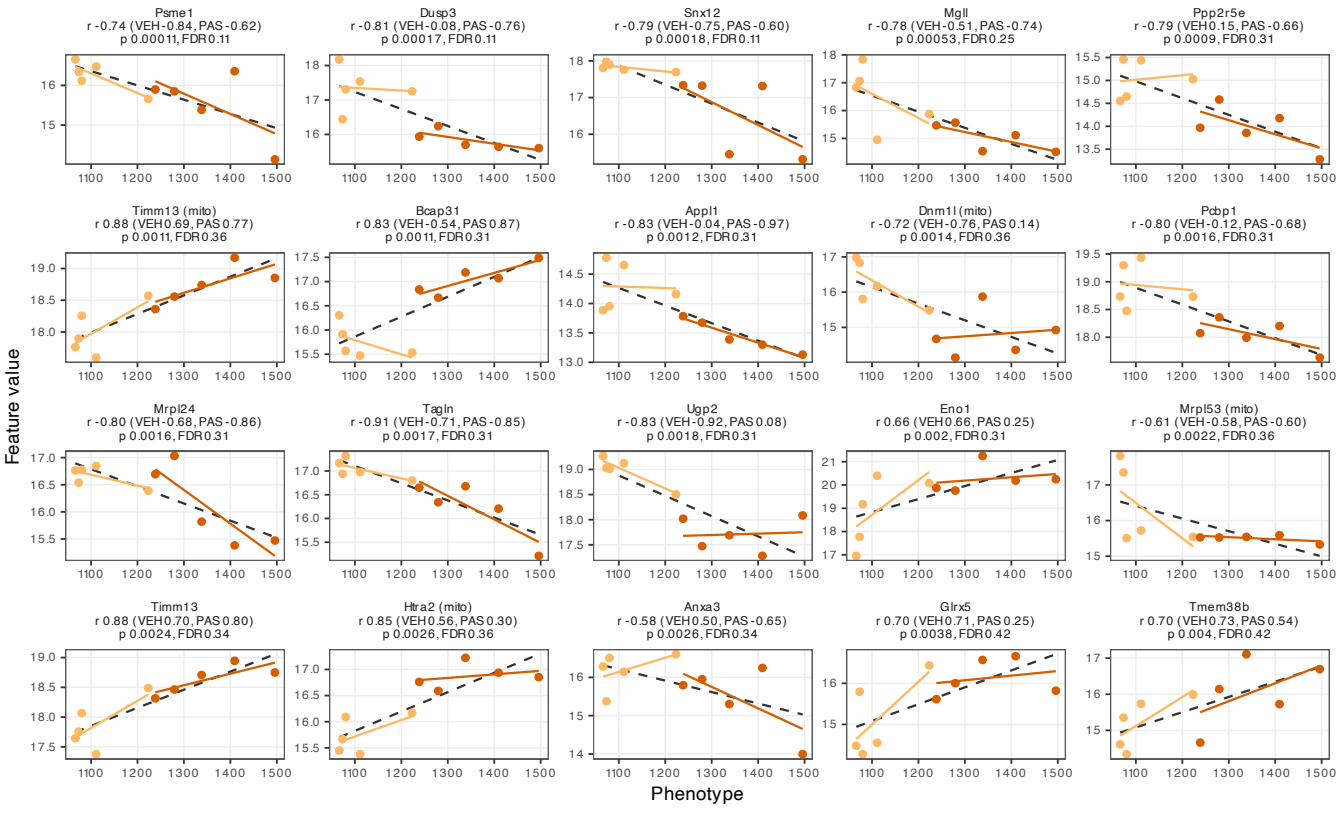
Female: Proteins associated with Capillary density, every association at p ≤ 0.05 (n = 170)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



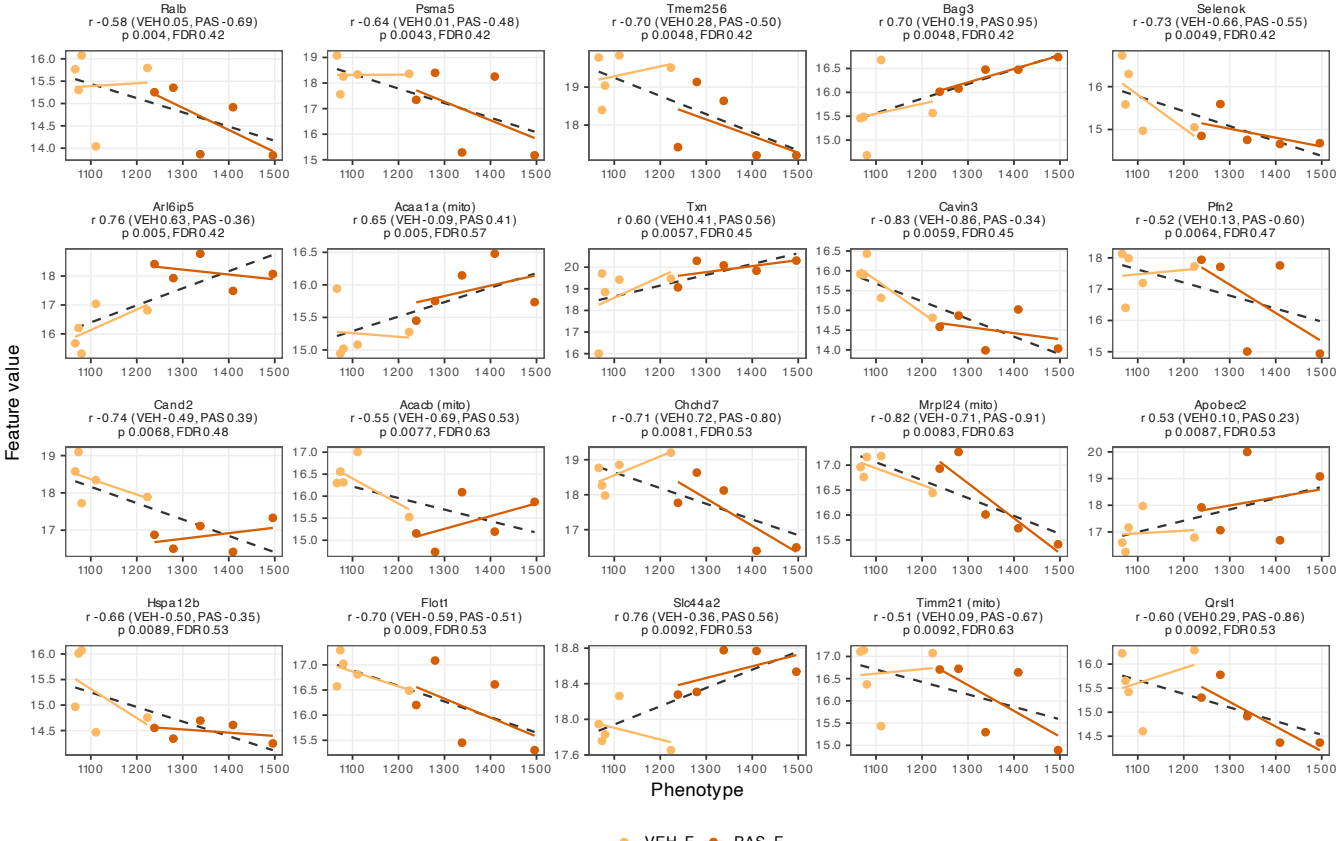
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



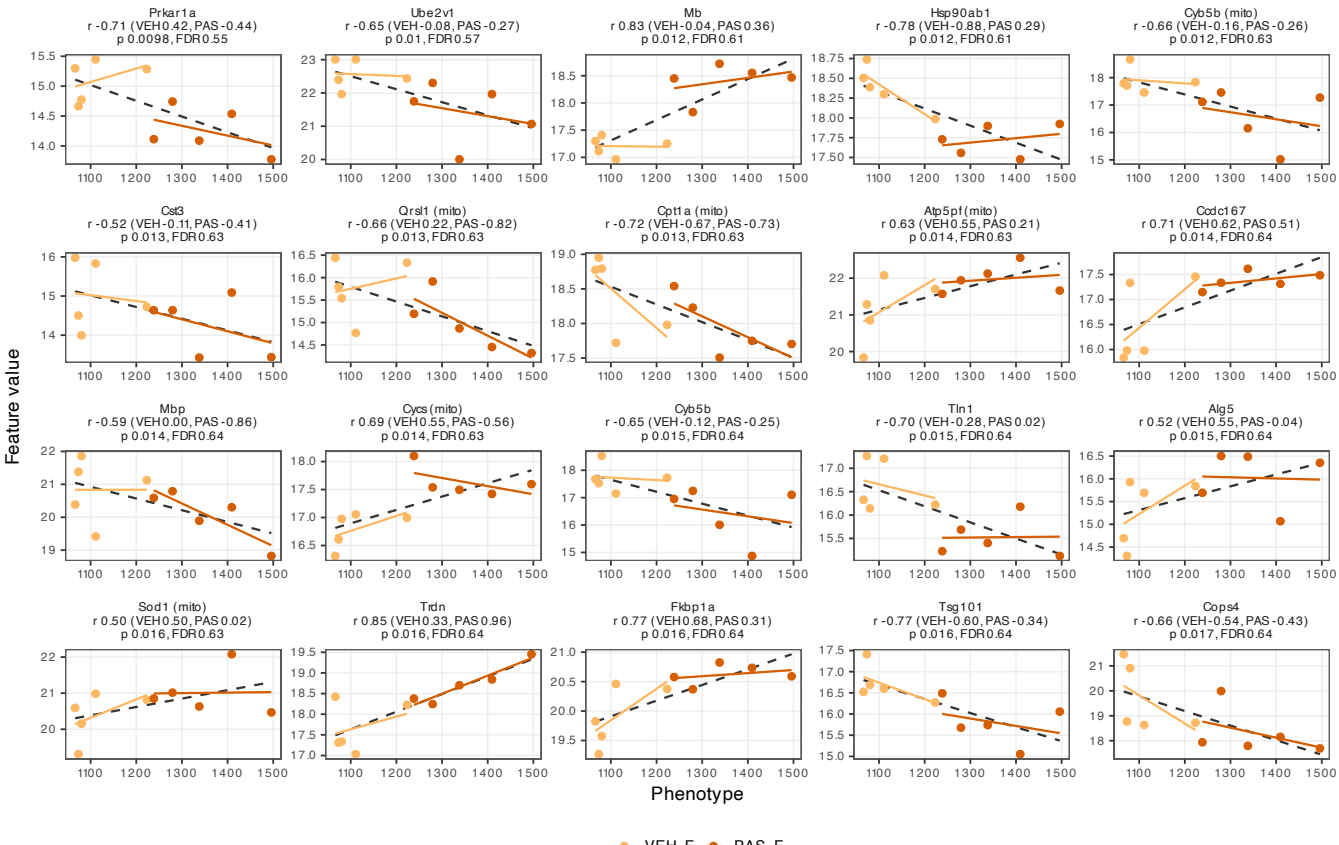
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



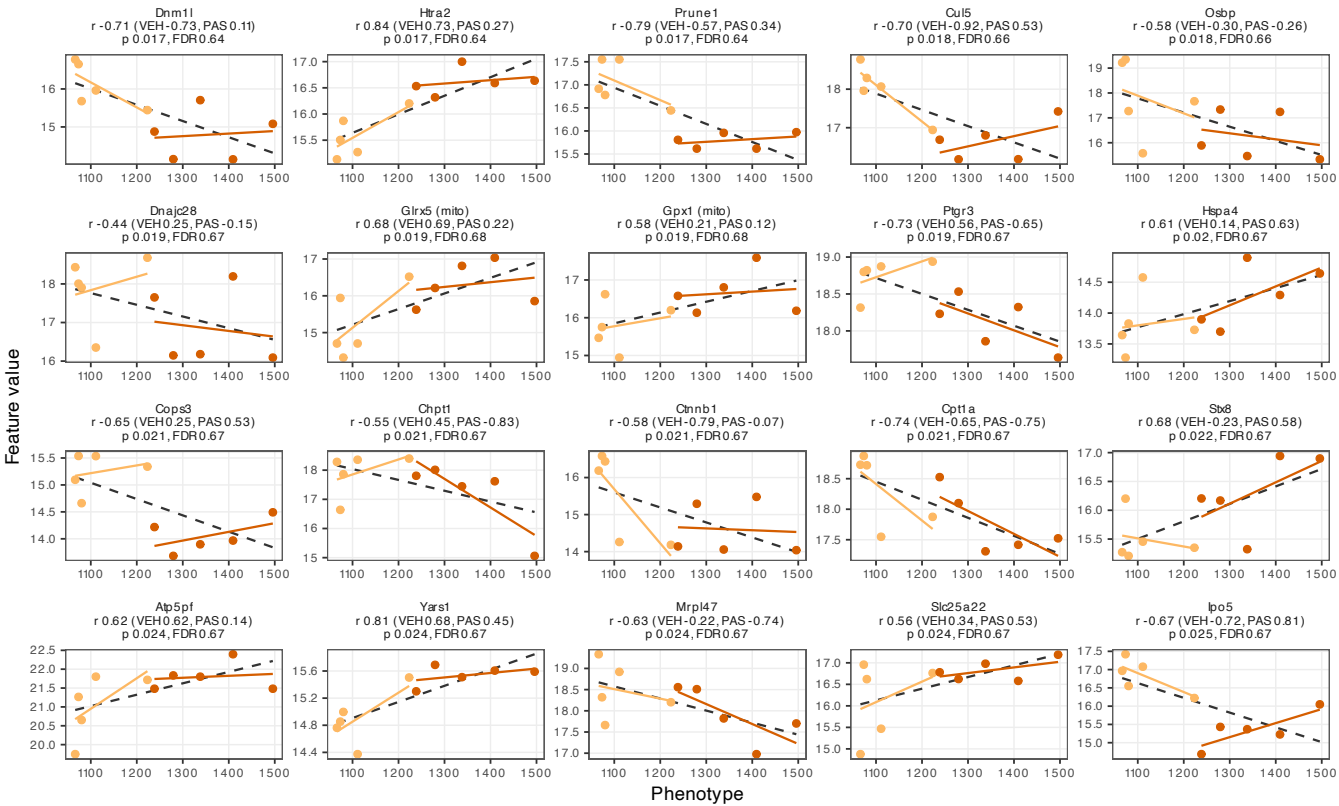
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



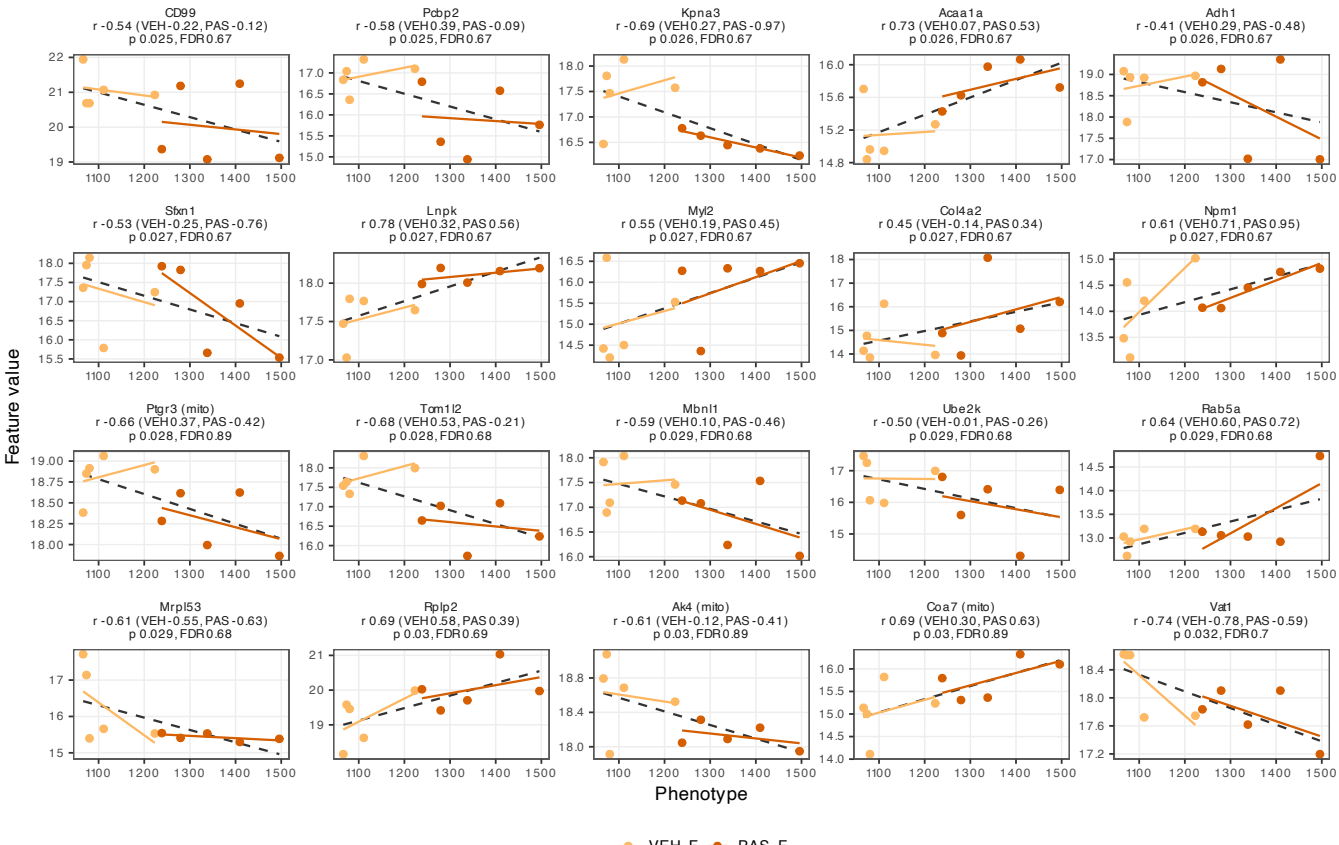
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



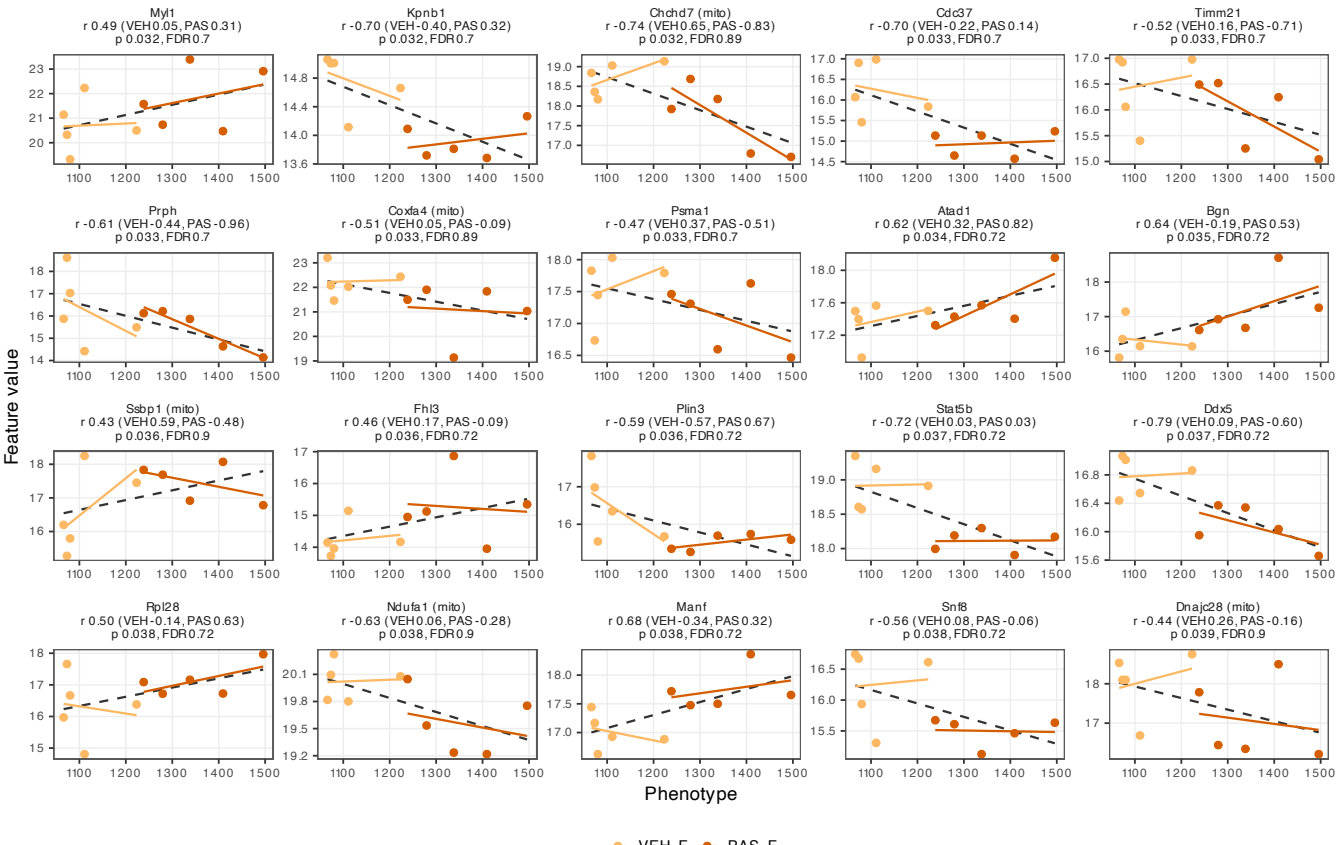
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



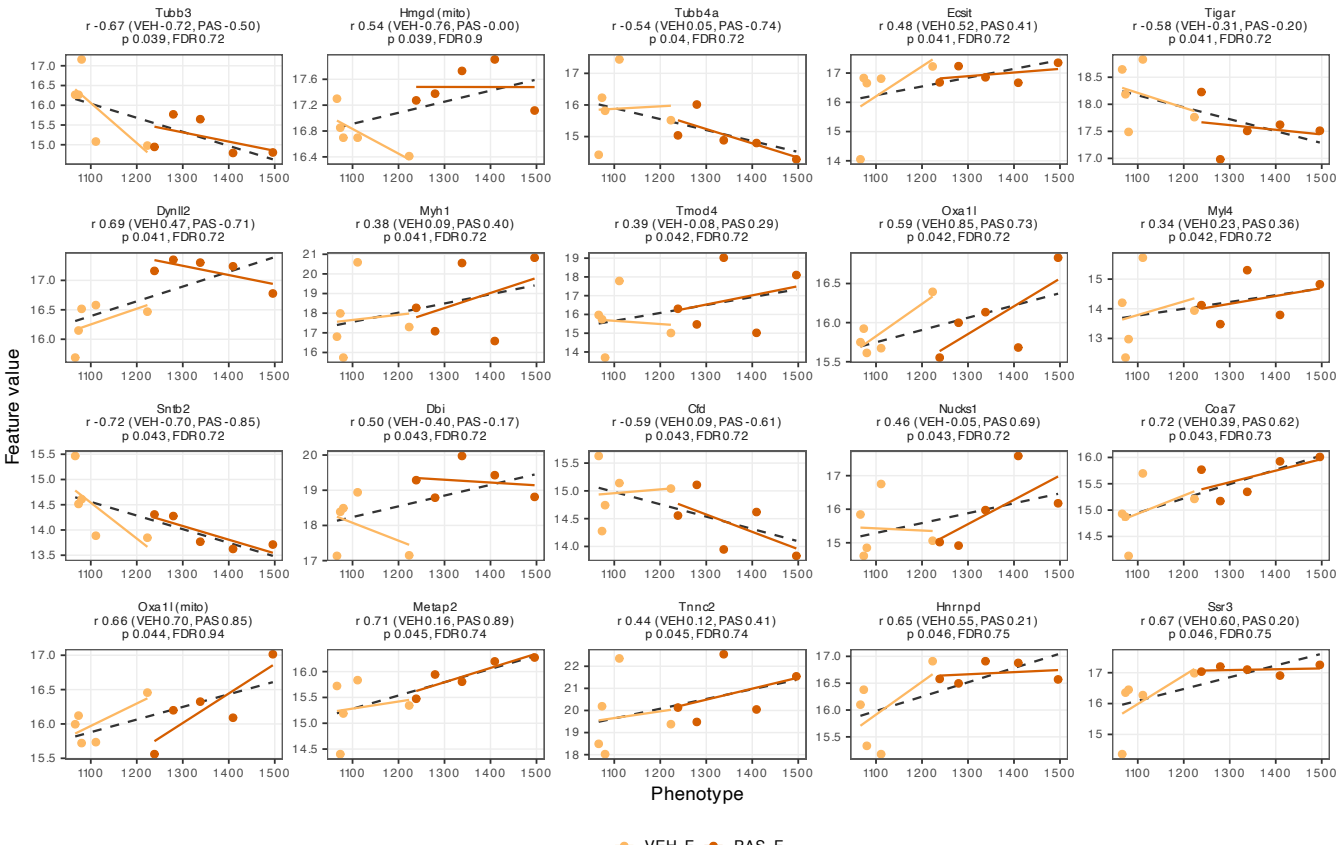
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



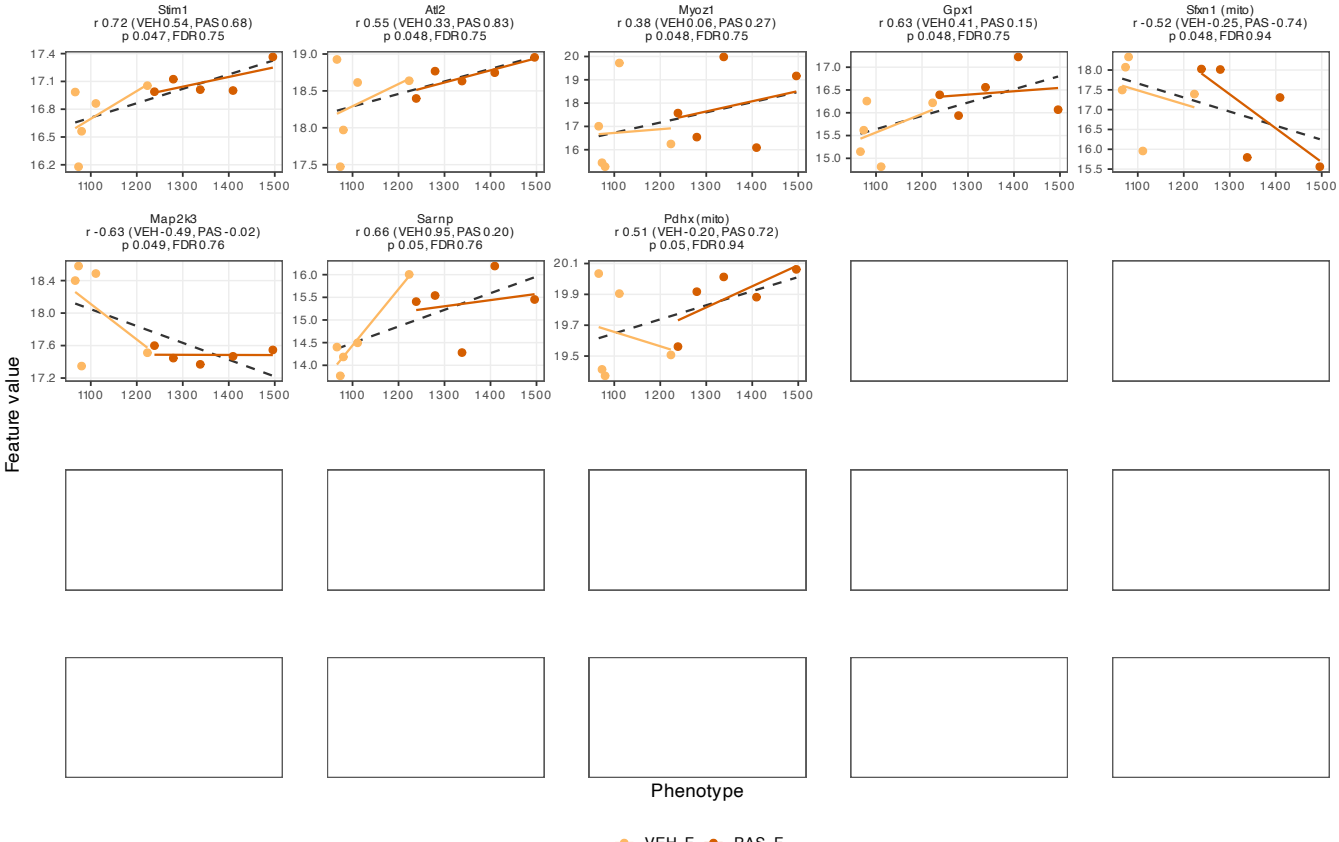
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



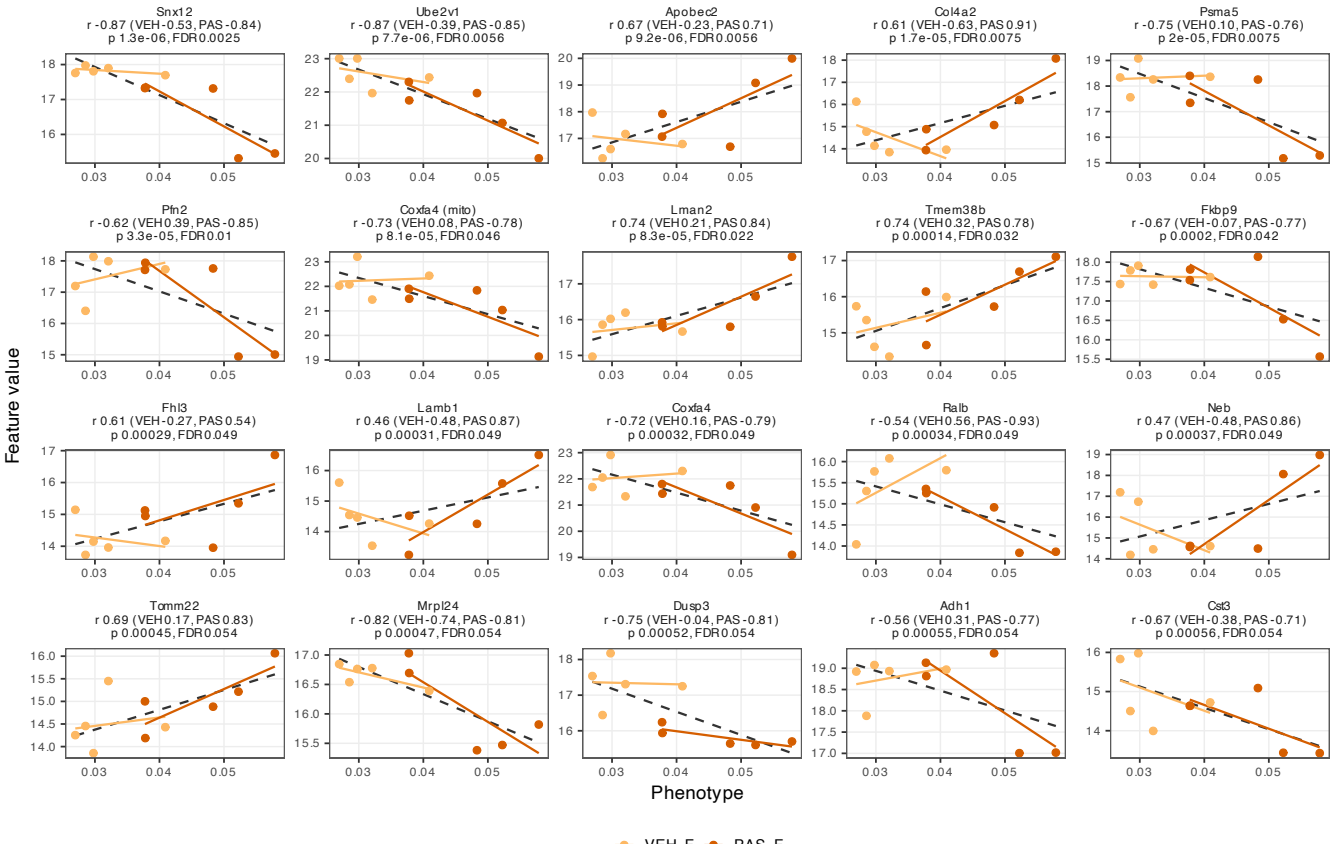
Female: Proteins associated with CSA, every association at p ≤ 0.05 (n = 148)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



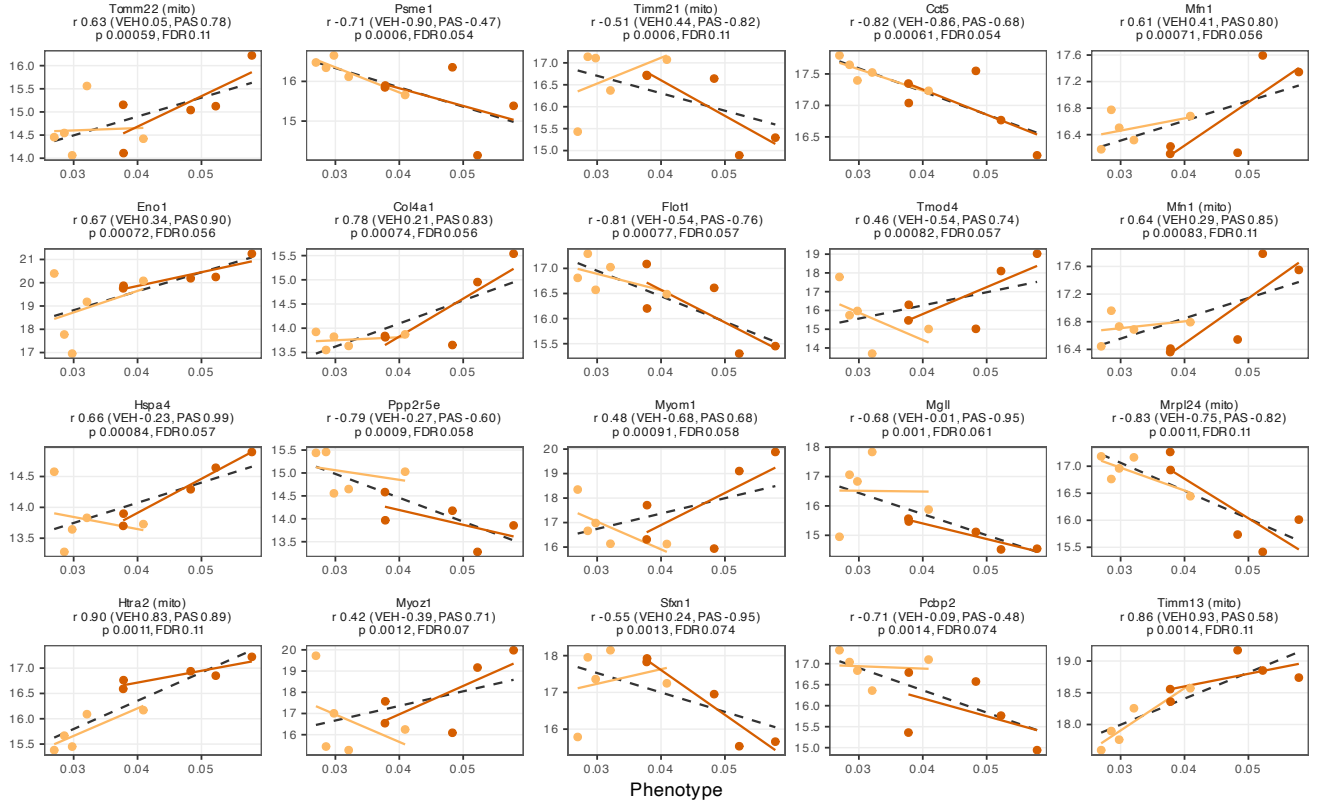
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



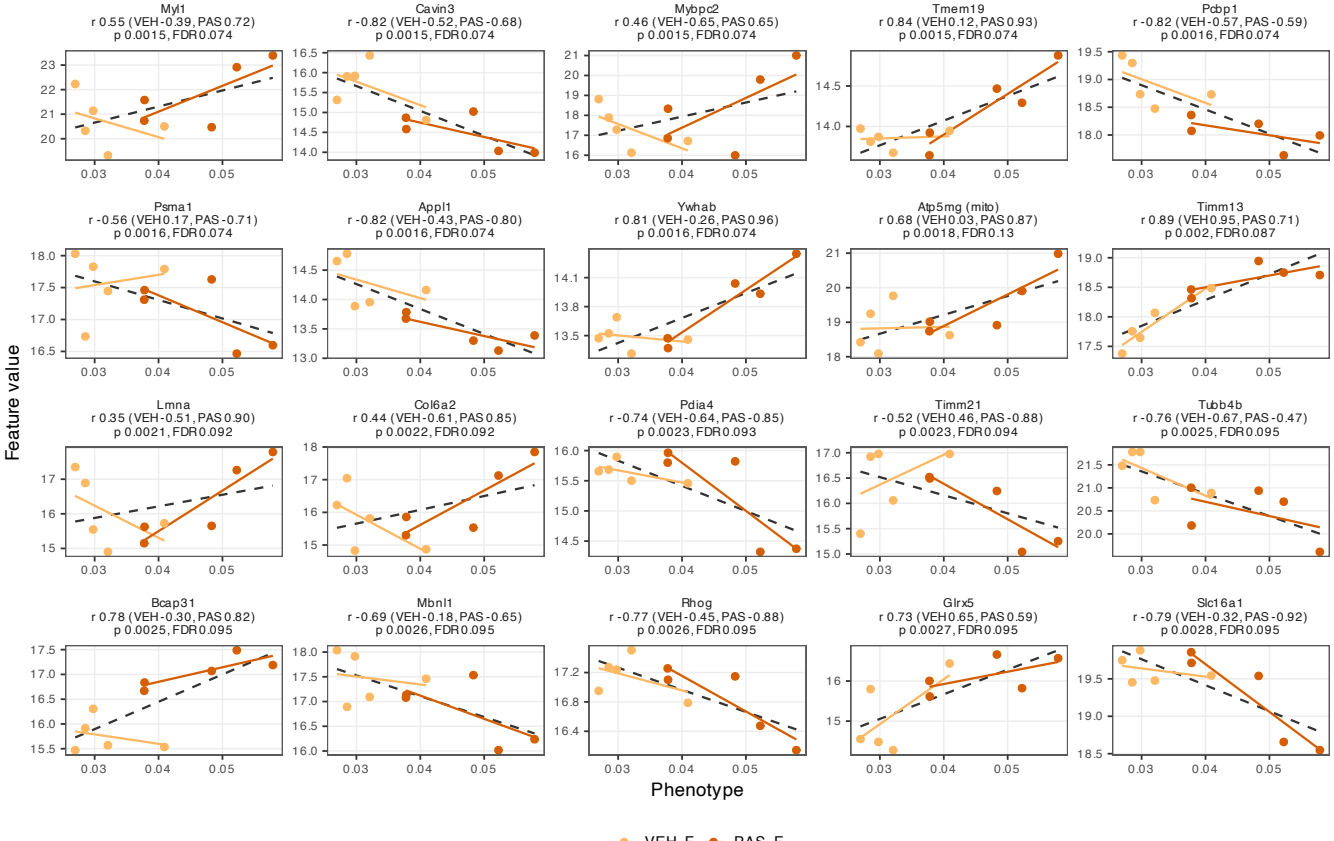
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



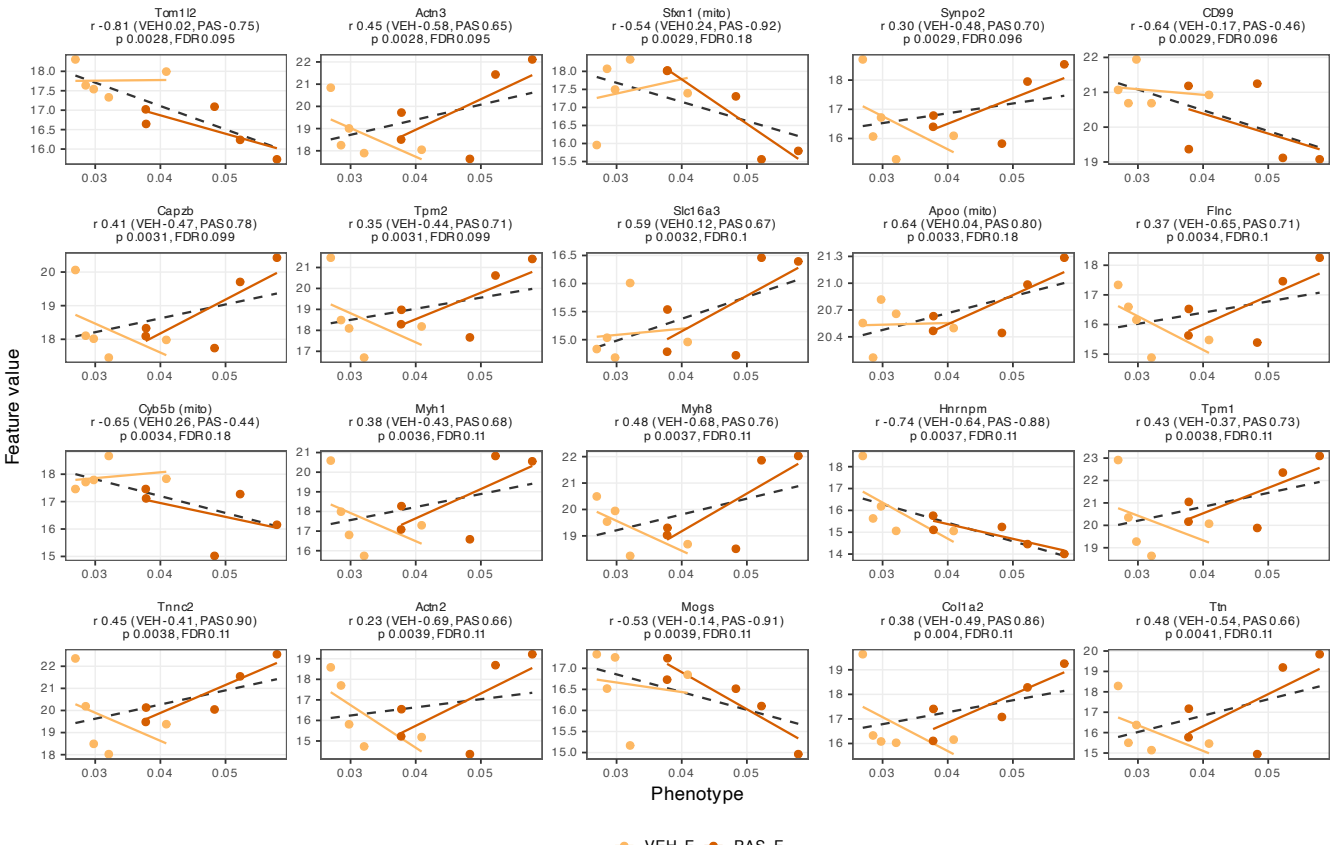
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



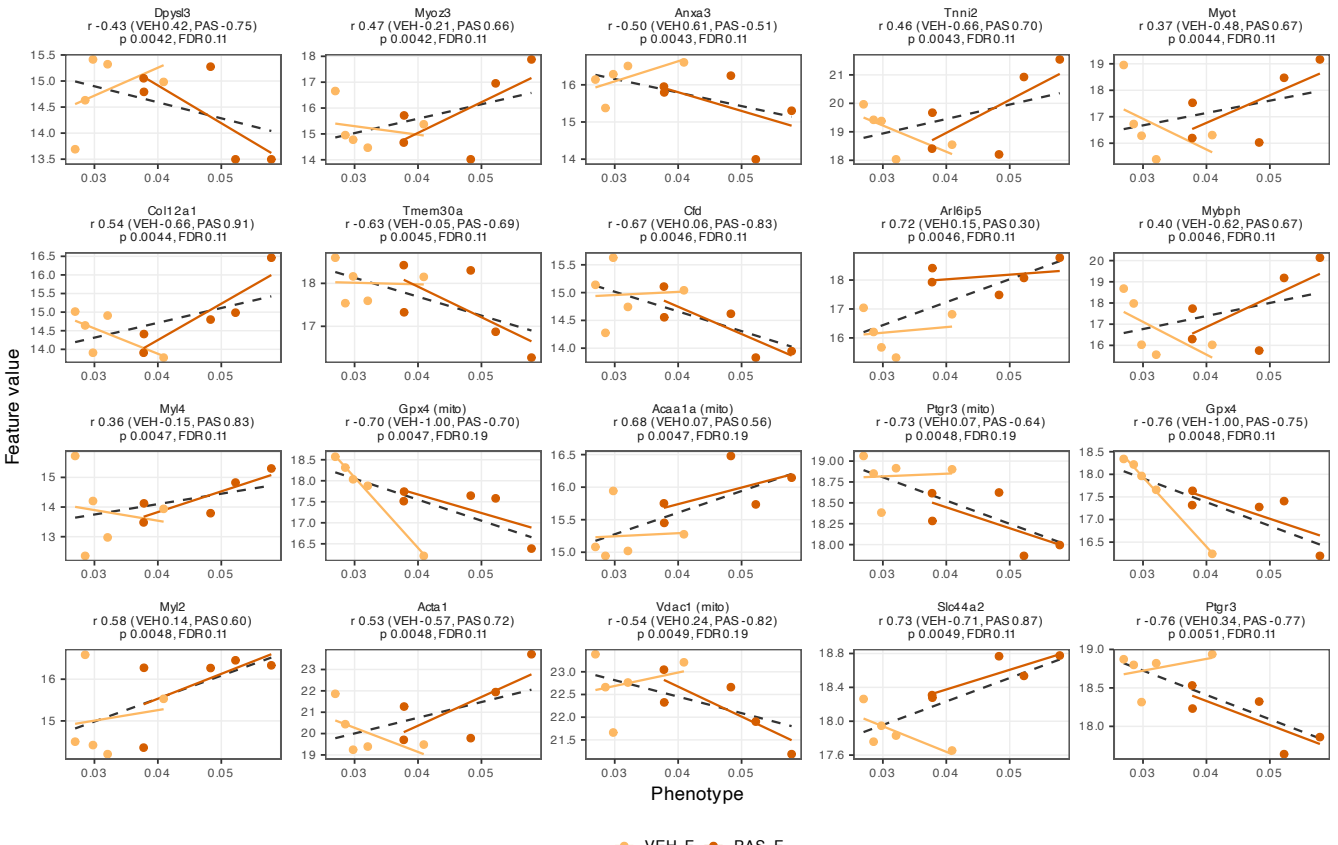
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



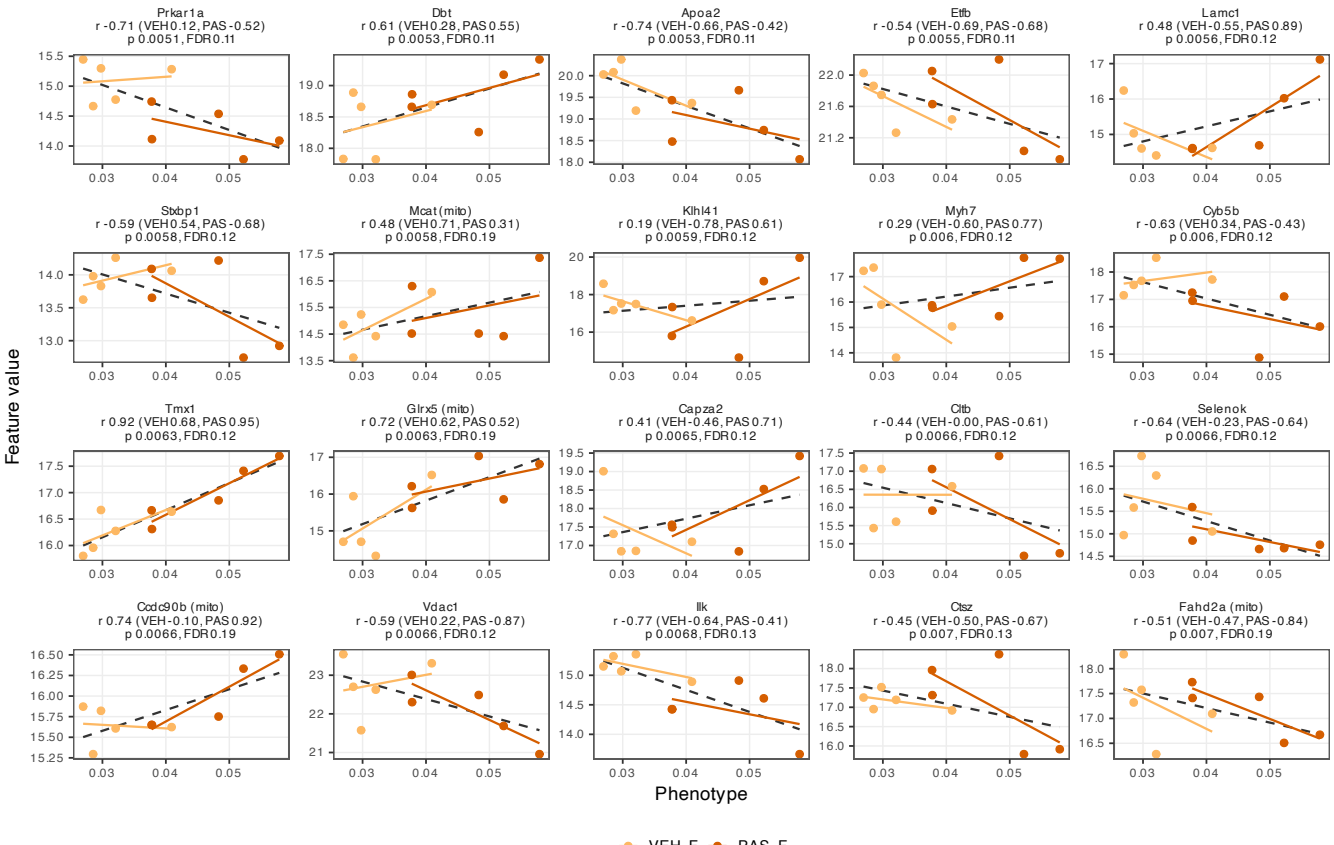
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



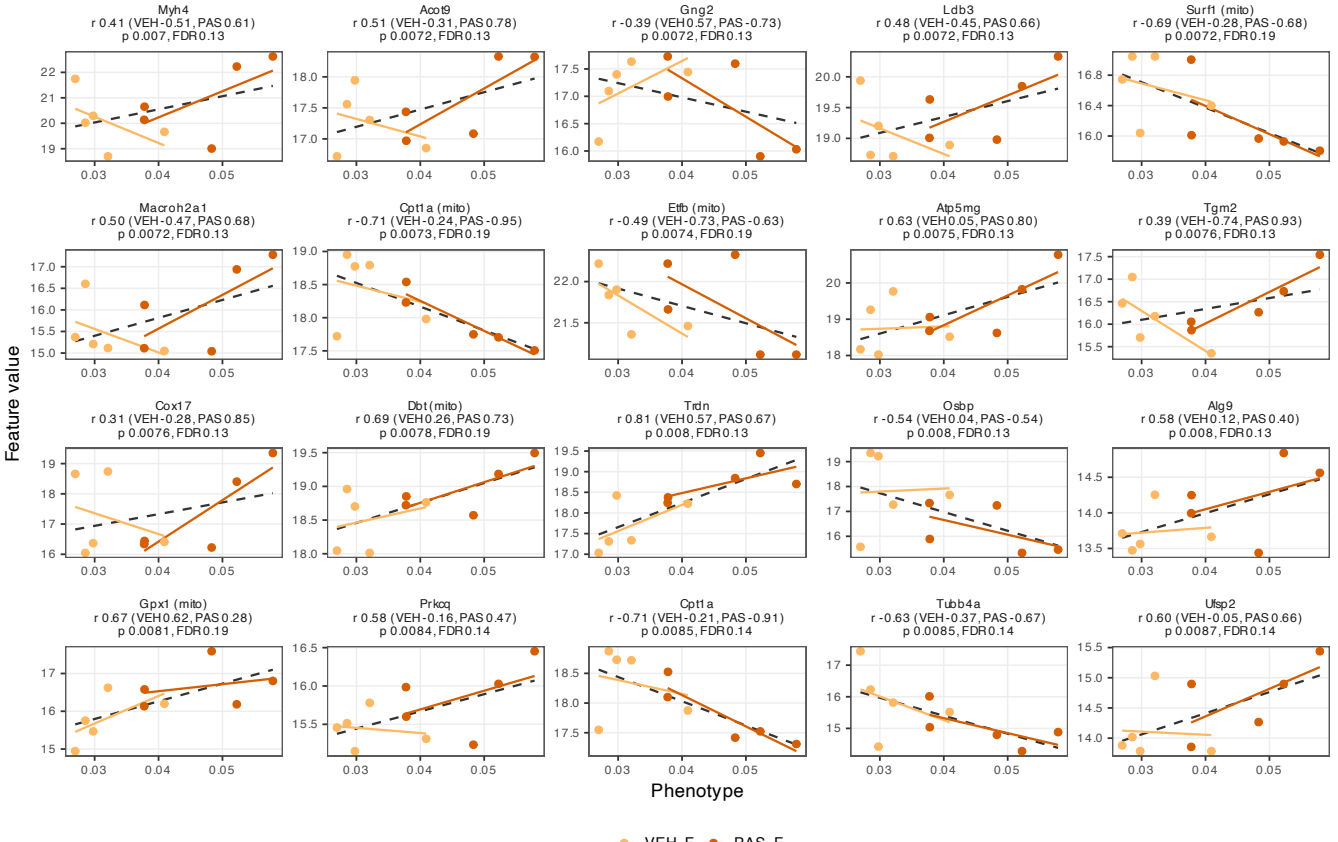
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



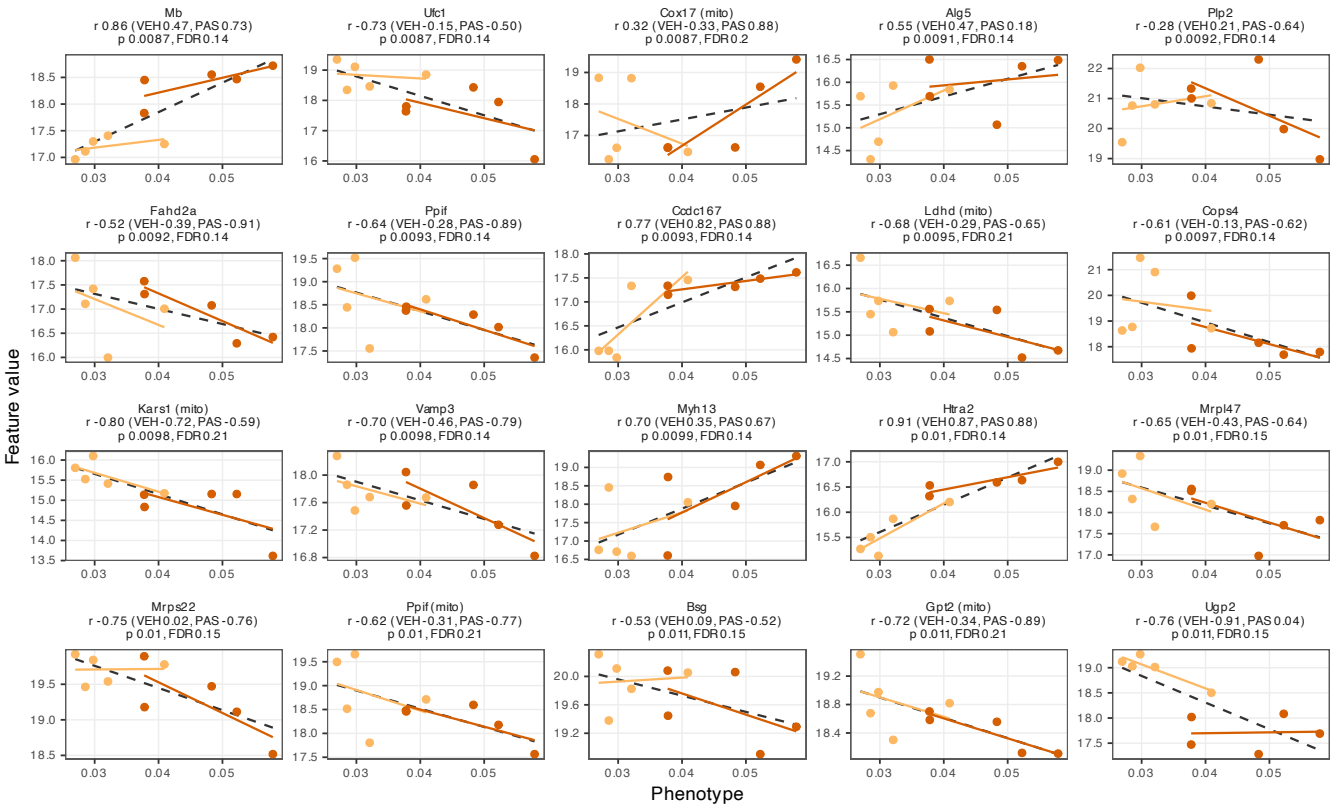
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



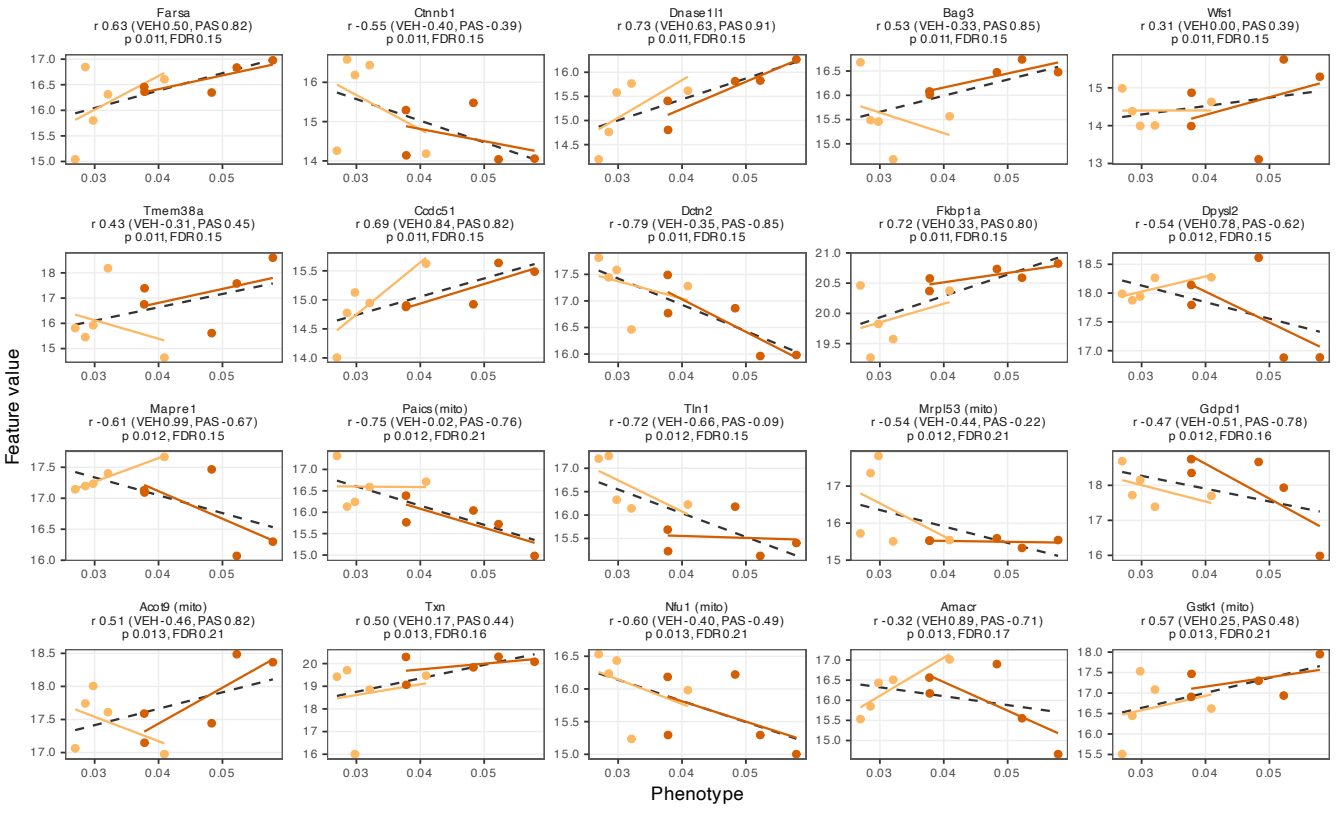
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



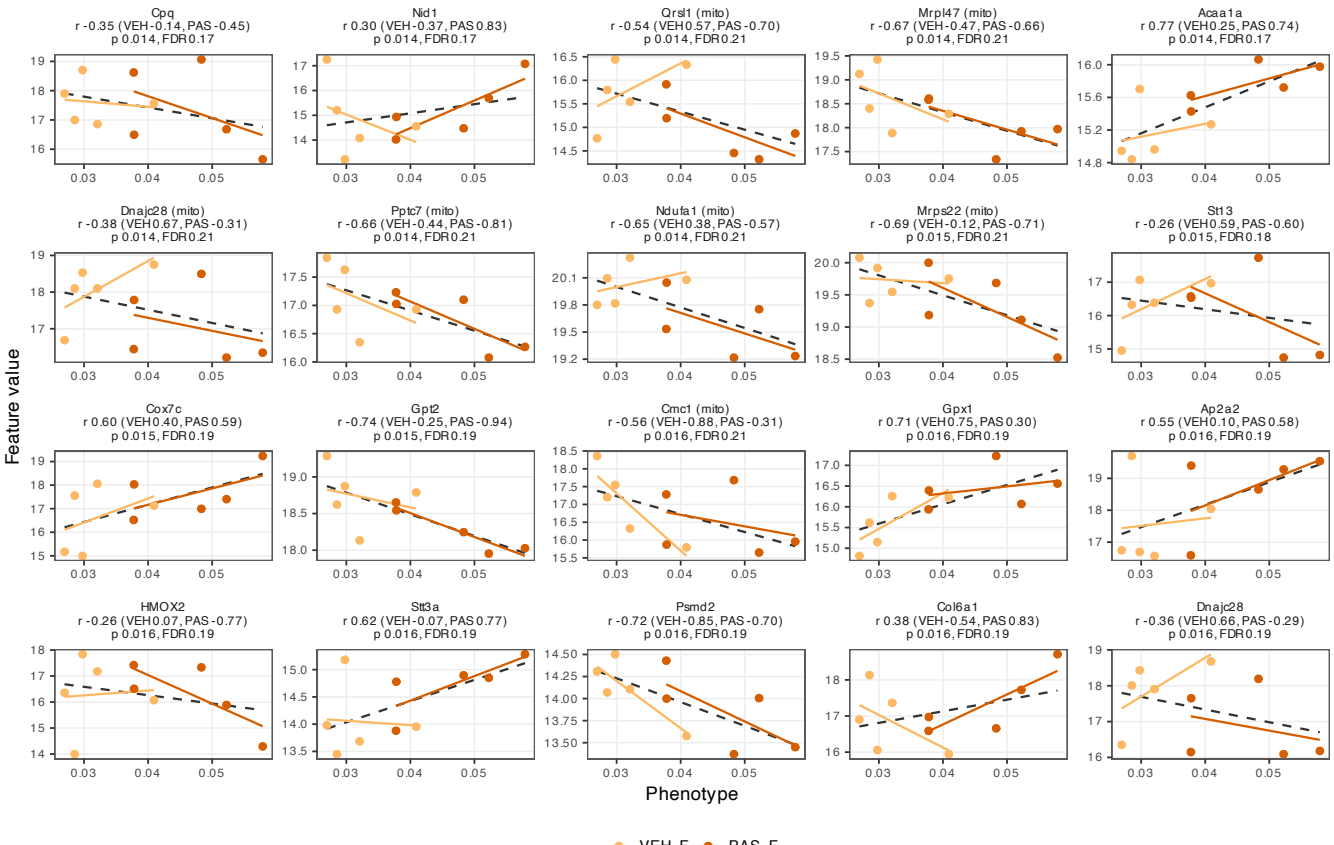
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



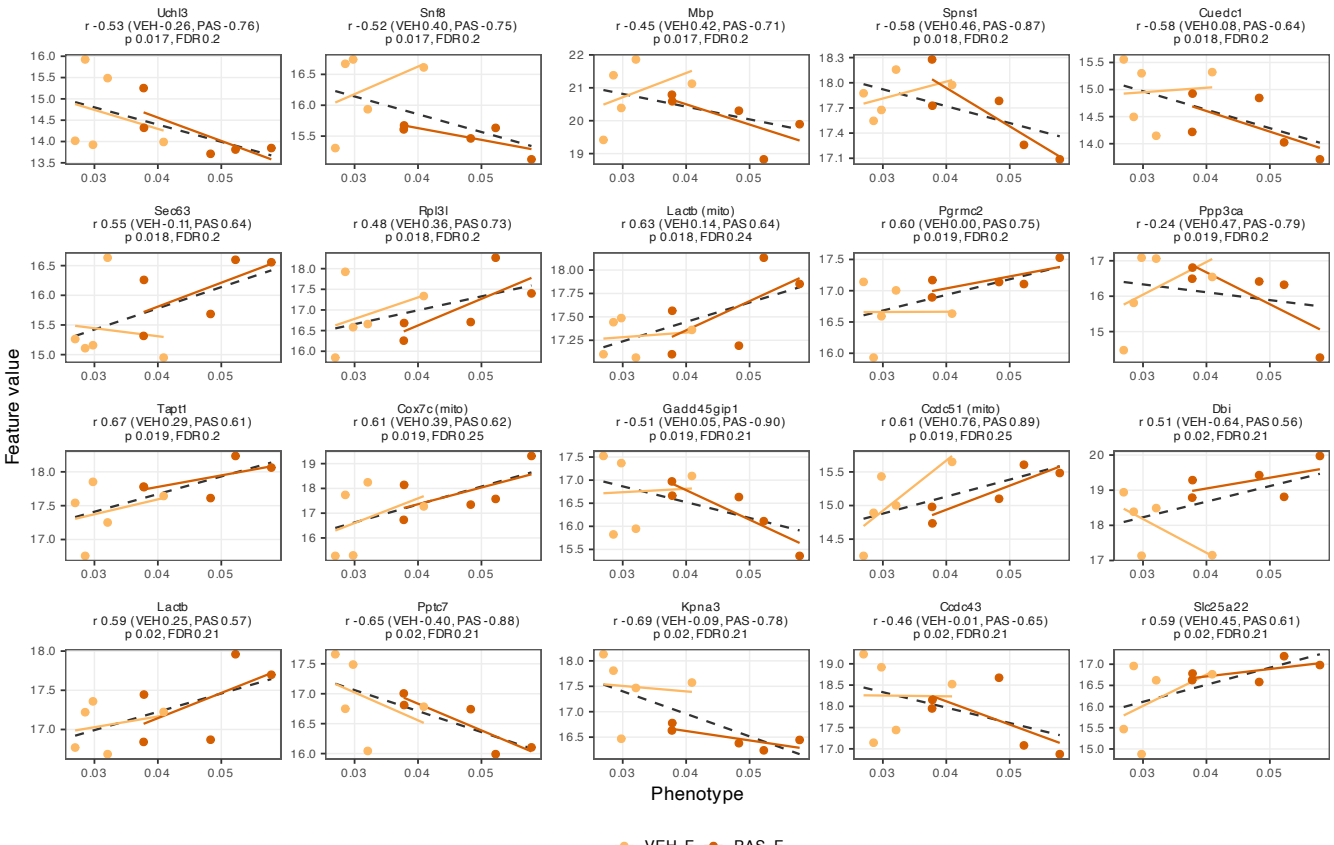
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



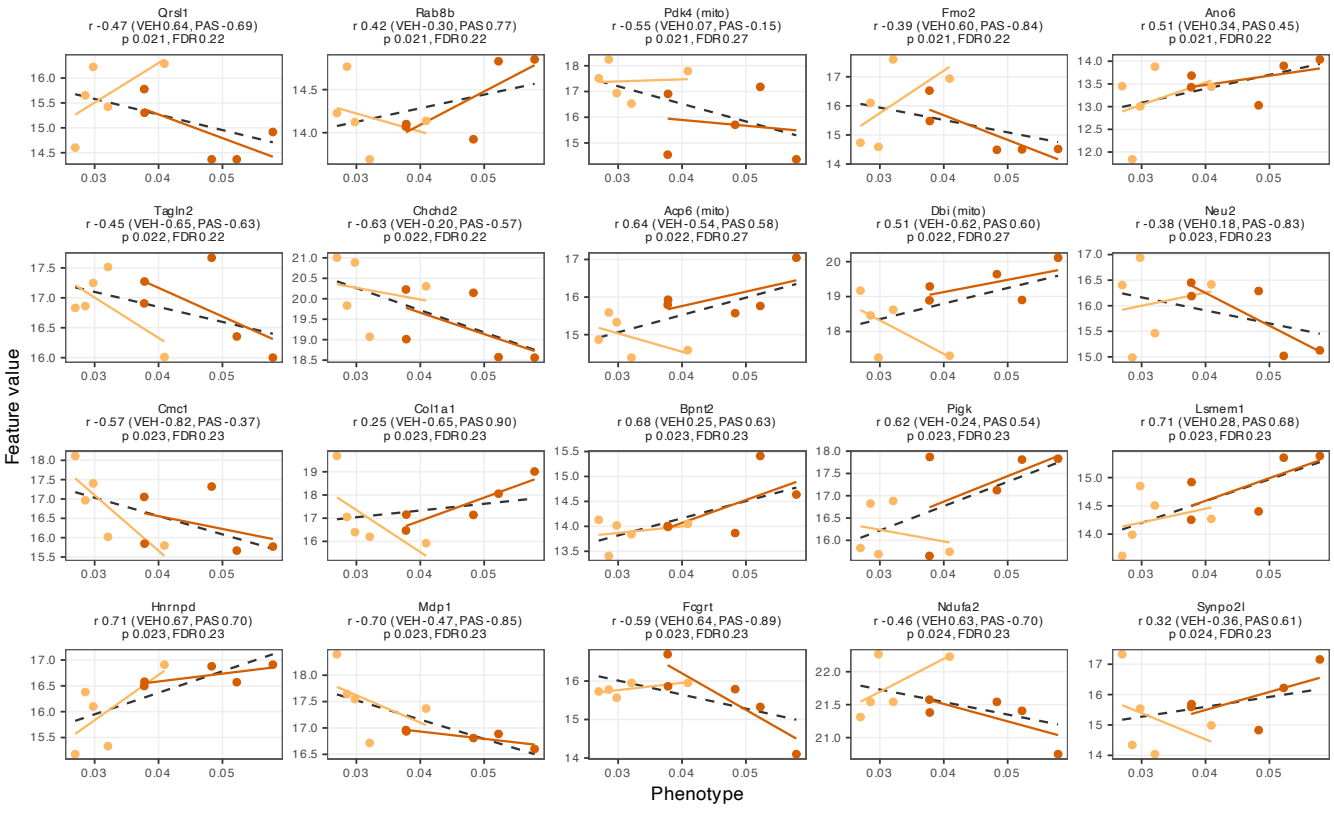
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



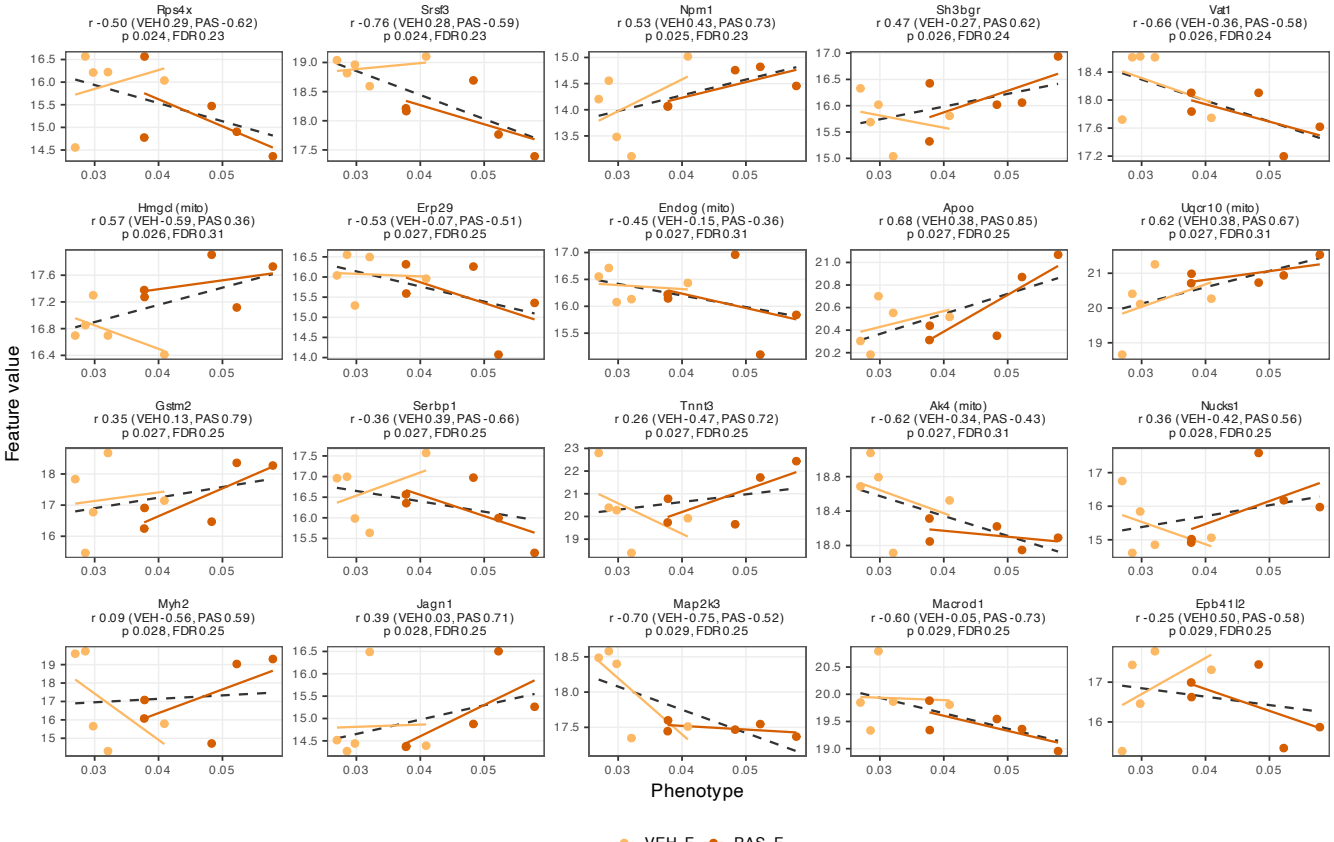
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



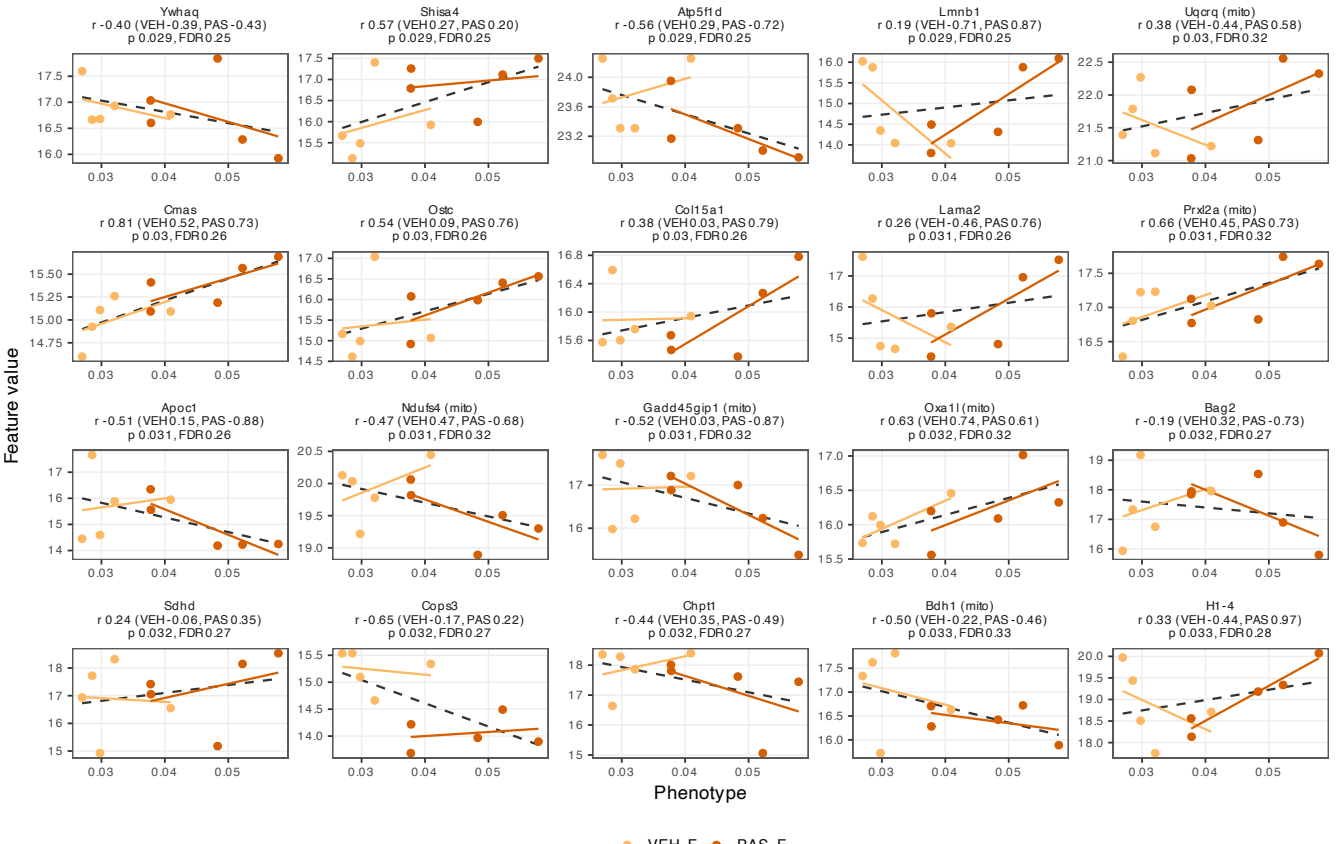
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



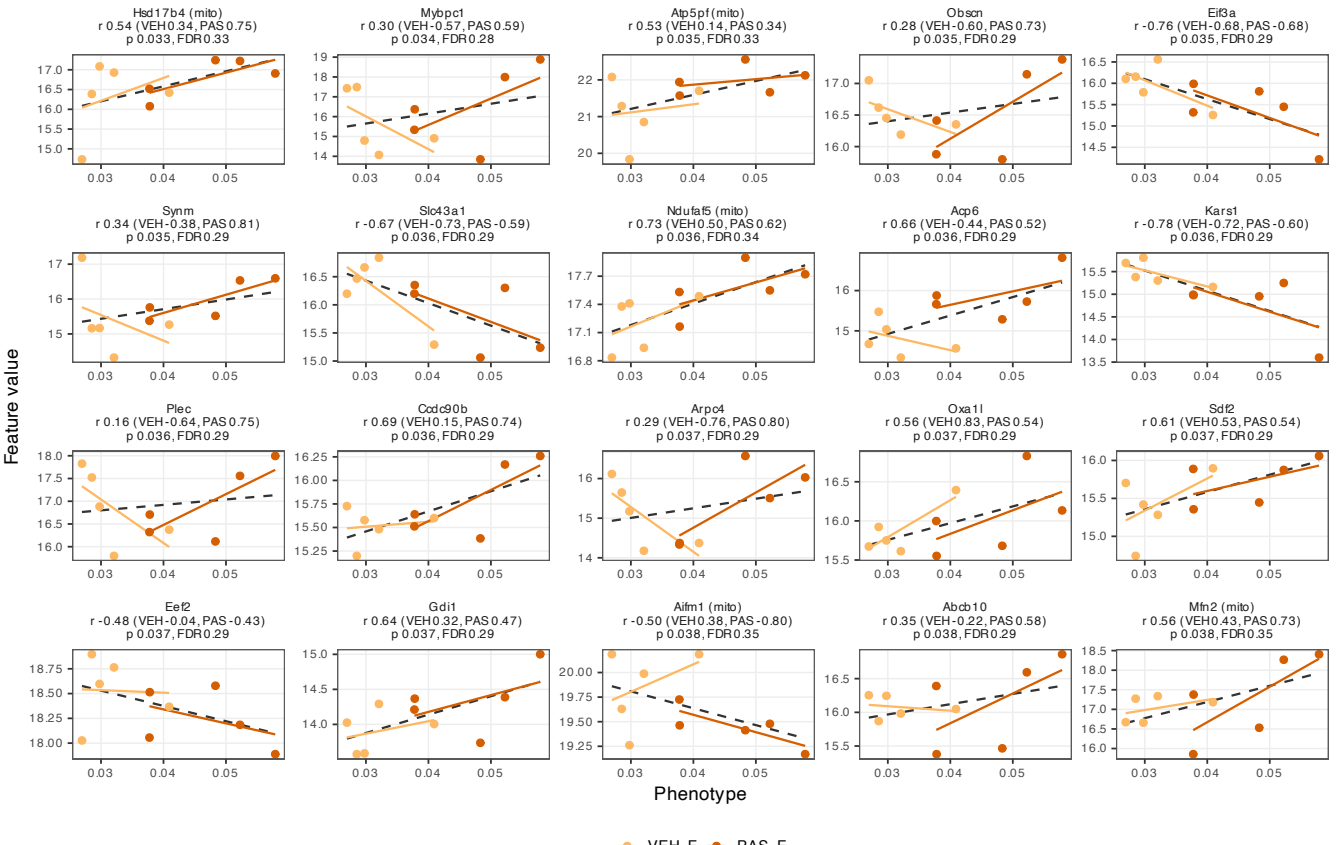
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



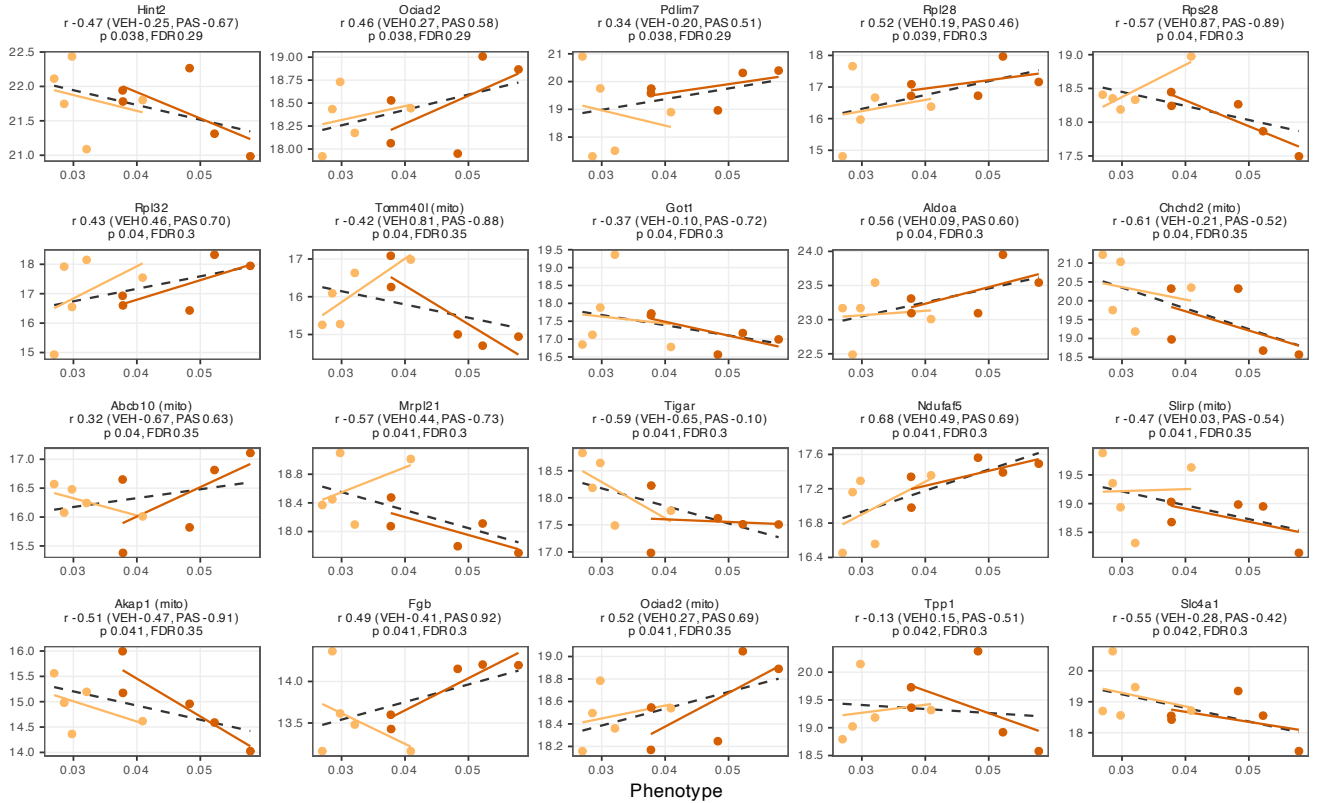
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



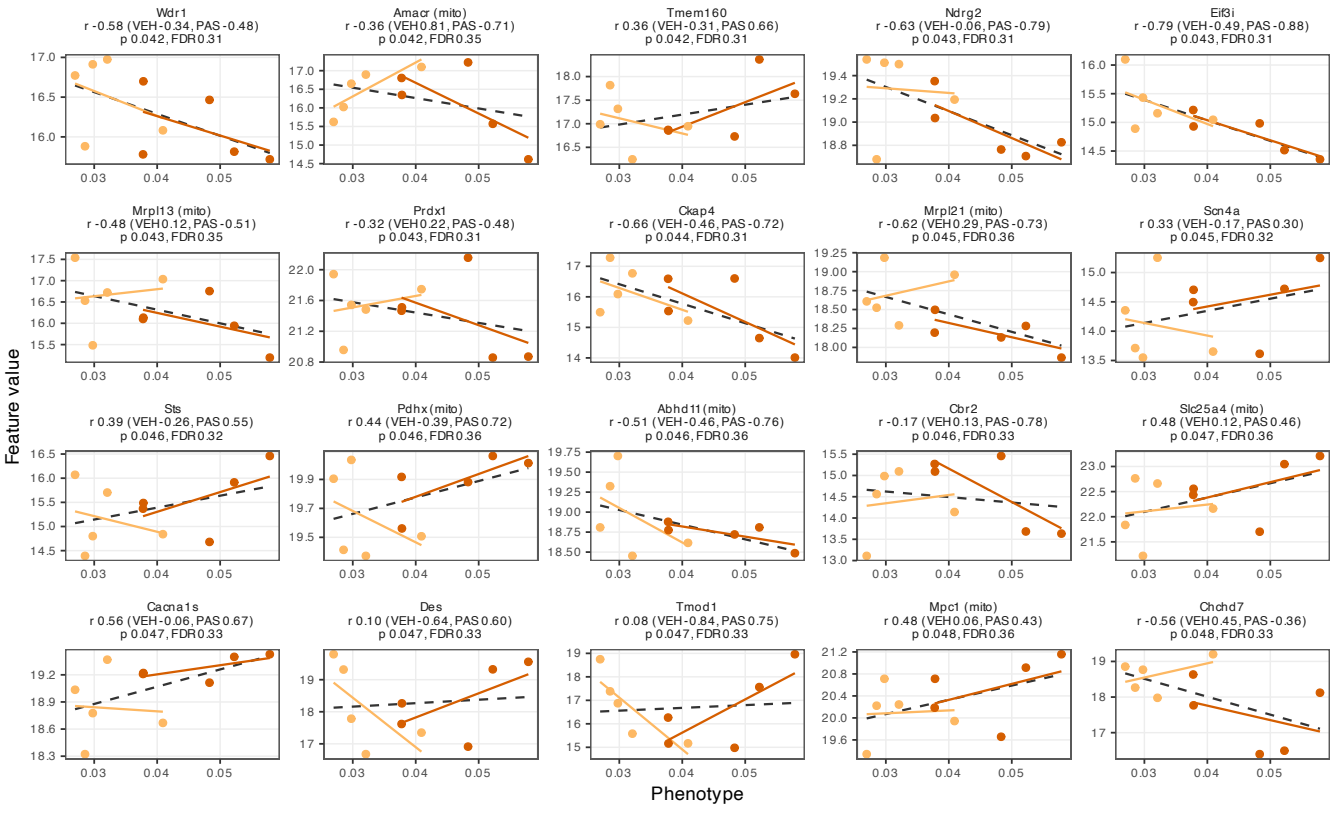
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



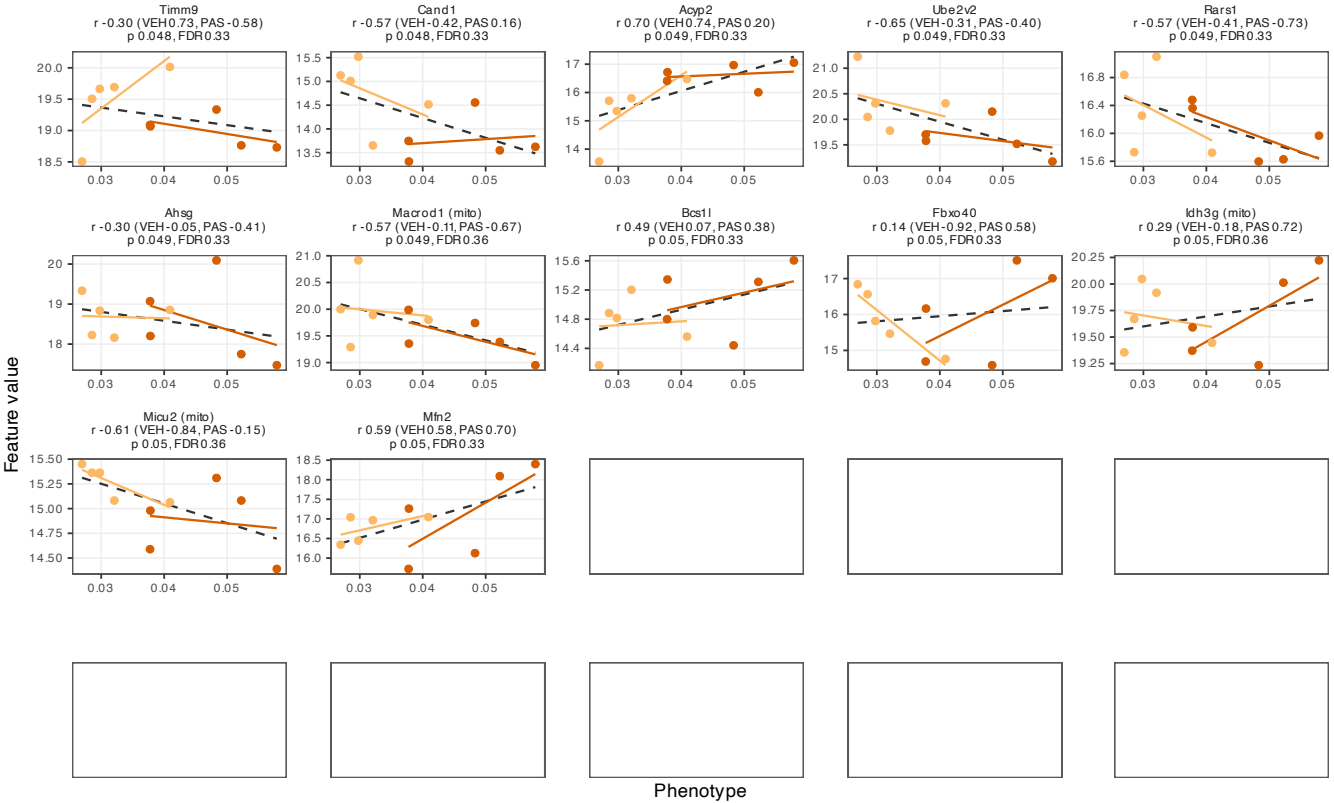
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



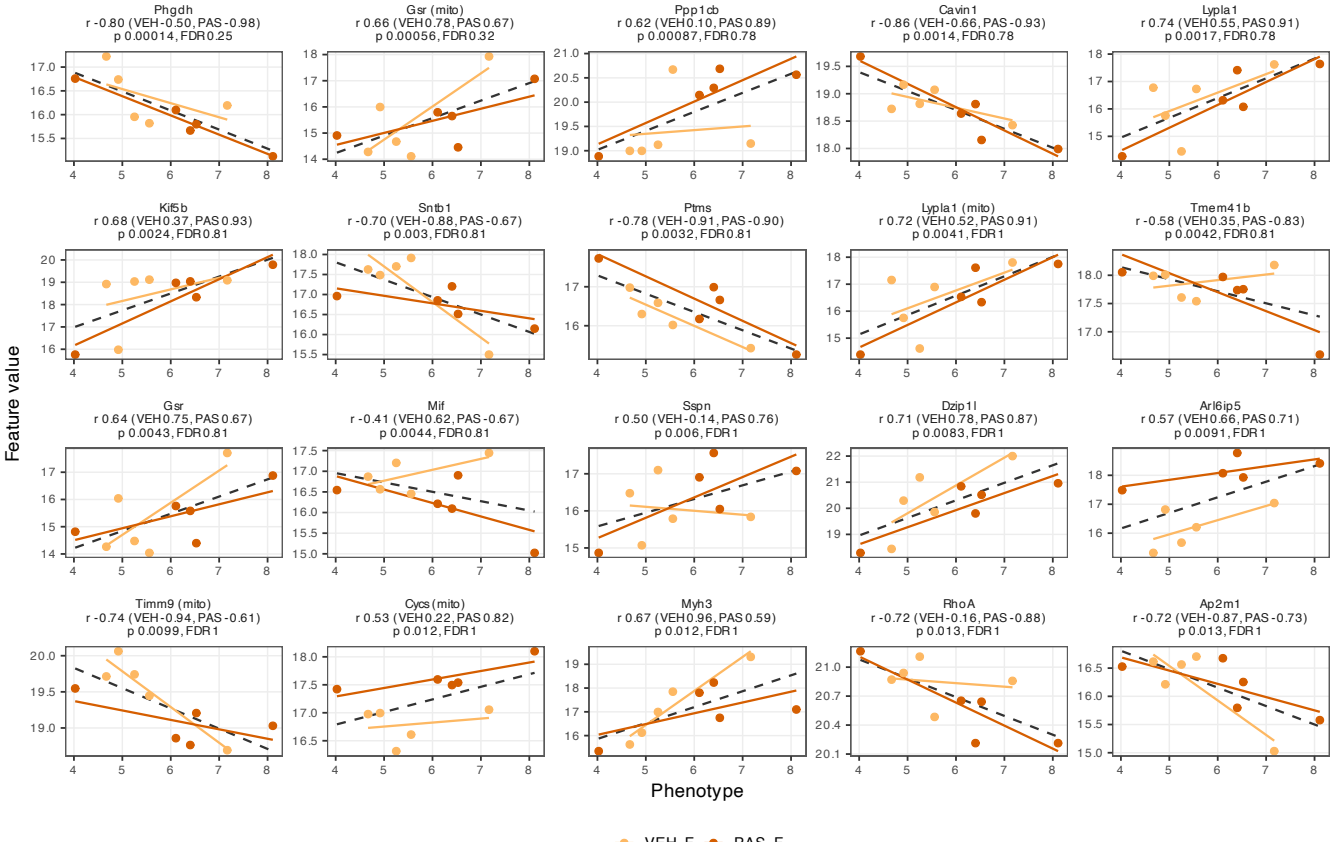
Female: Proteins associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 352)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



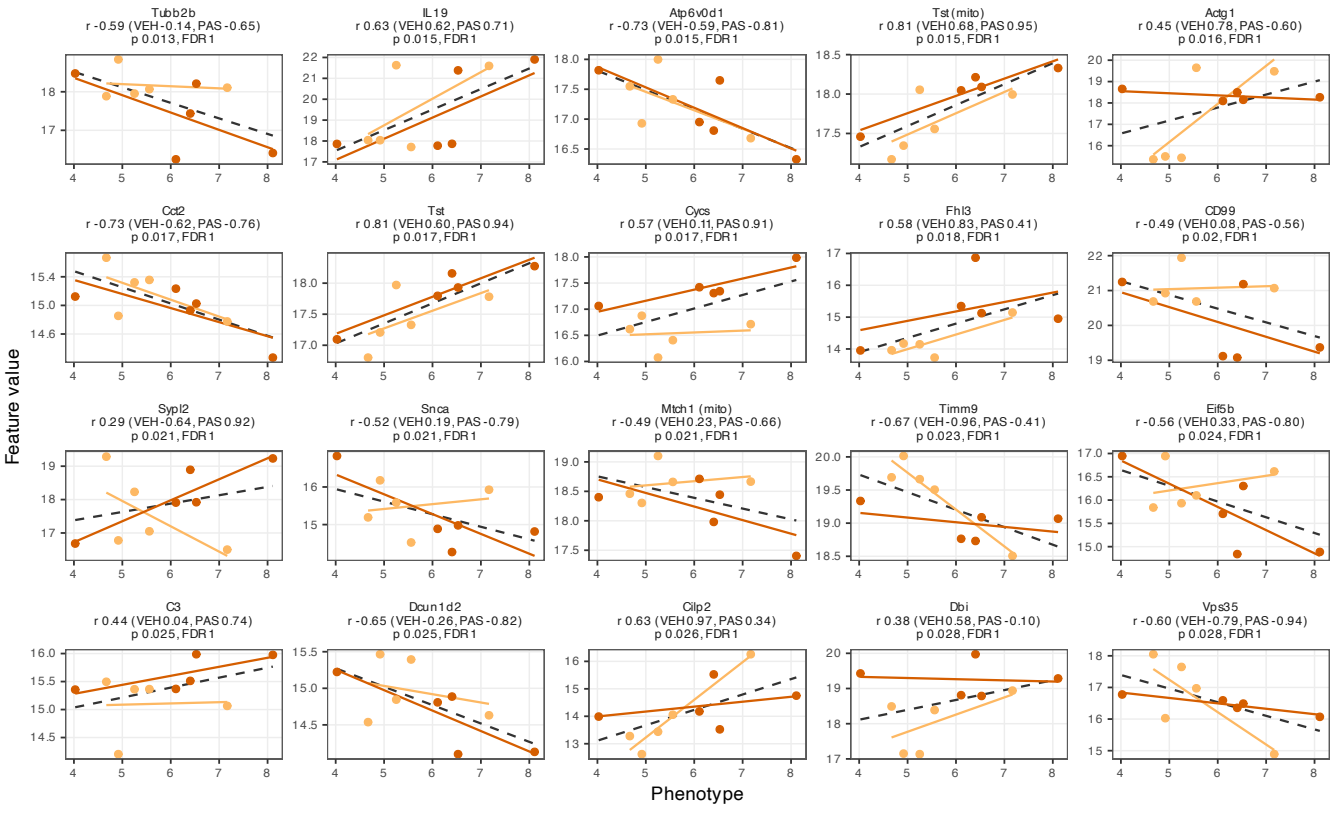
Female: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 68)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



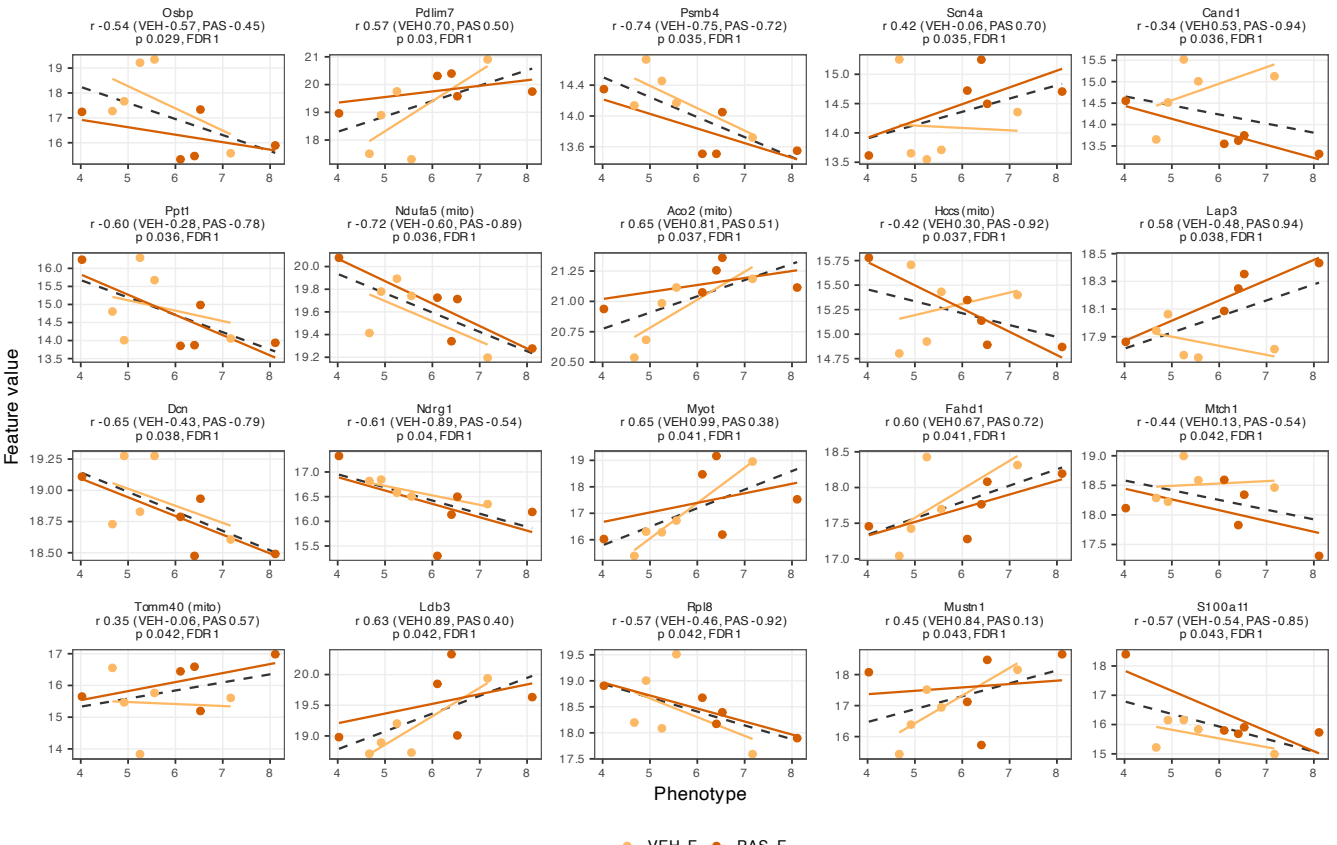
Female: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 68)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



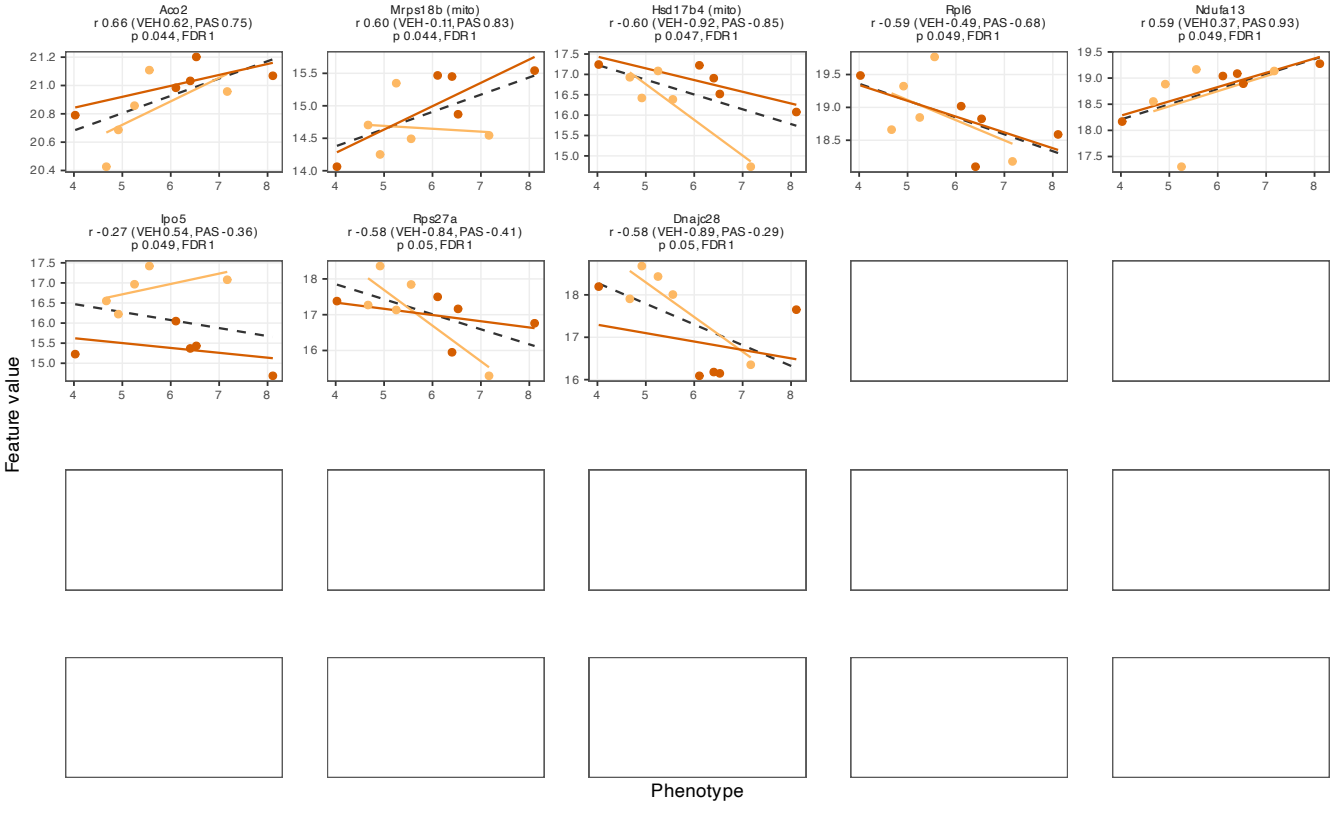
Female: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 68)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



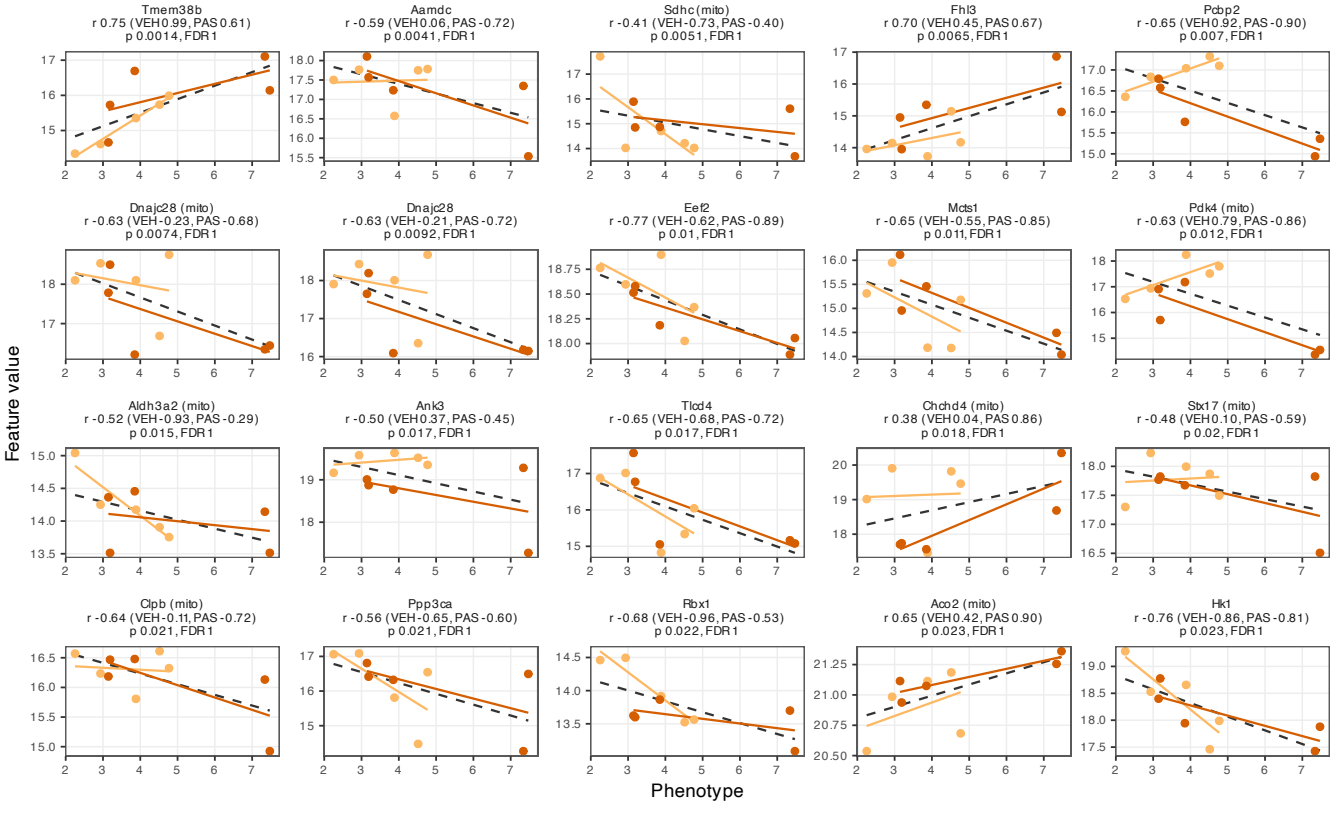
Female: Proteins associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 68)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



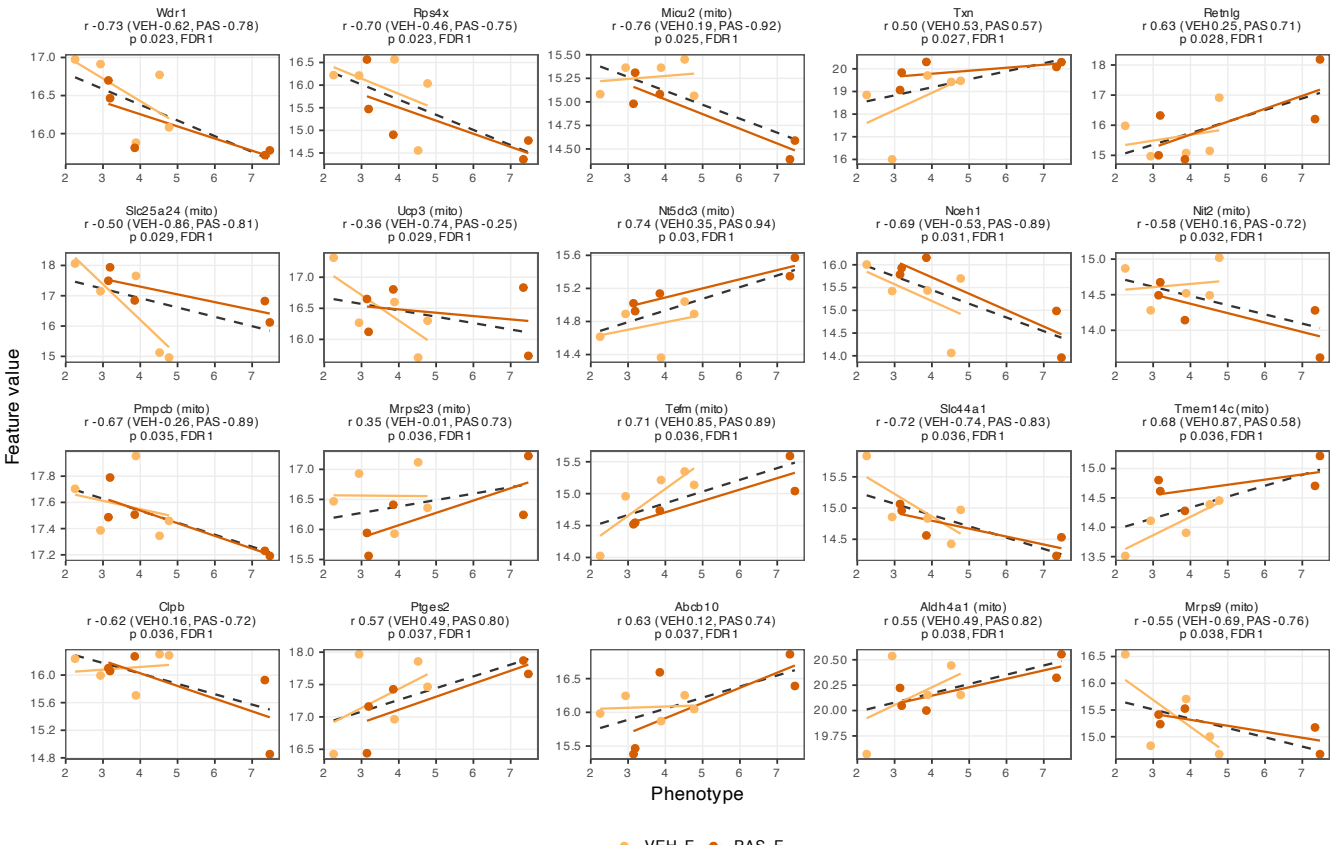
Female: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 65)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



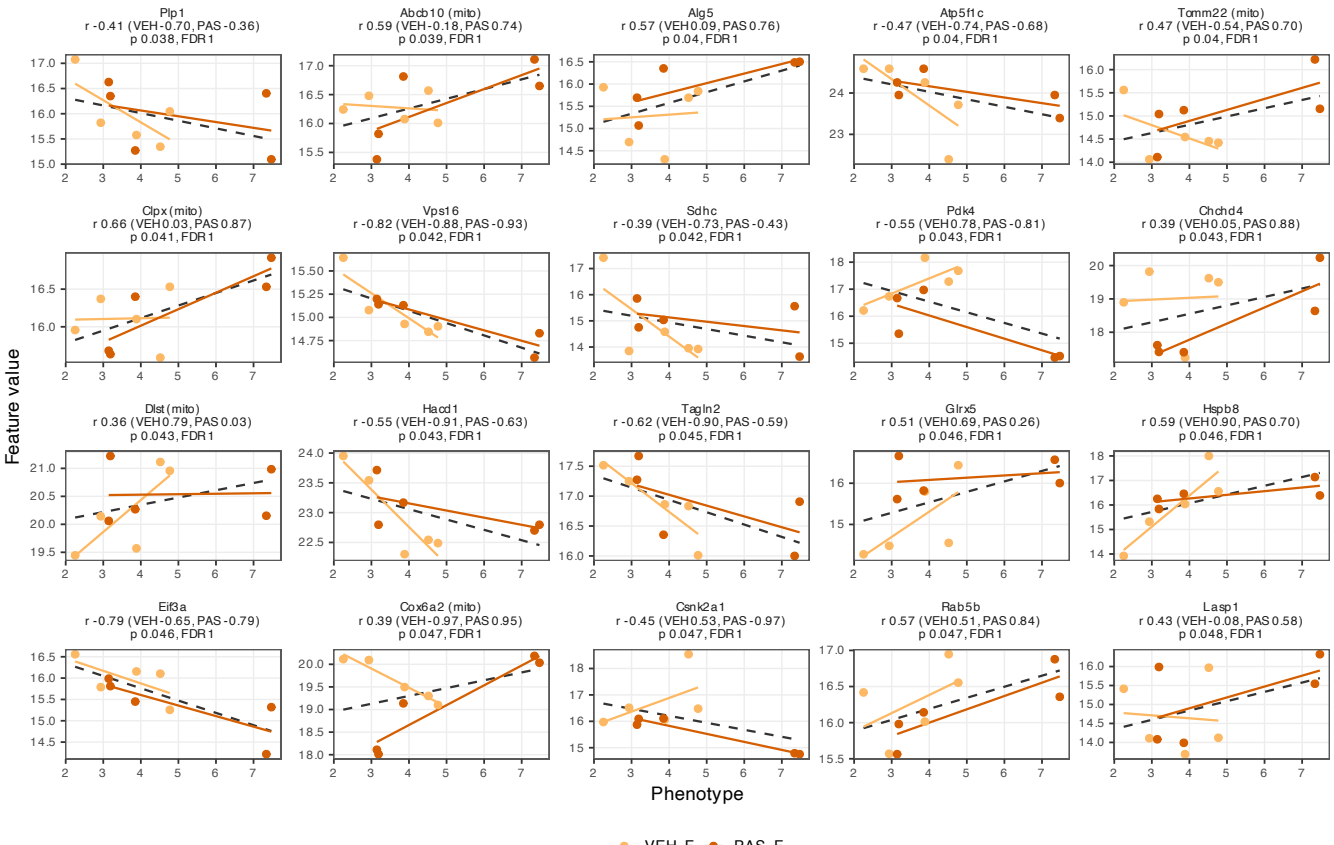
Female: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 65)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



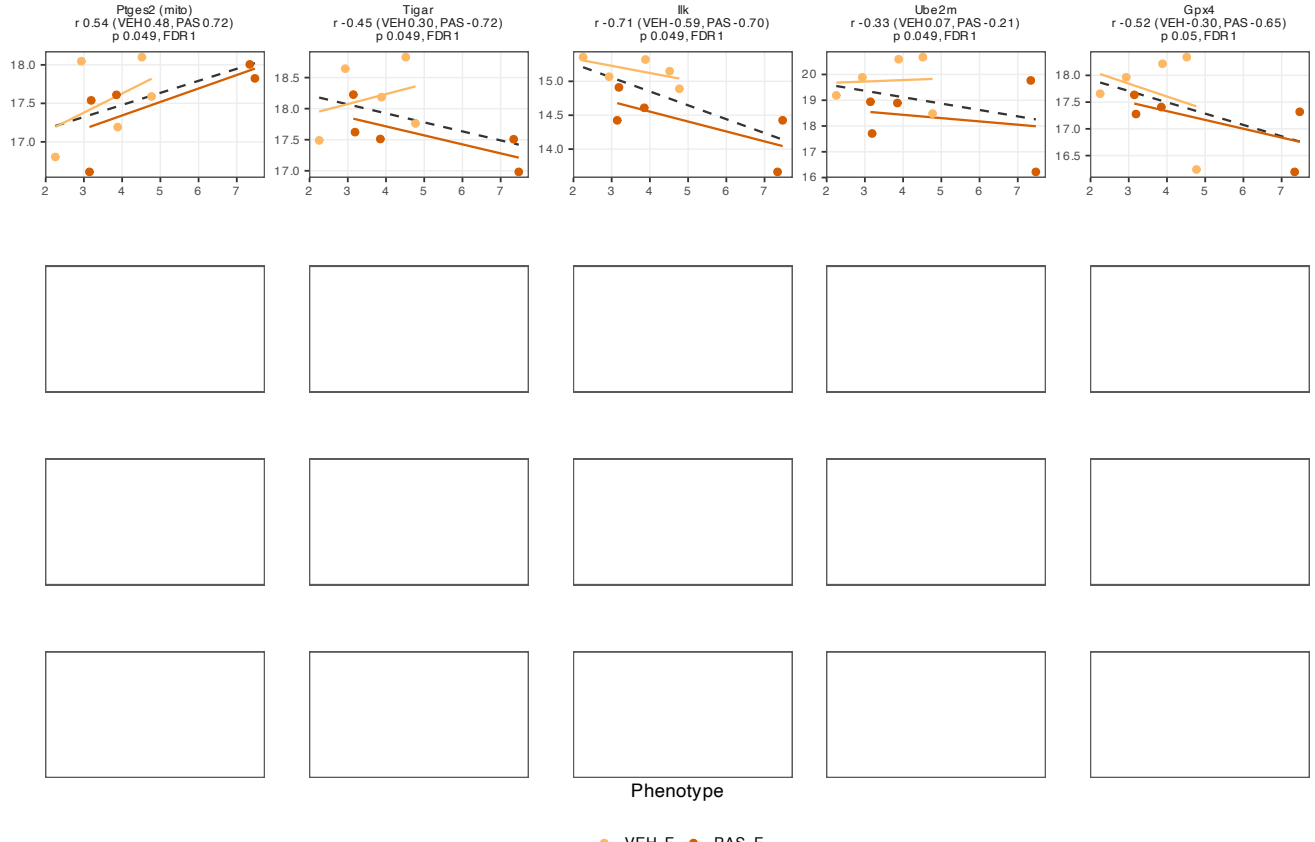
Female: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 65)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



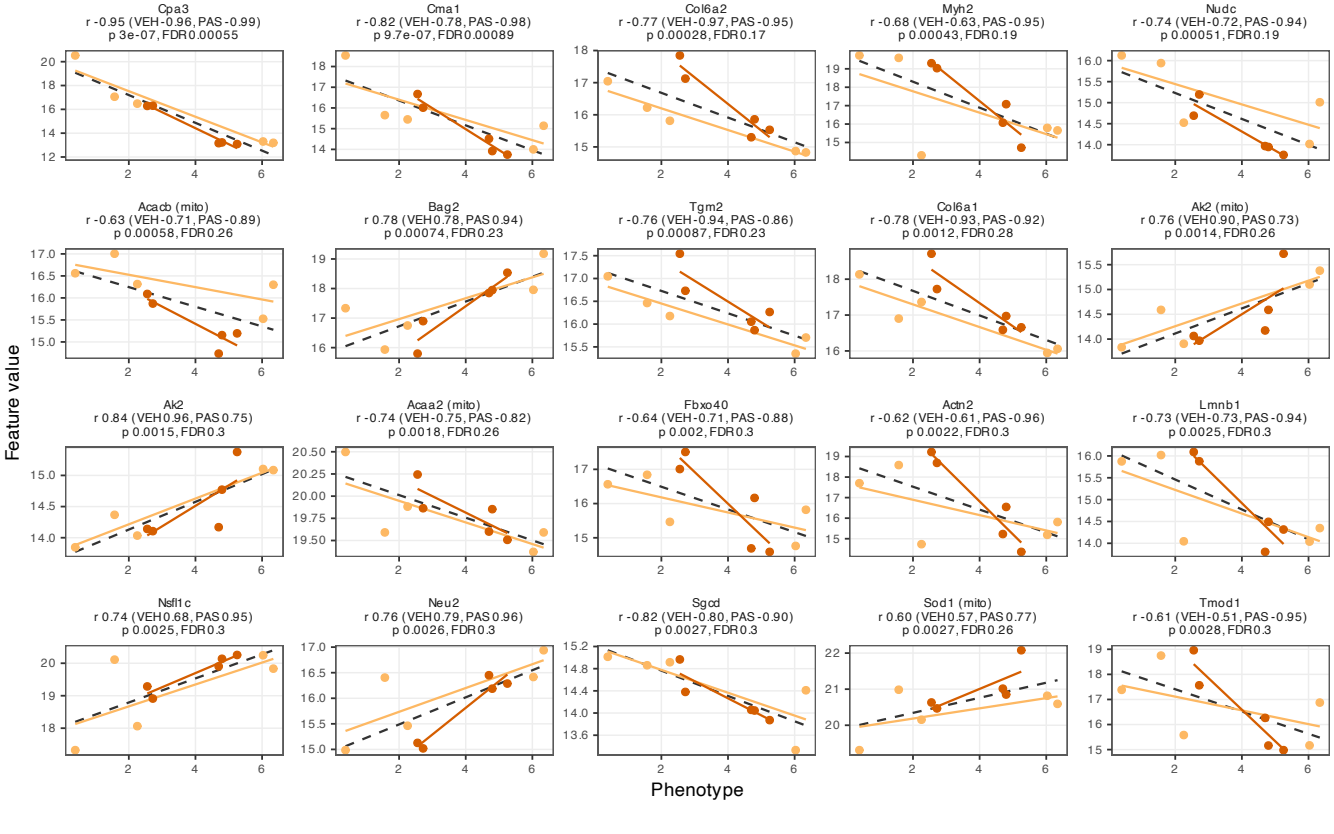
Female: Proteins associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 65)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



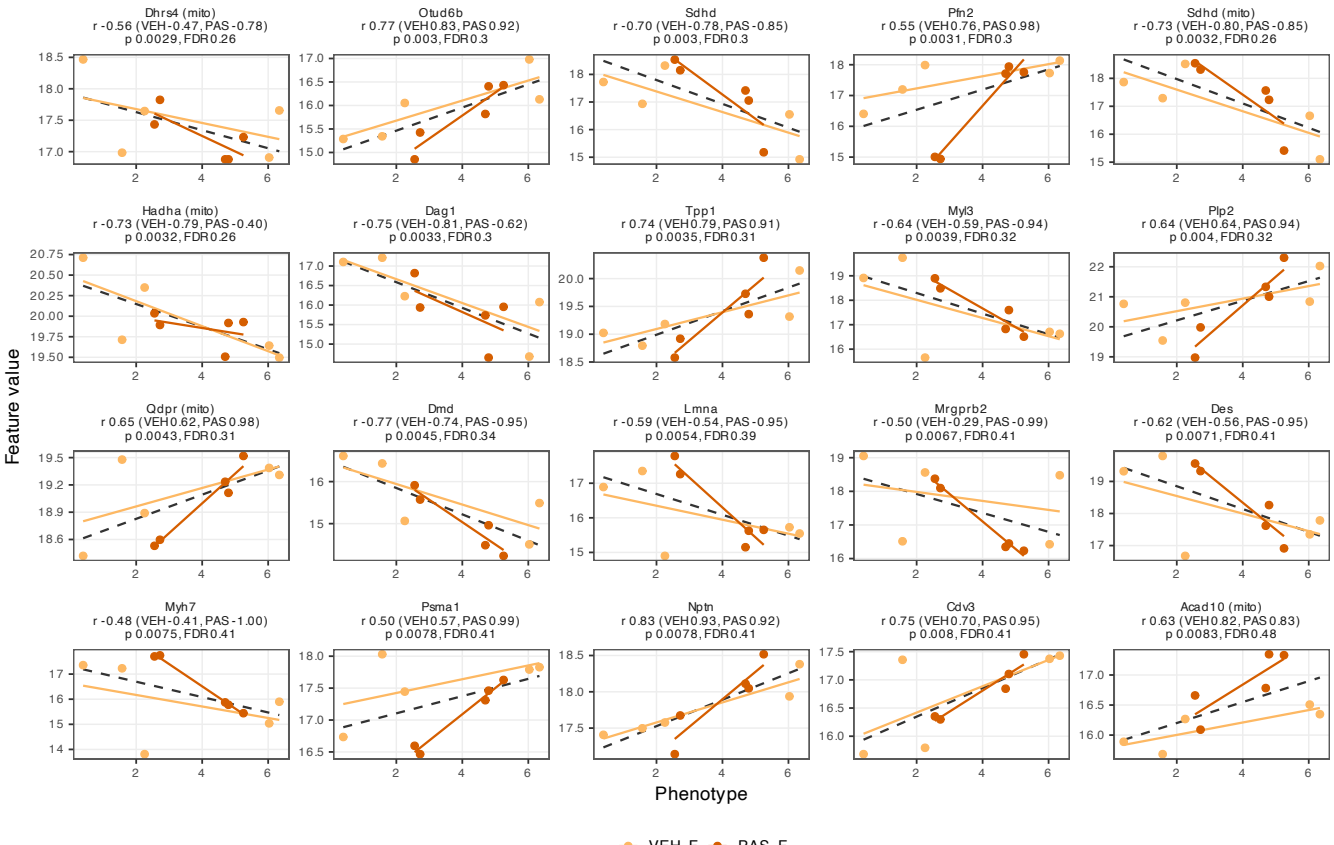
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



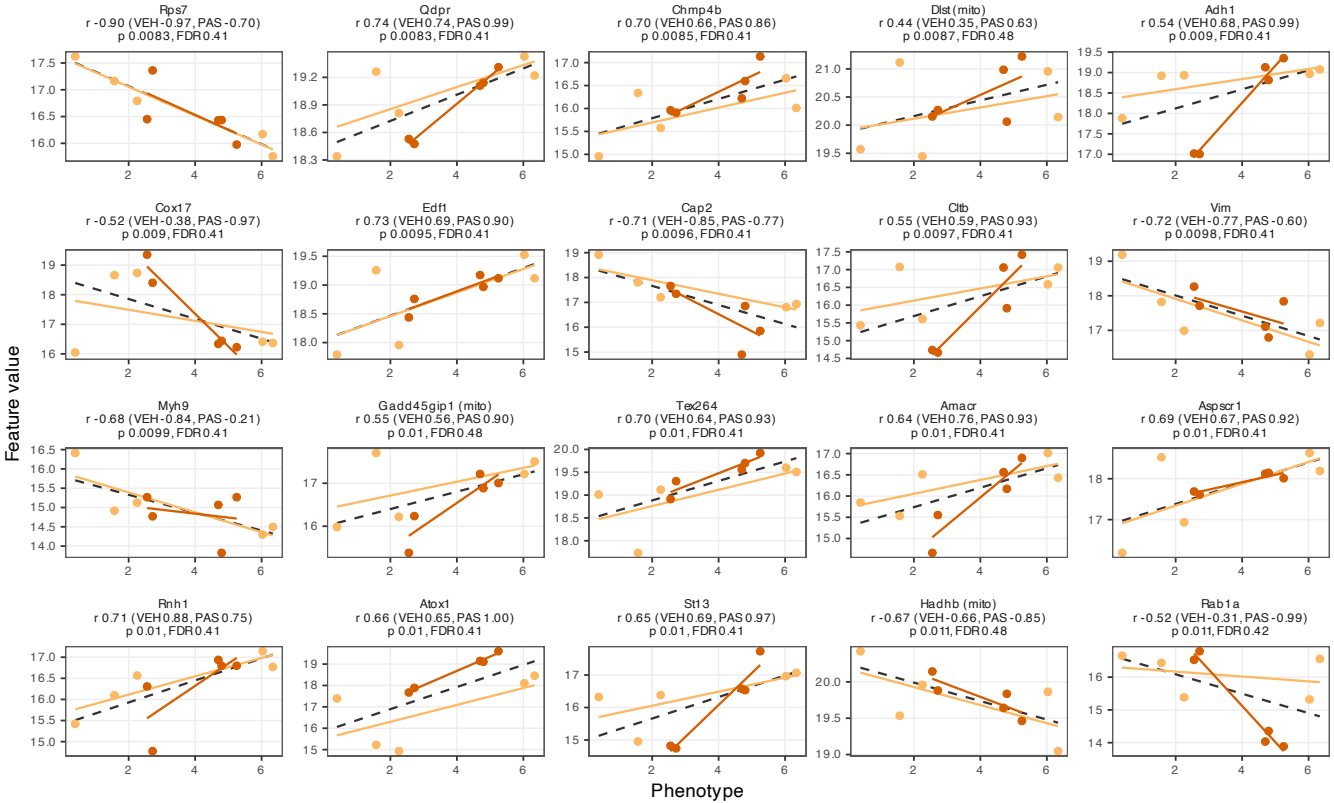
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



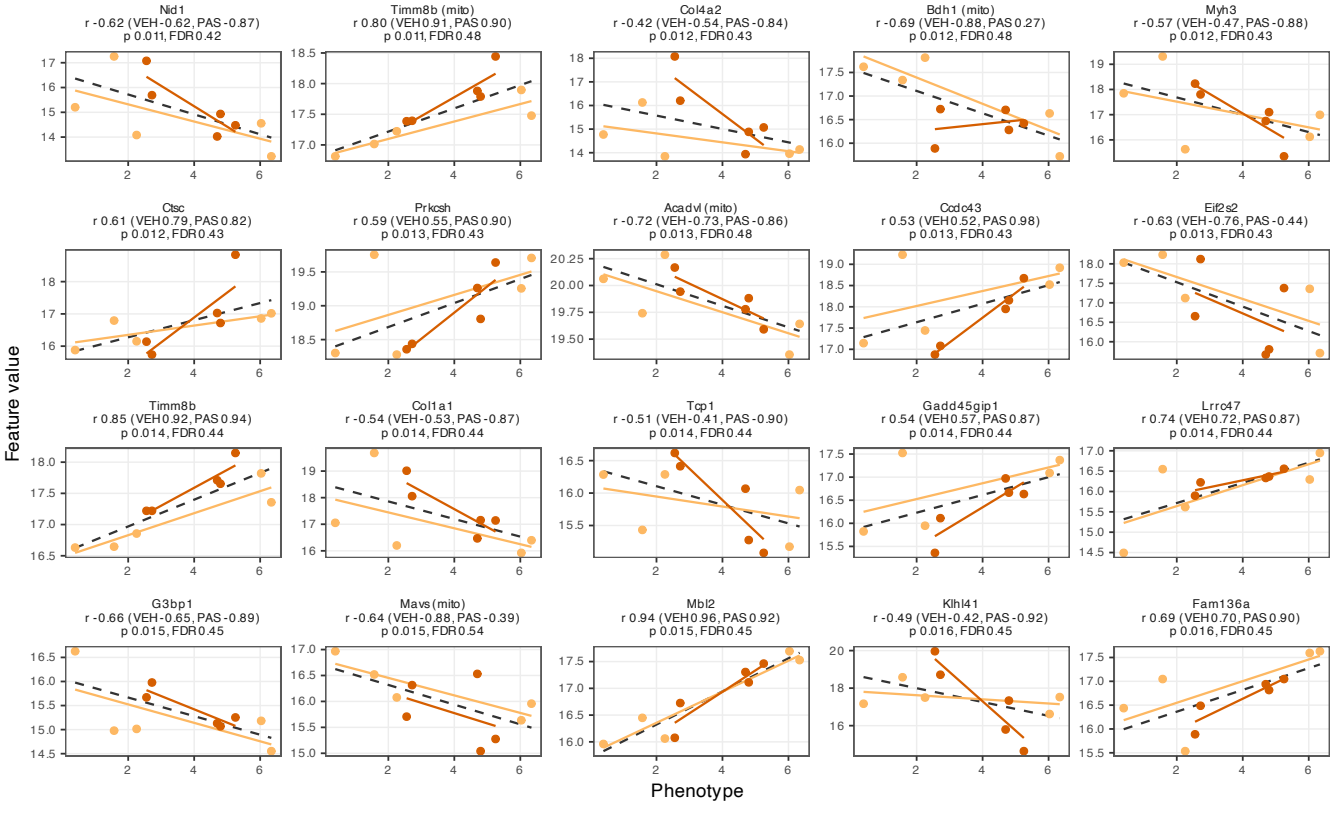
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



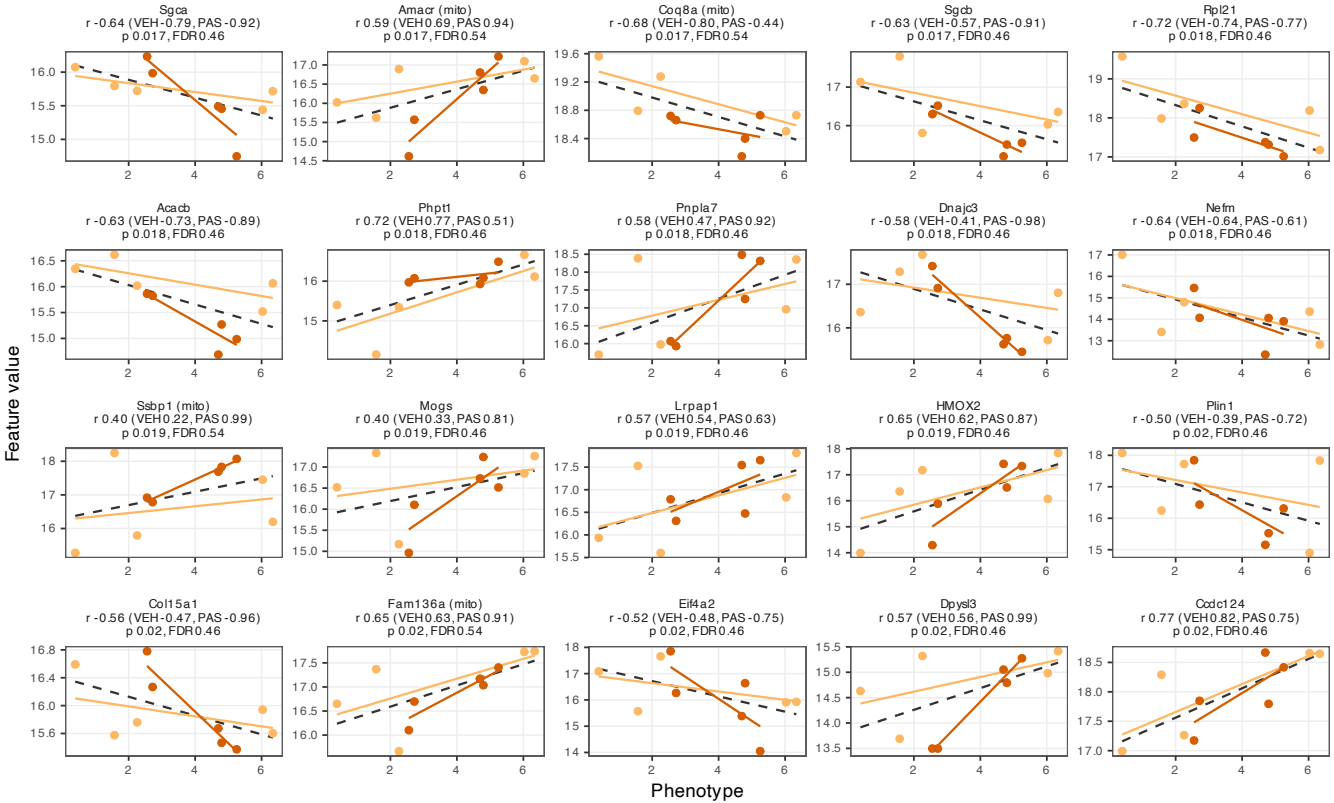
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



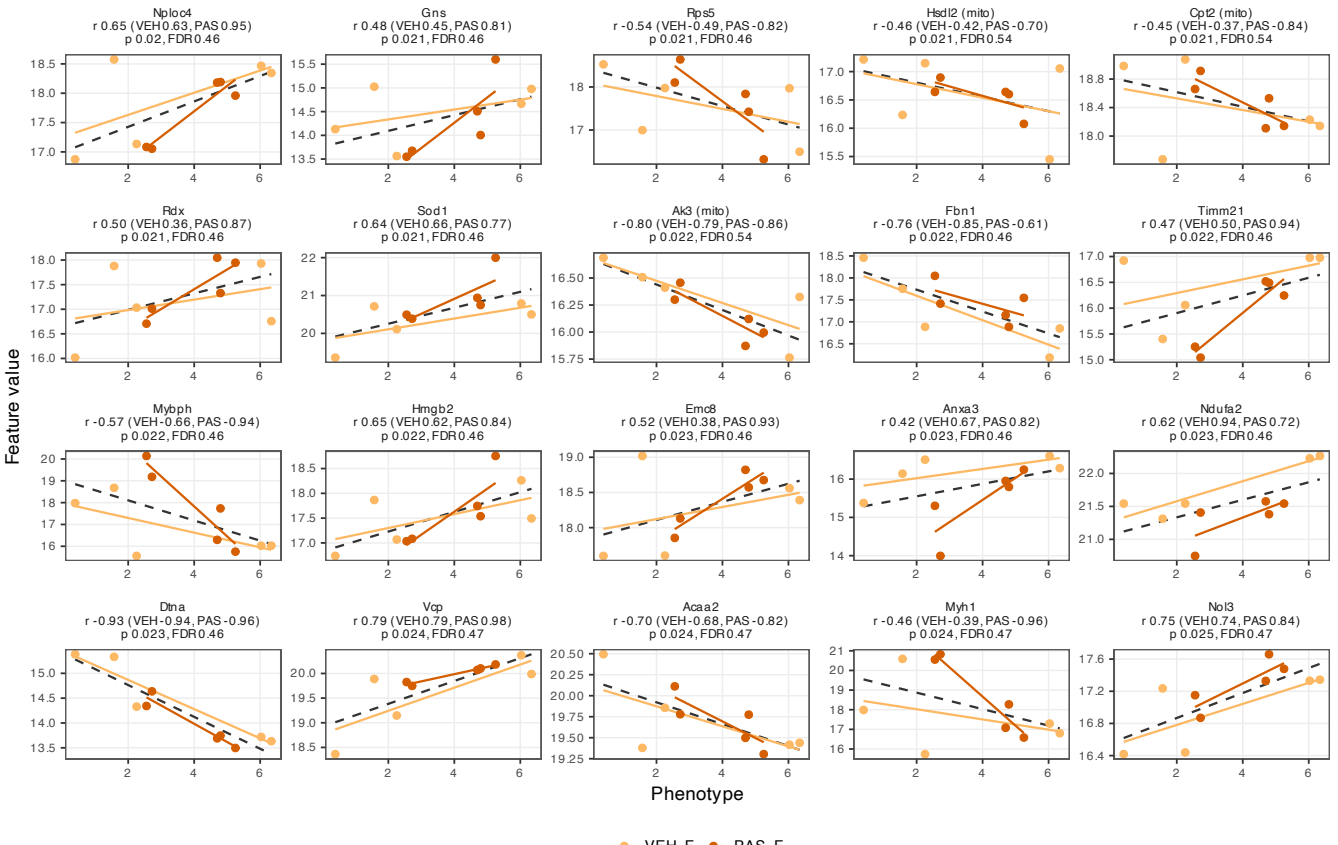
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



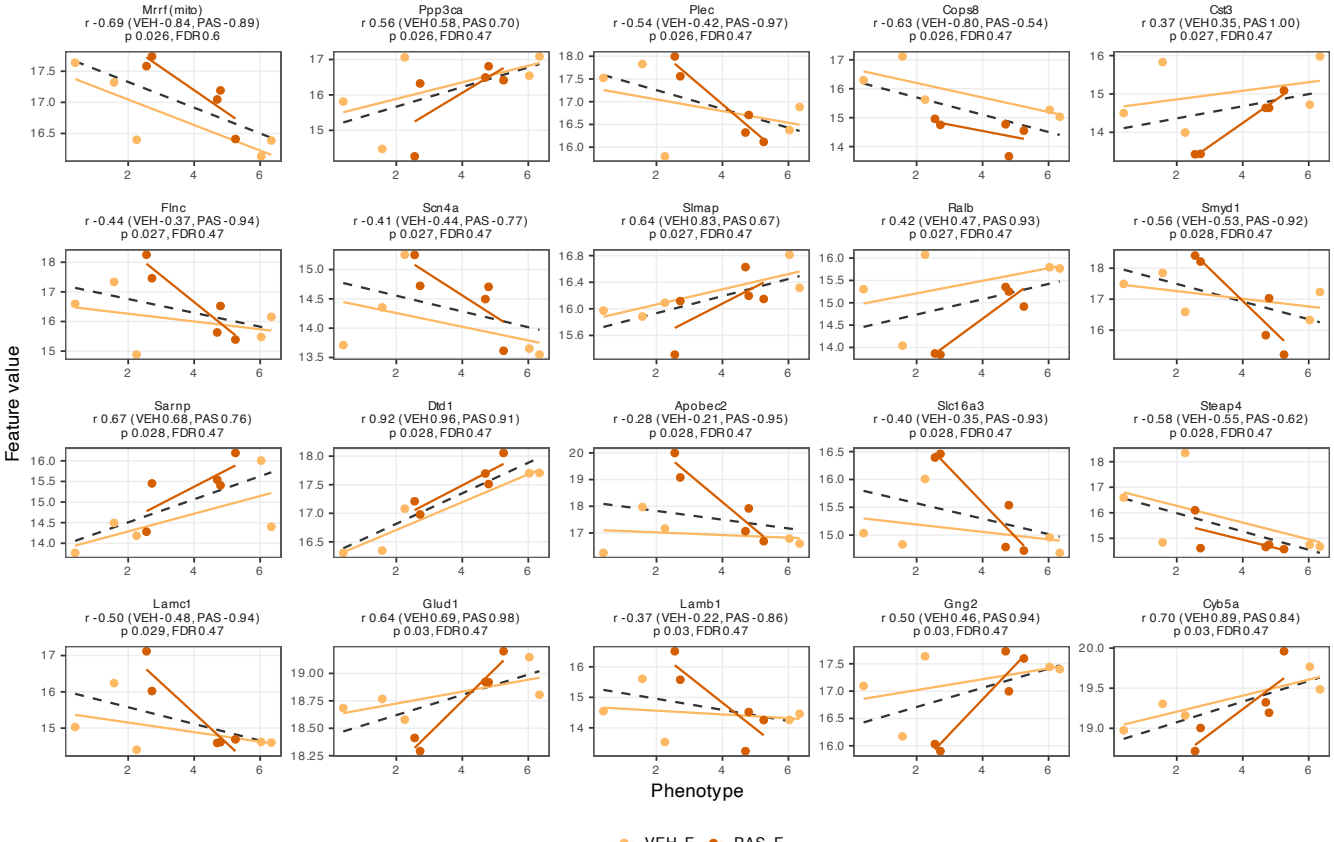
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



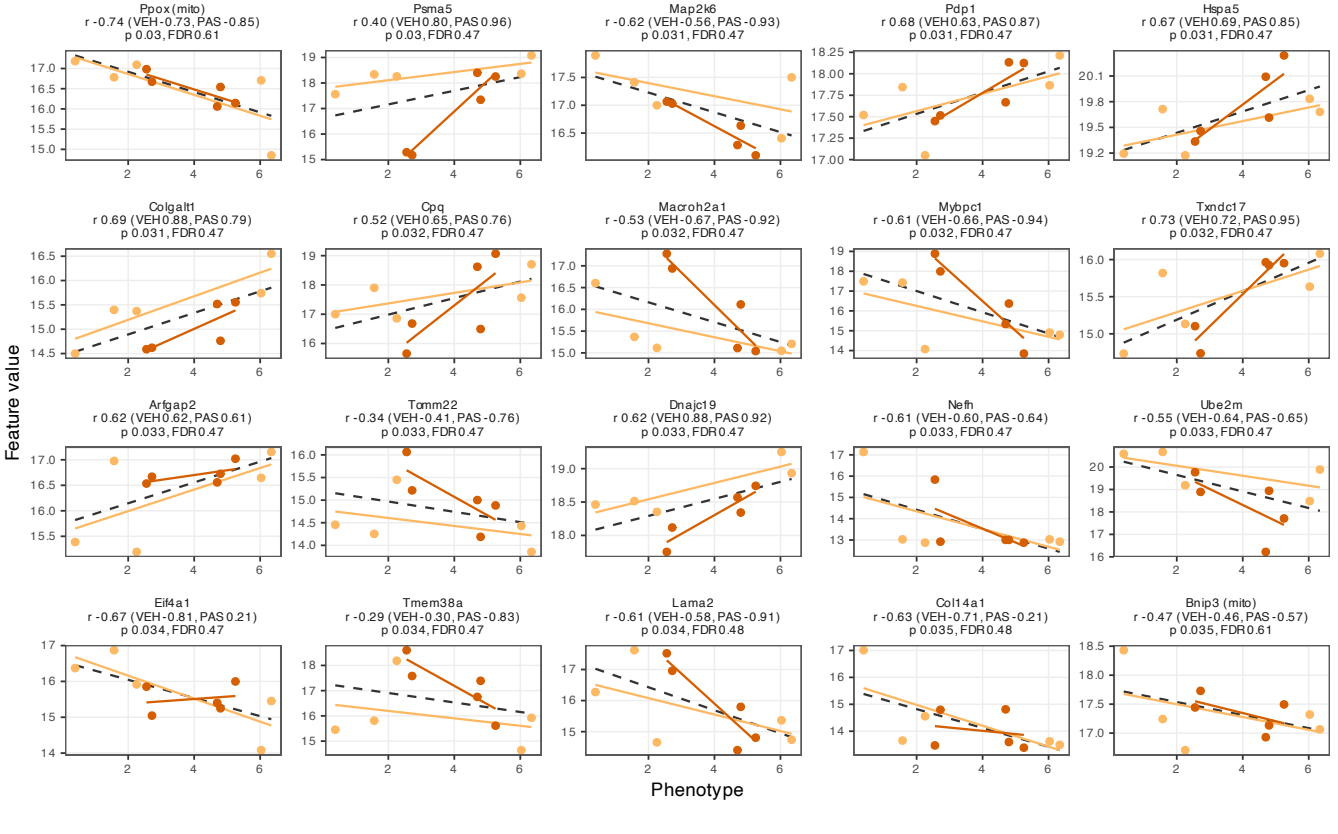
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



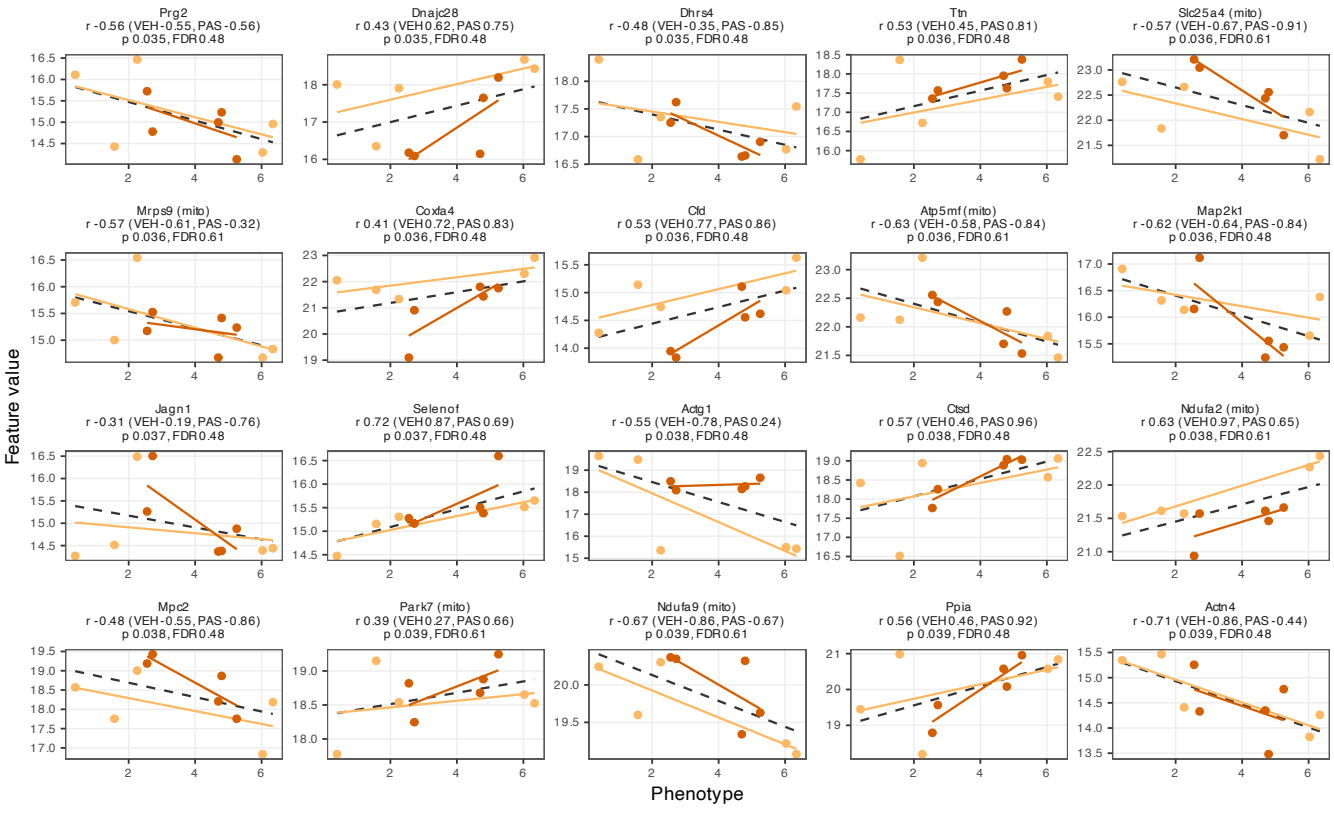
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



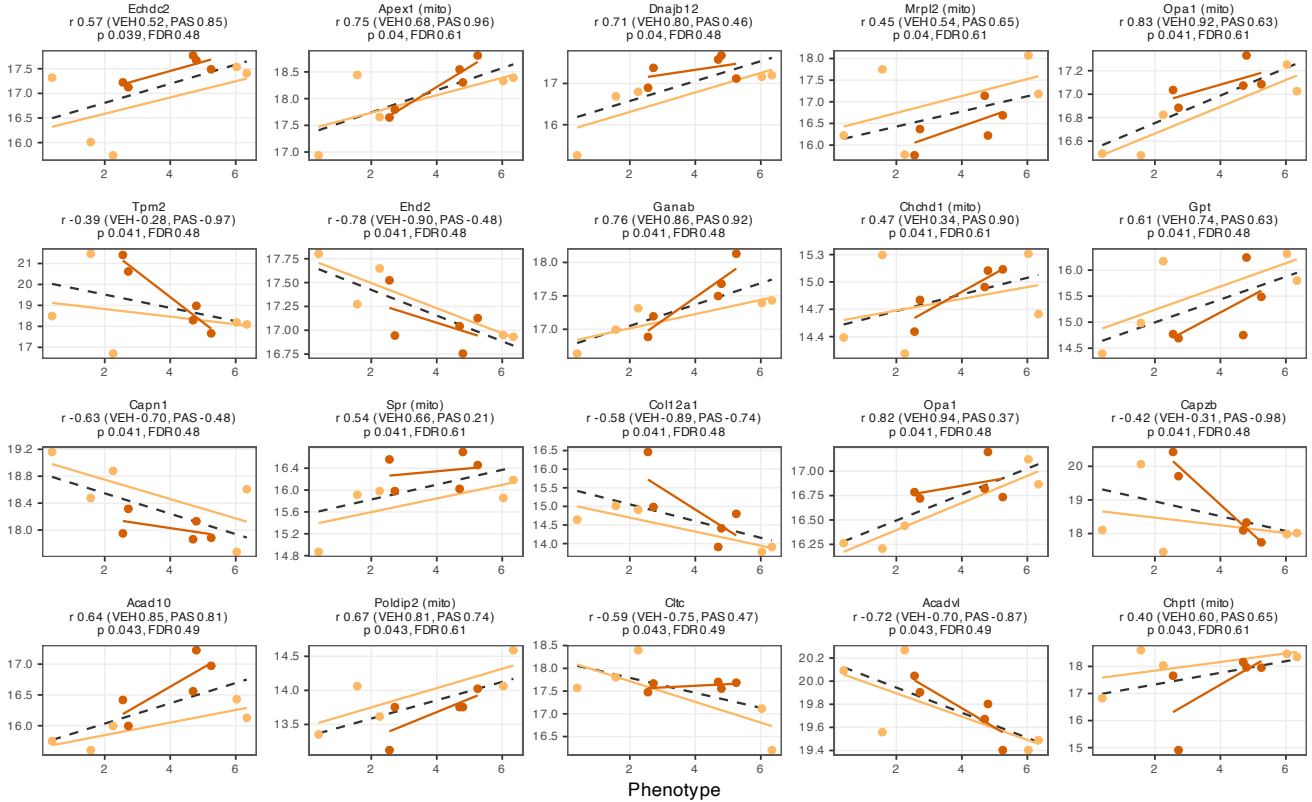
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



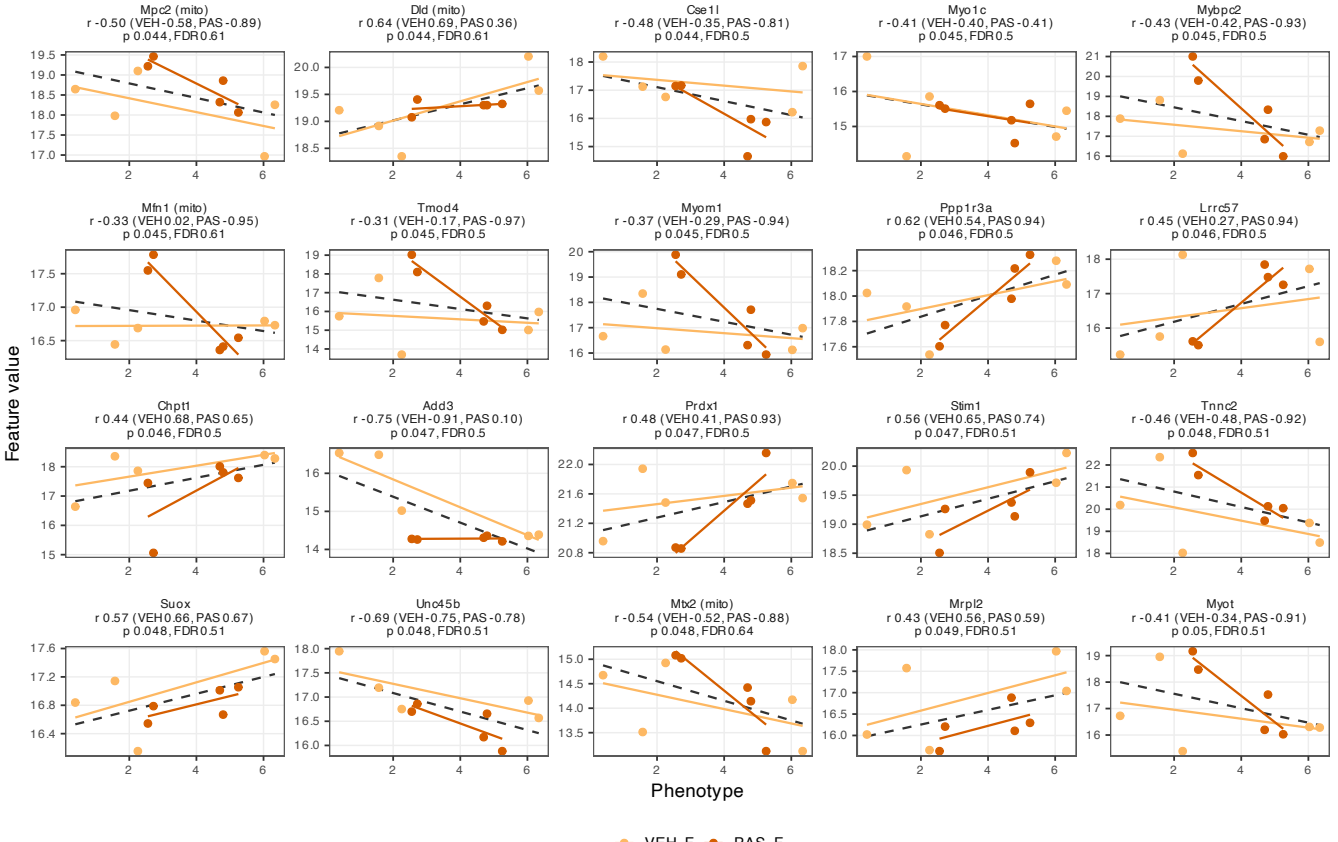
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



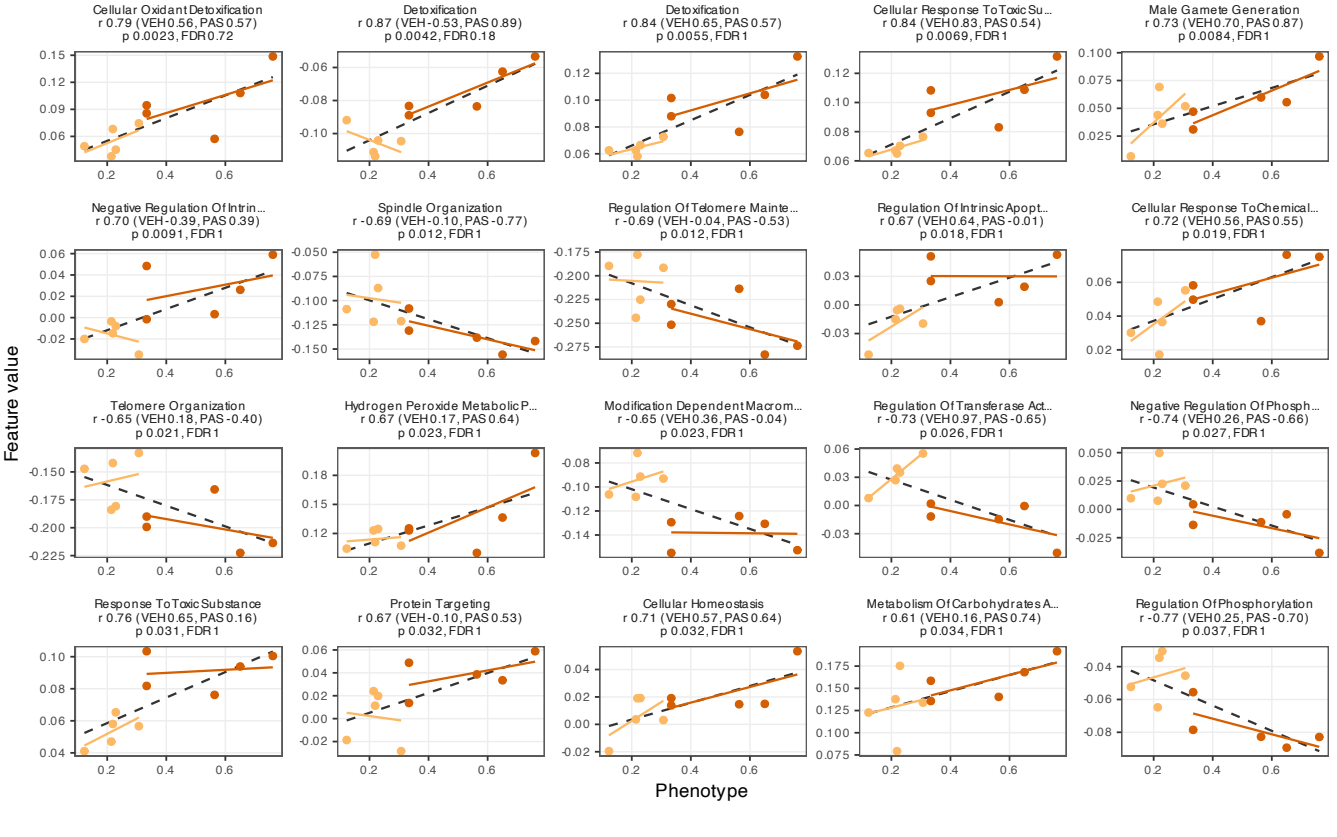
Female: Proteins associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 220)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



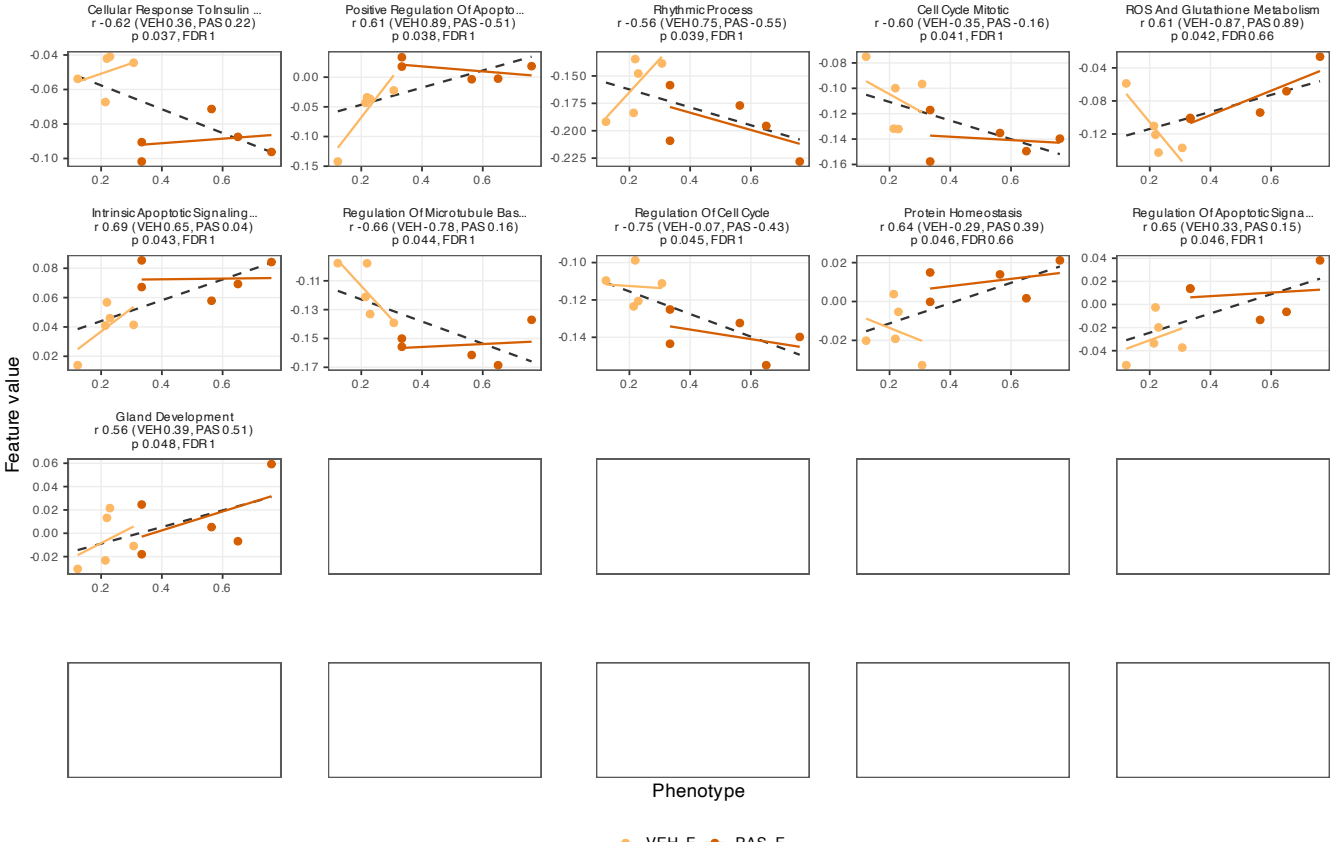
Female: Pathways associated with Capillary density, every association at p ≤ 0.05 (n = 31)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



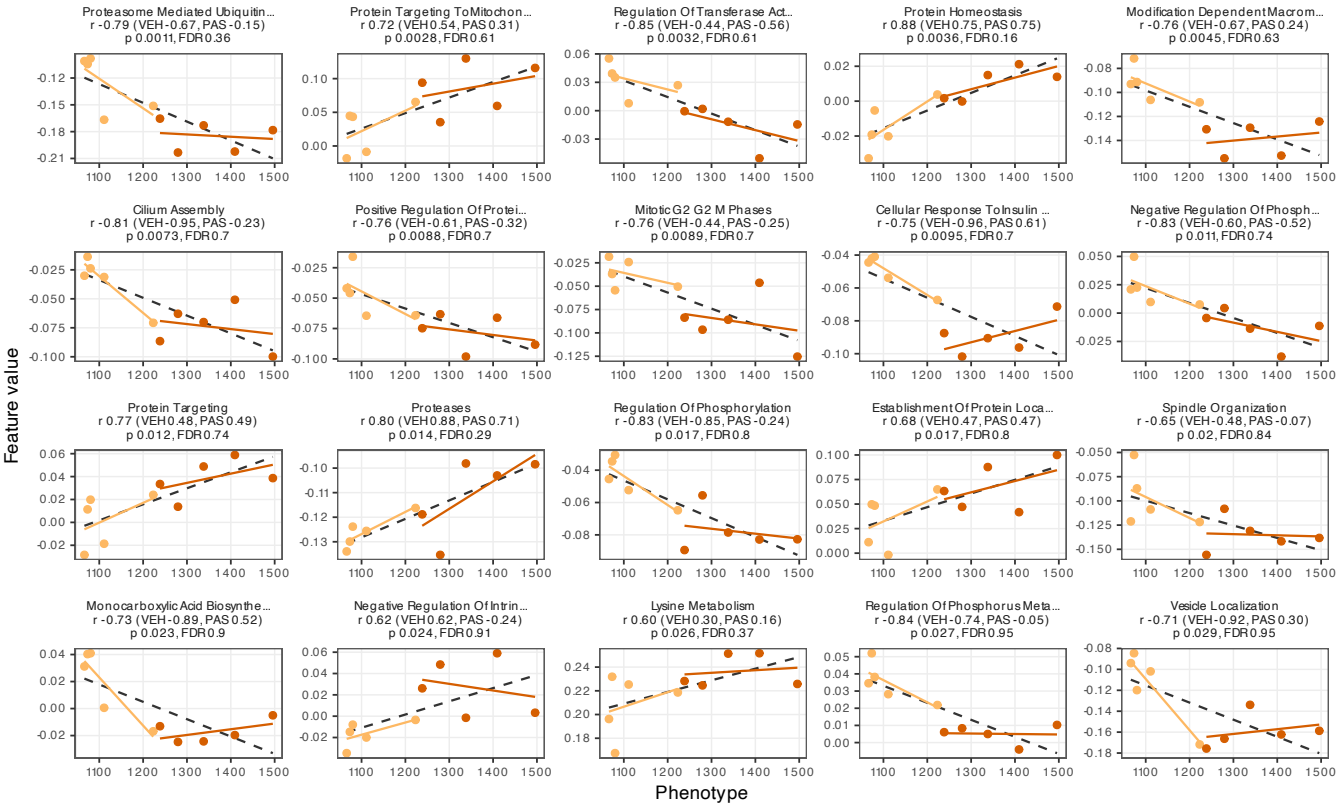
Female: Pathways associated with Capillary density, every association at p ≤ 0.05 (n = 31)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



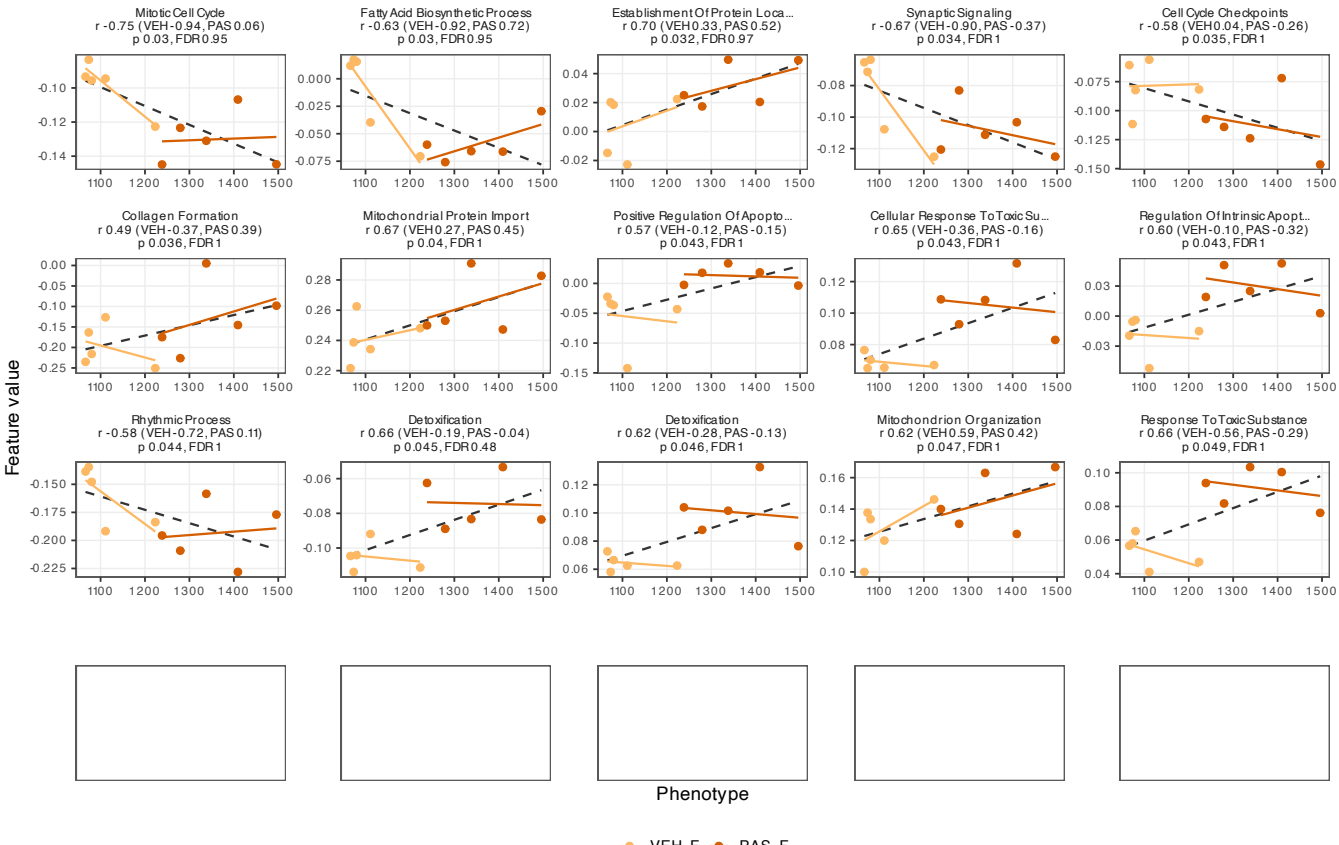
Female: Pathways associated with CSA, every association at p ≤ 0.05 (n = 35)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



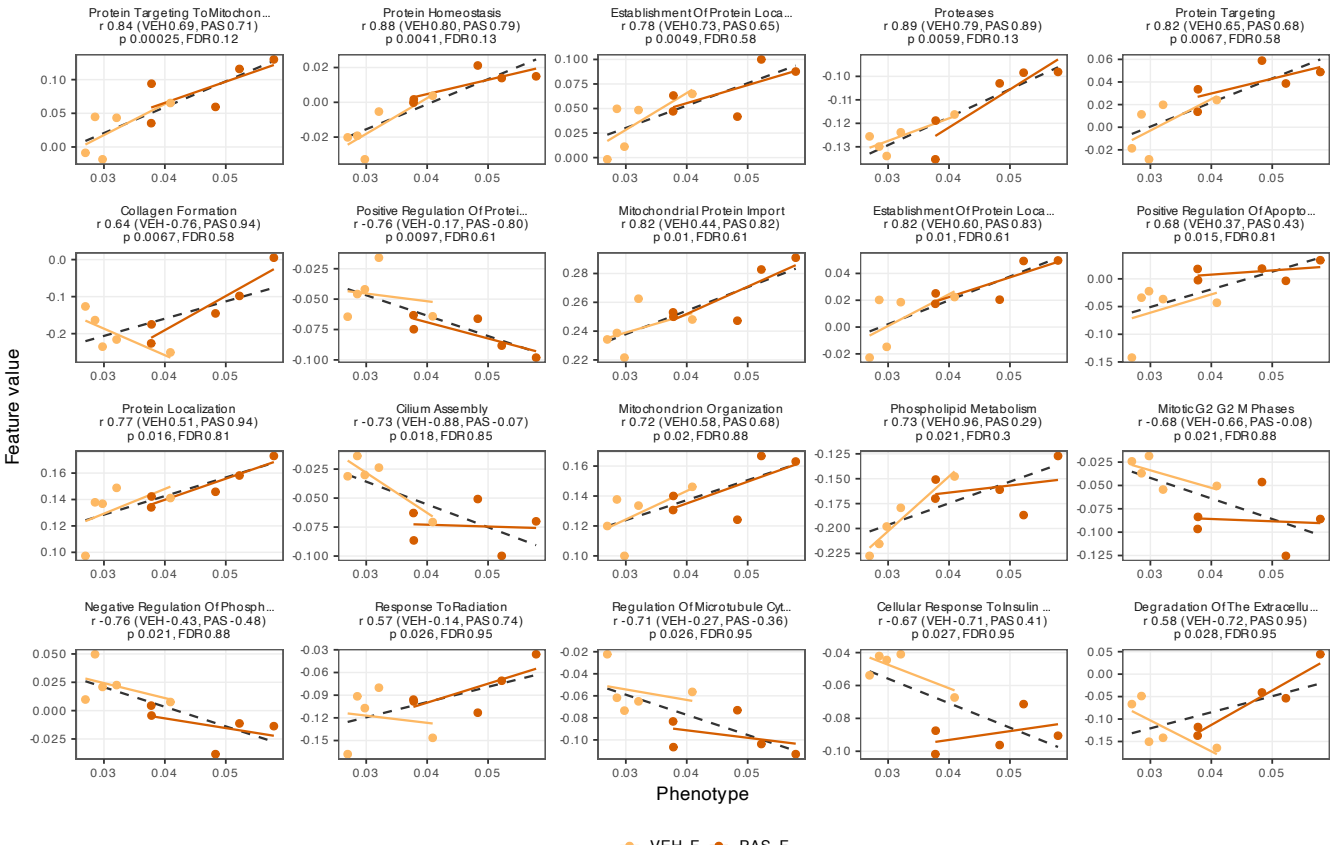
Female: Pathways associated with CSA, every association at p ≤ 0.05 (n = 35)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



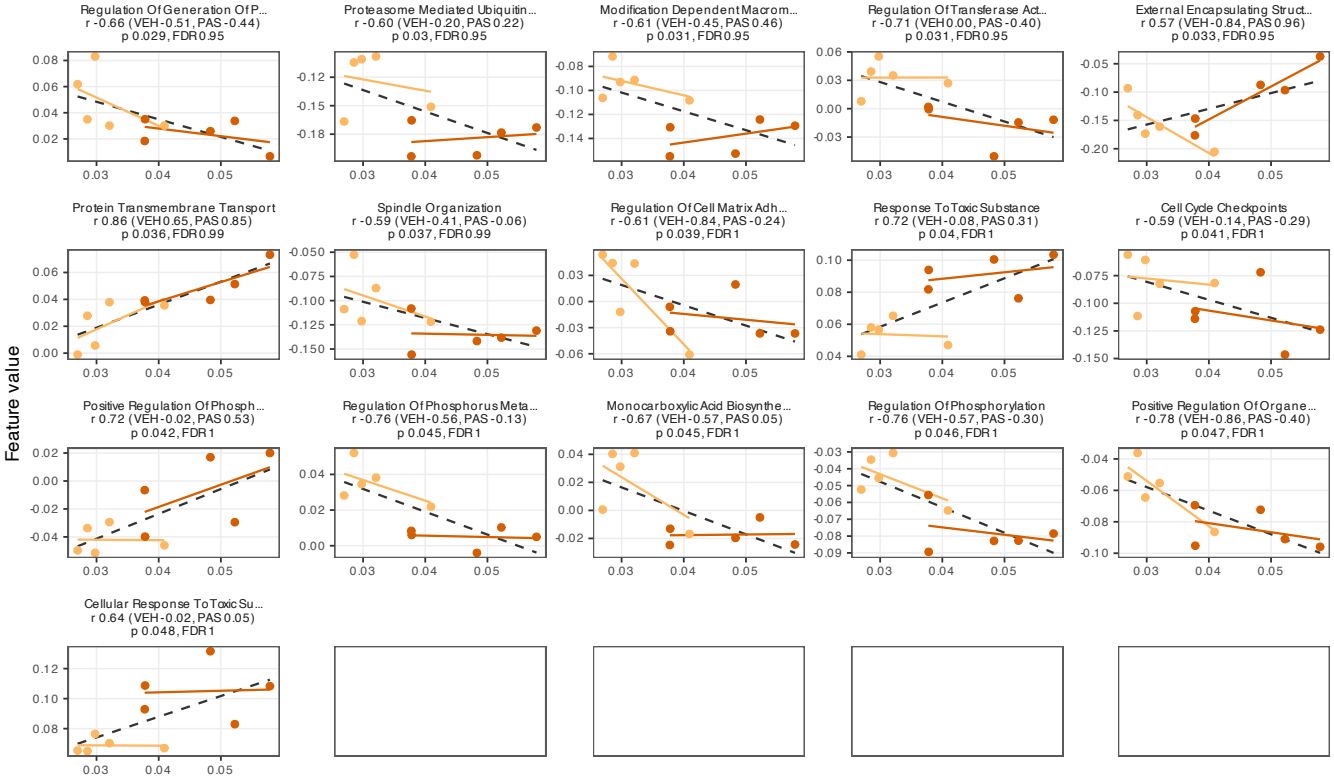
Female: Pathways associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 36)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



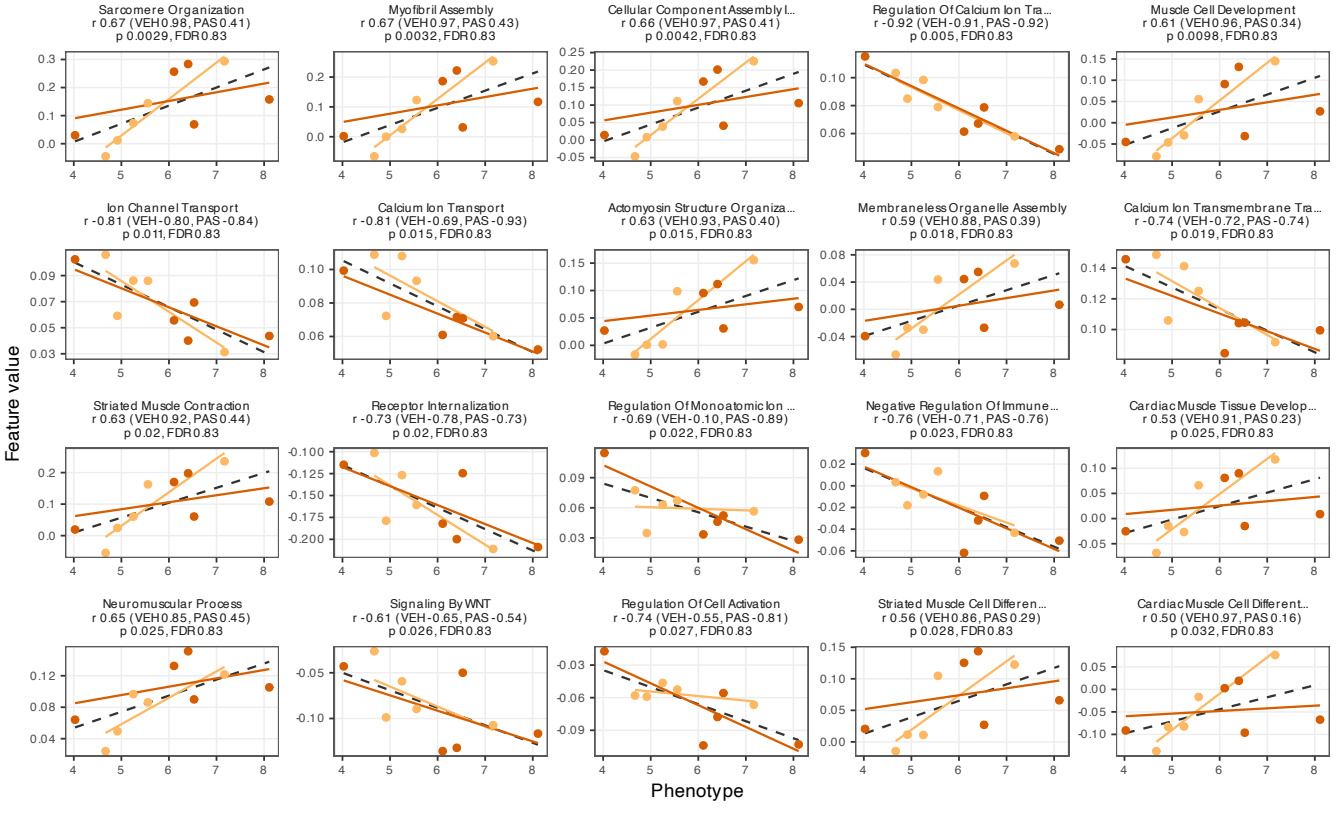
Female: Pathways associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 36)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



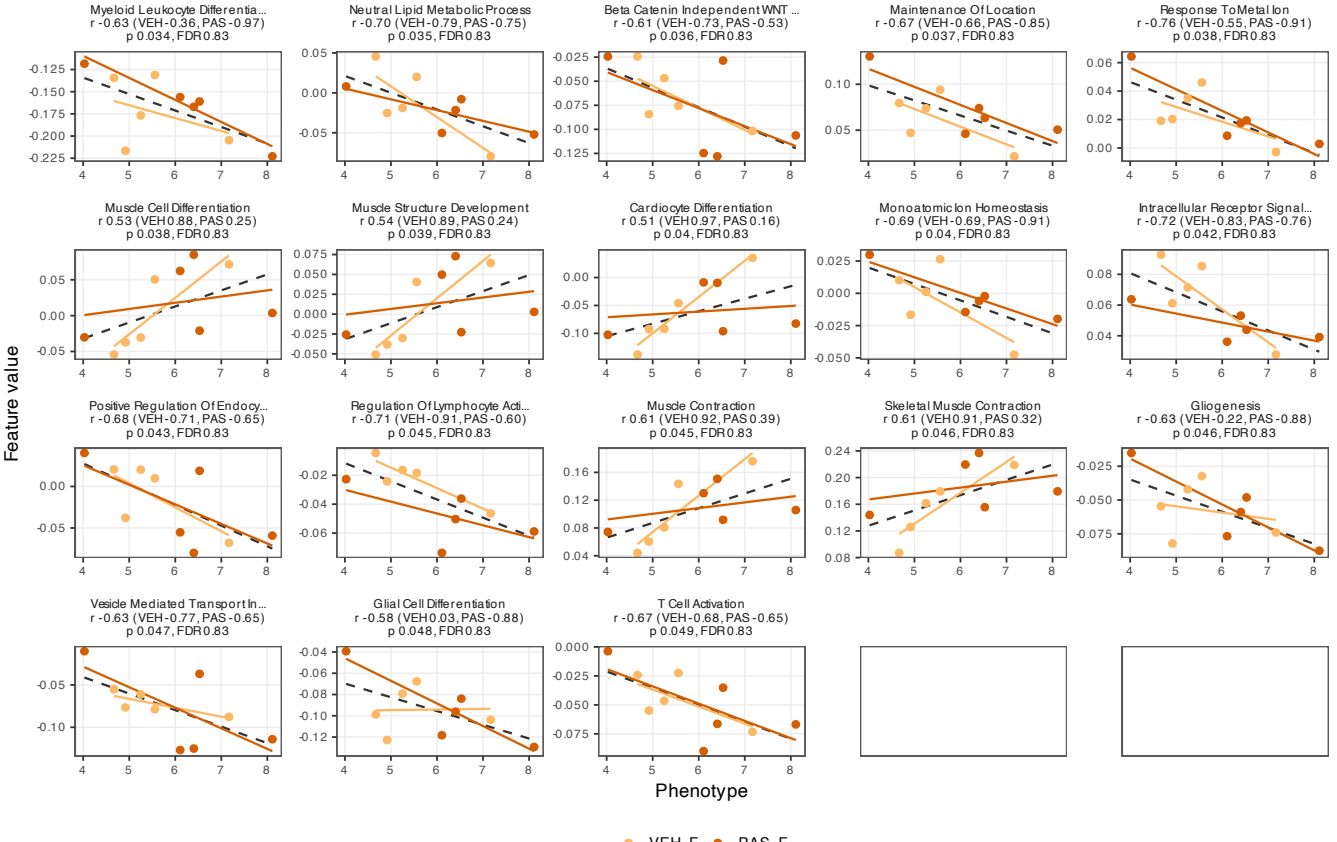
Female: Pathways associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 38)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



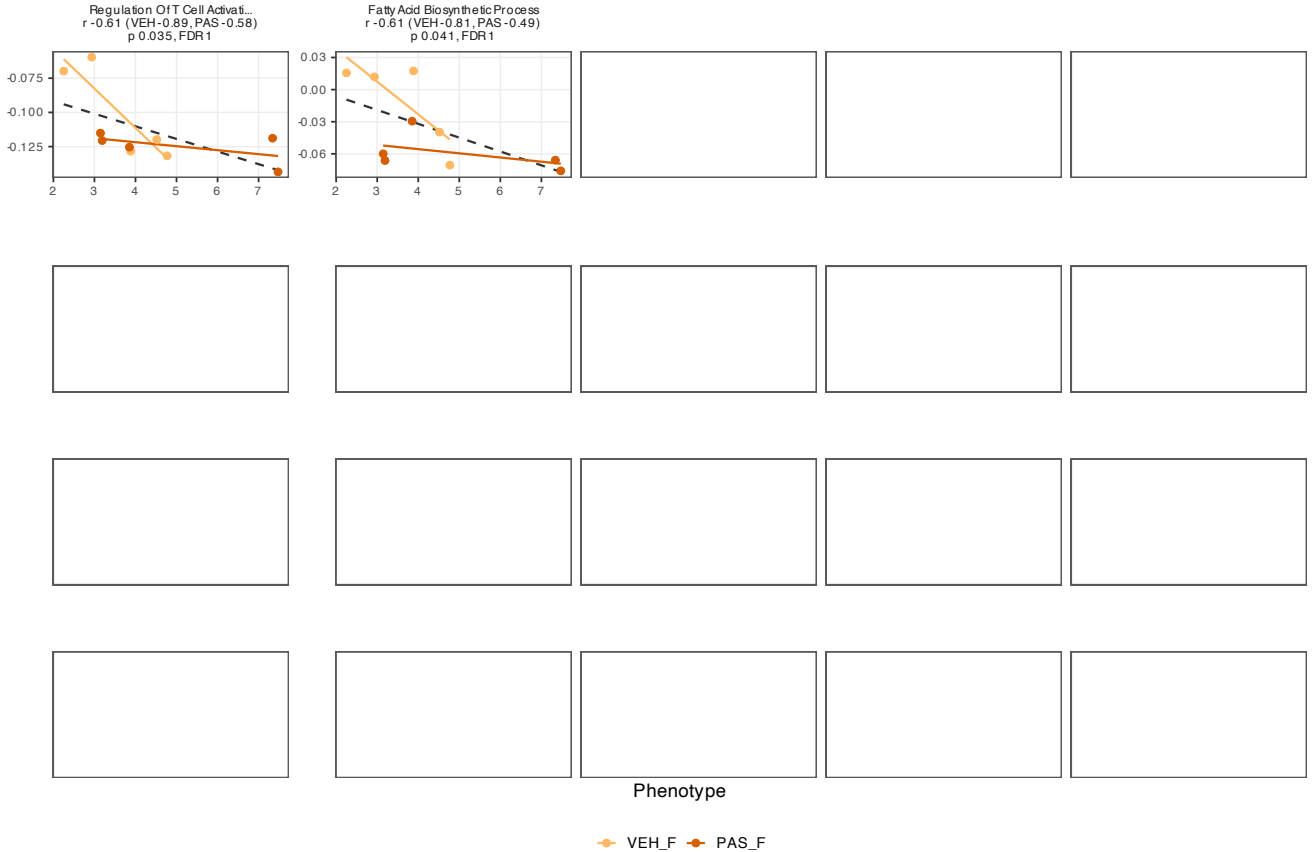
Female: Pathways associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 38)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



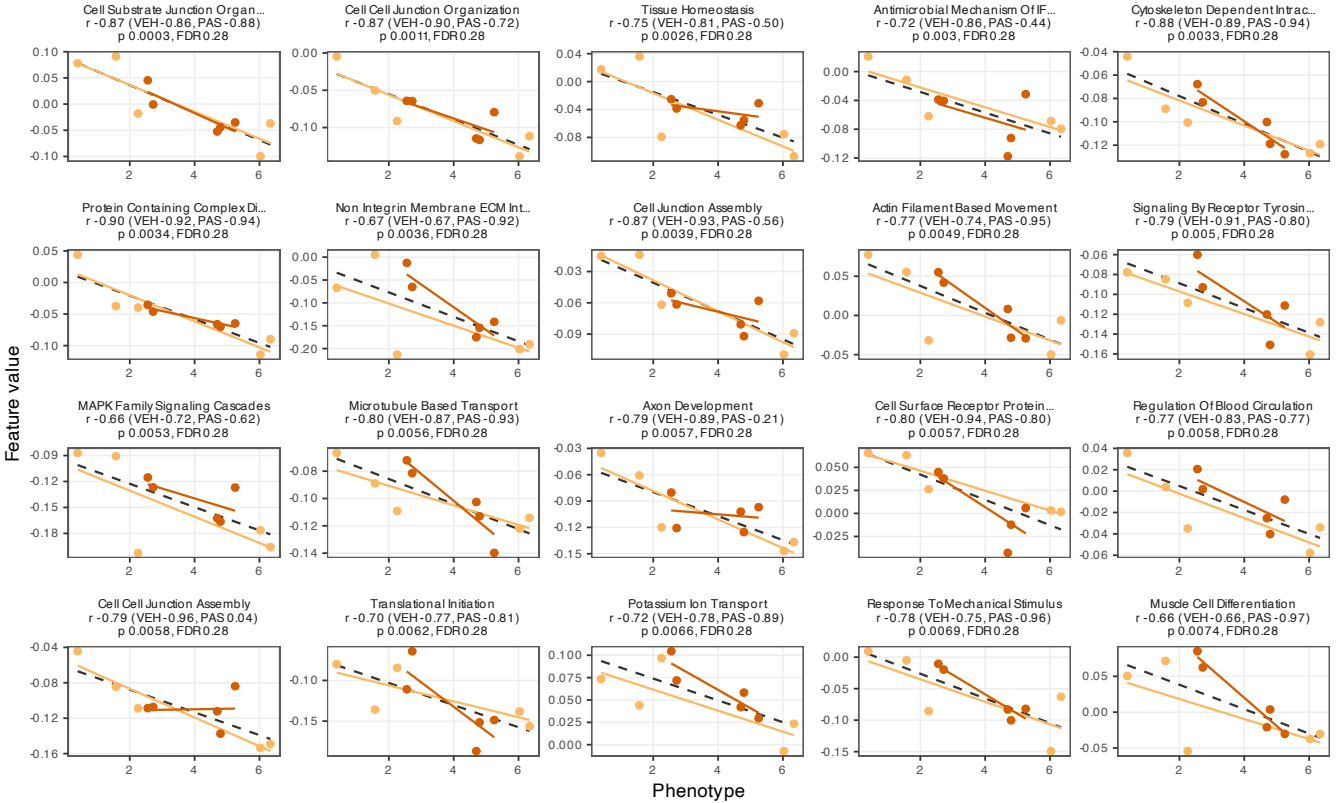
Female: Pathways associated with Plantaris pyruvate, every association at p ≤ 0.05 (n = 2)

Phenotype effect of PAS in this sex: +31% (p 0.3), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



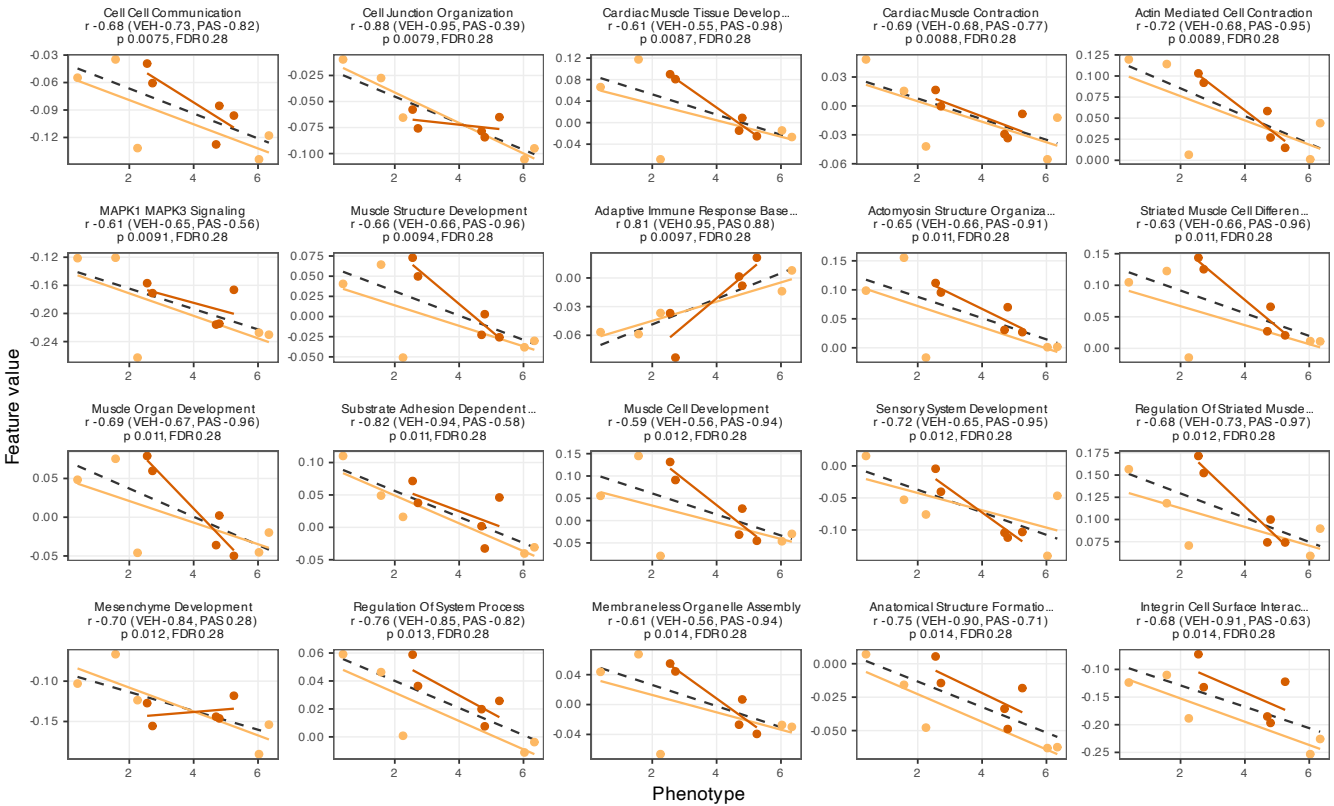
Female: Pathways associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 116)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



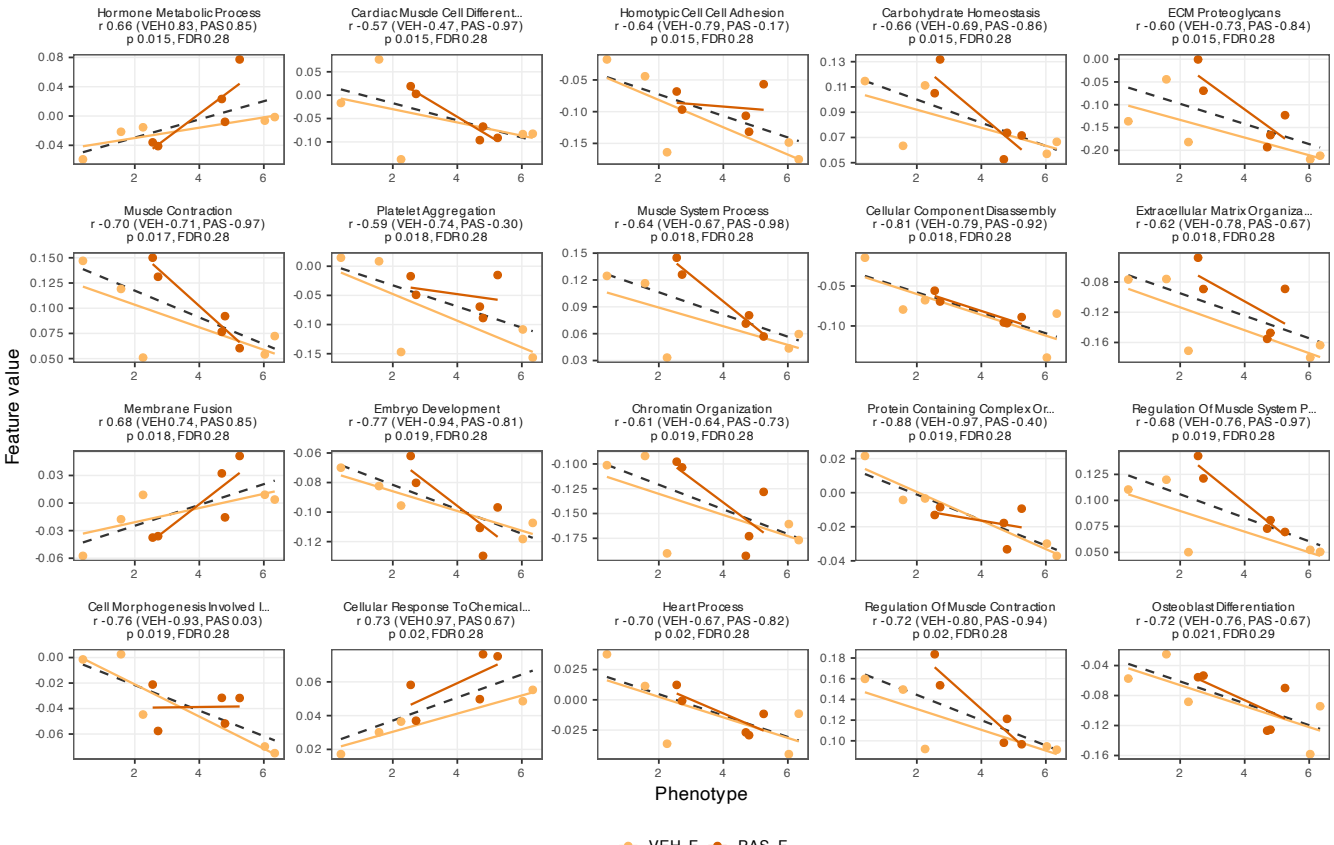
Female: Pathways associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 116)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



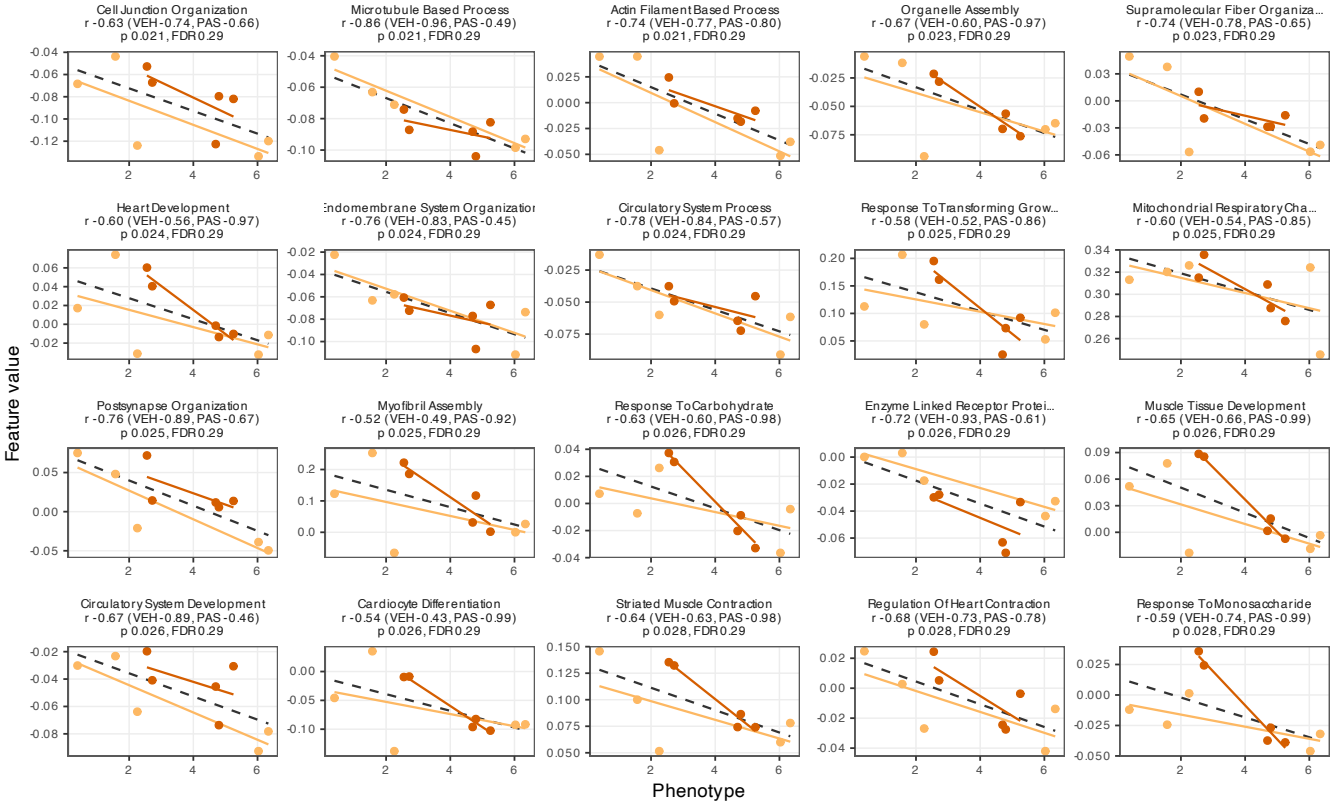
Female: Pathways associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 116)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



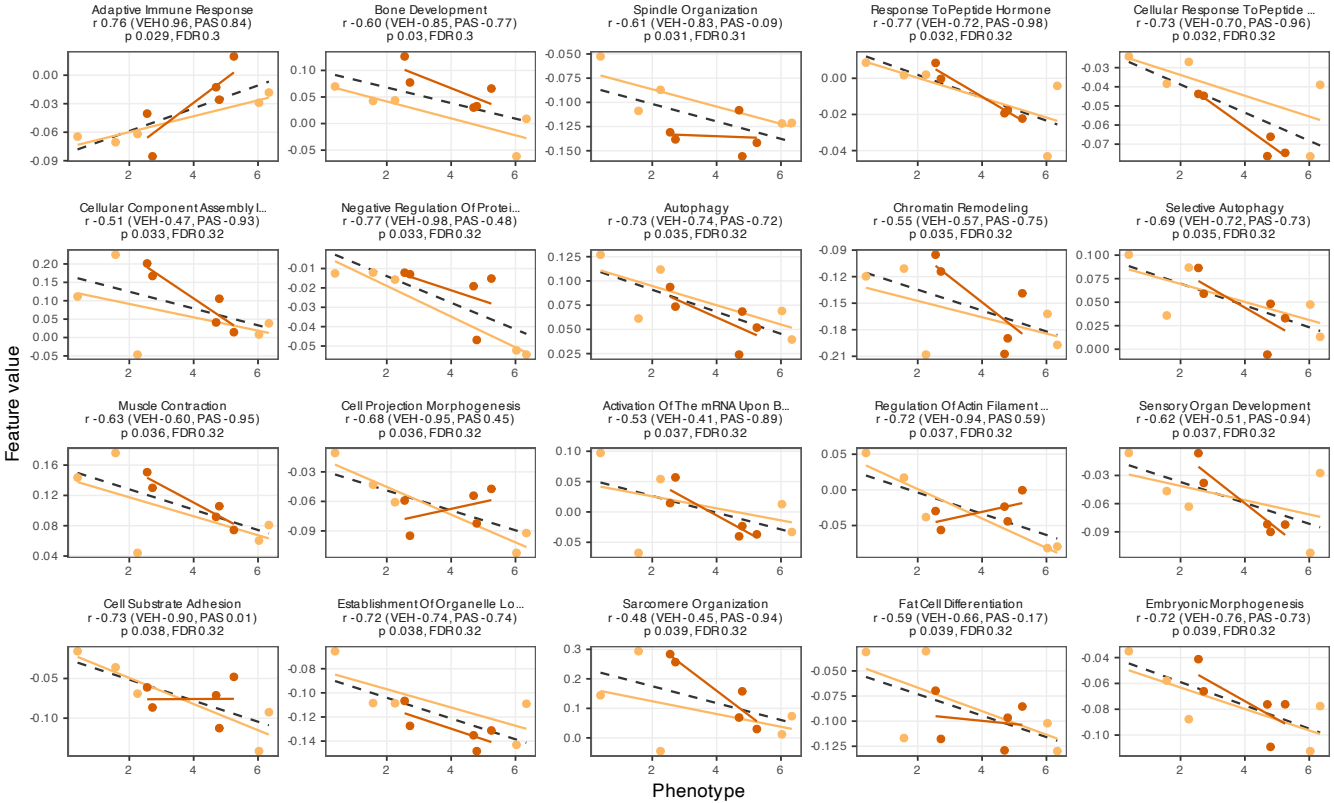
Female: Pathways associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 116)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



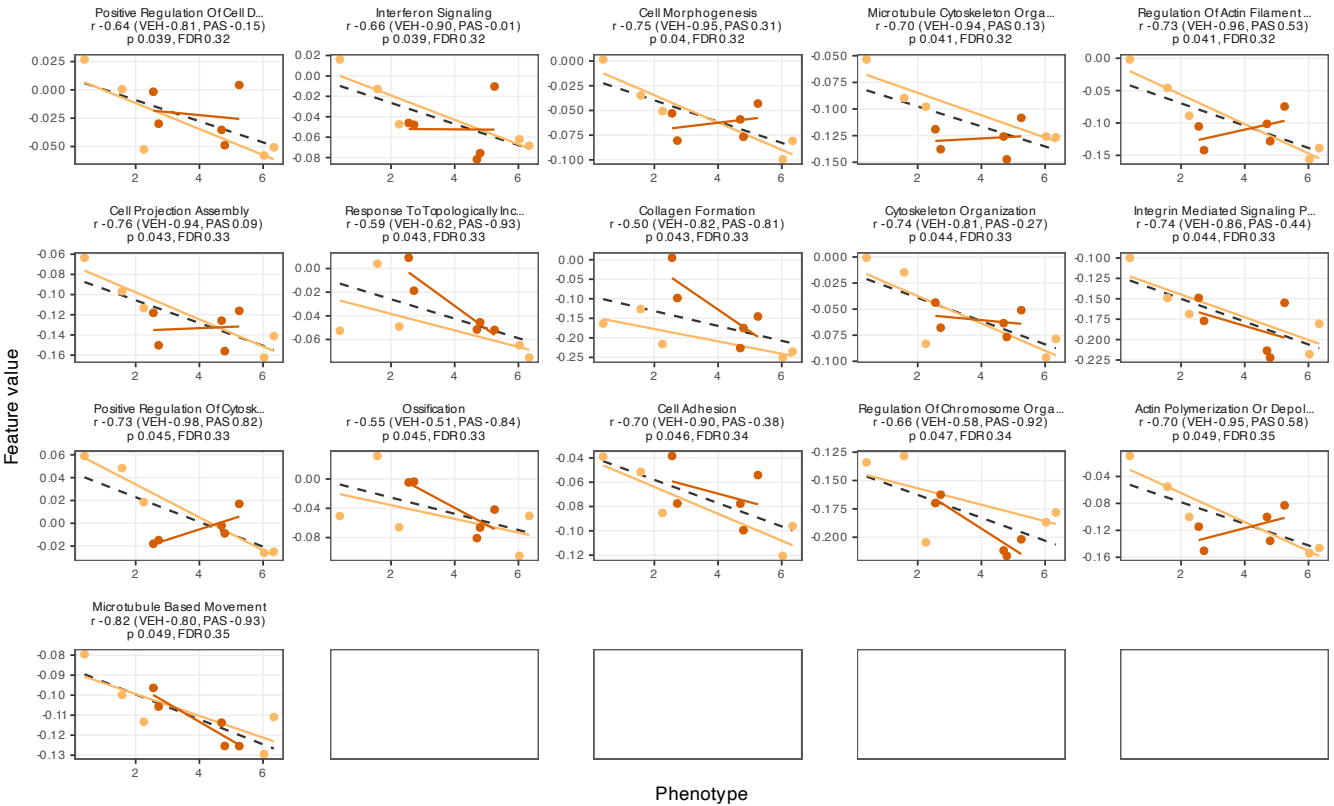
Female: Pathways associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 116)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



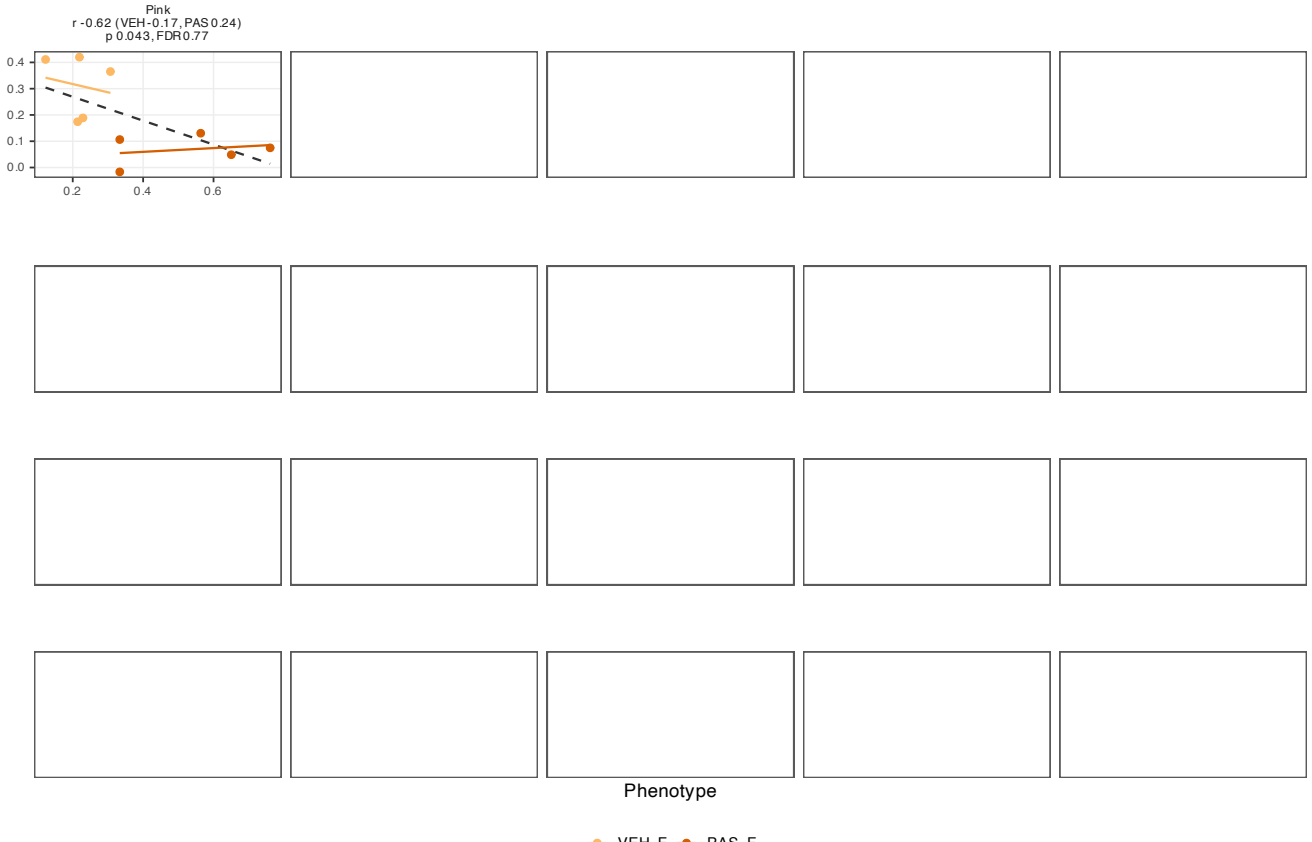
Female: Pathways associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 116)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



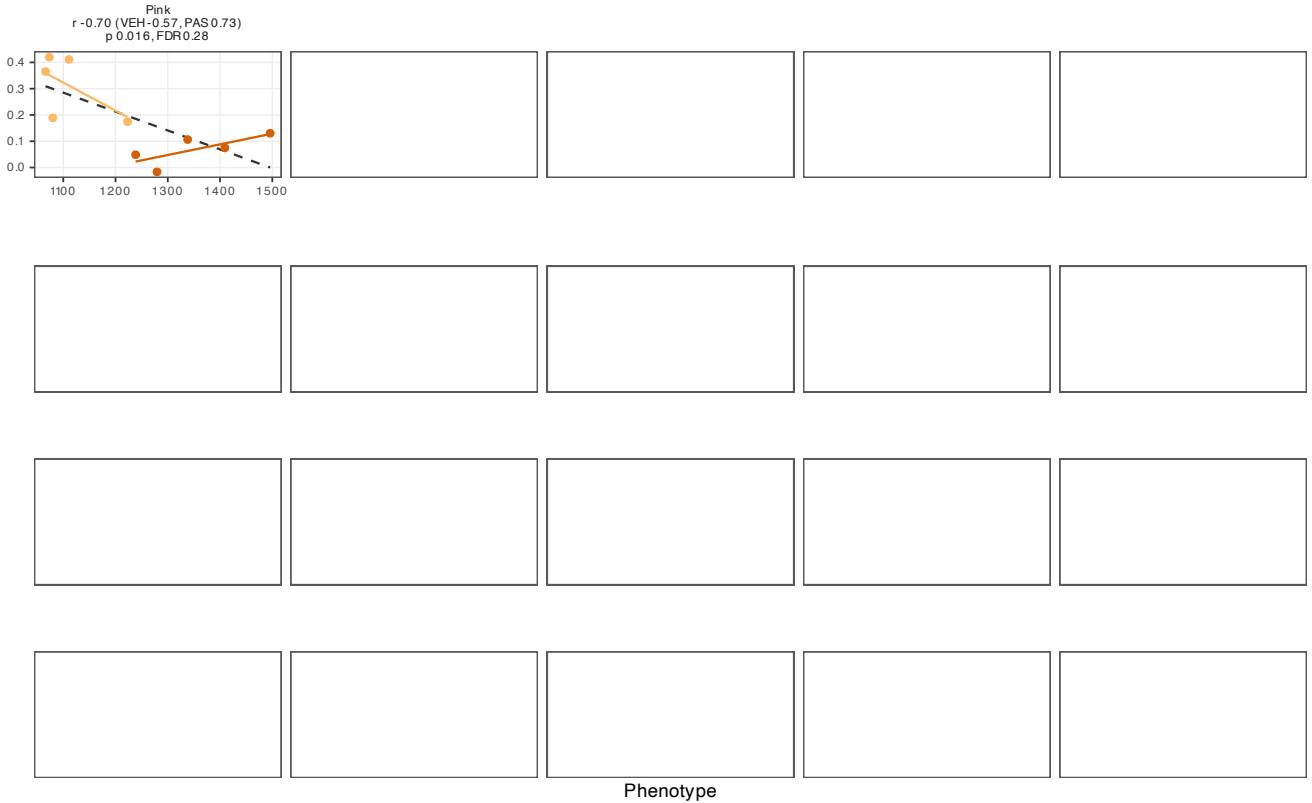
Female: Modules associated with Capillary density, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



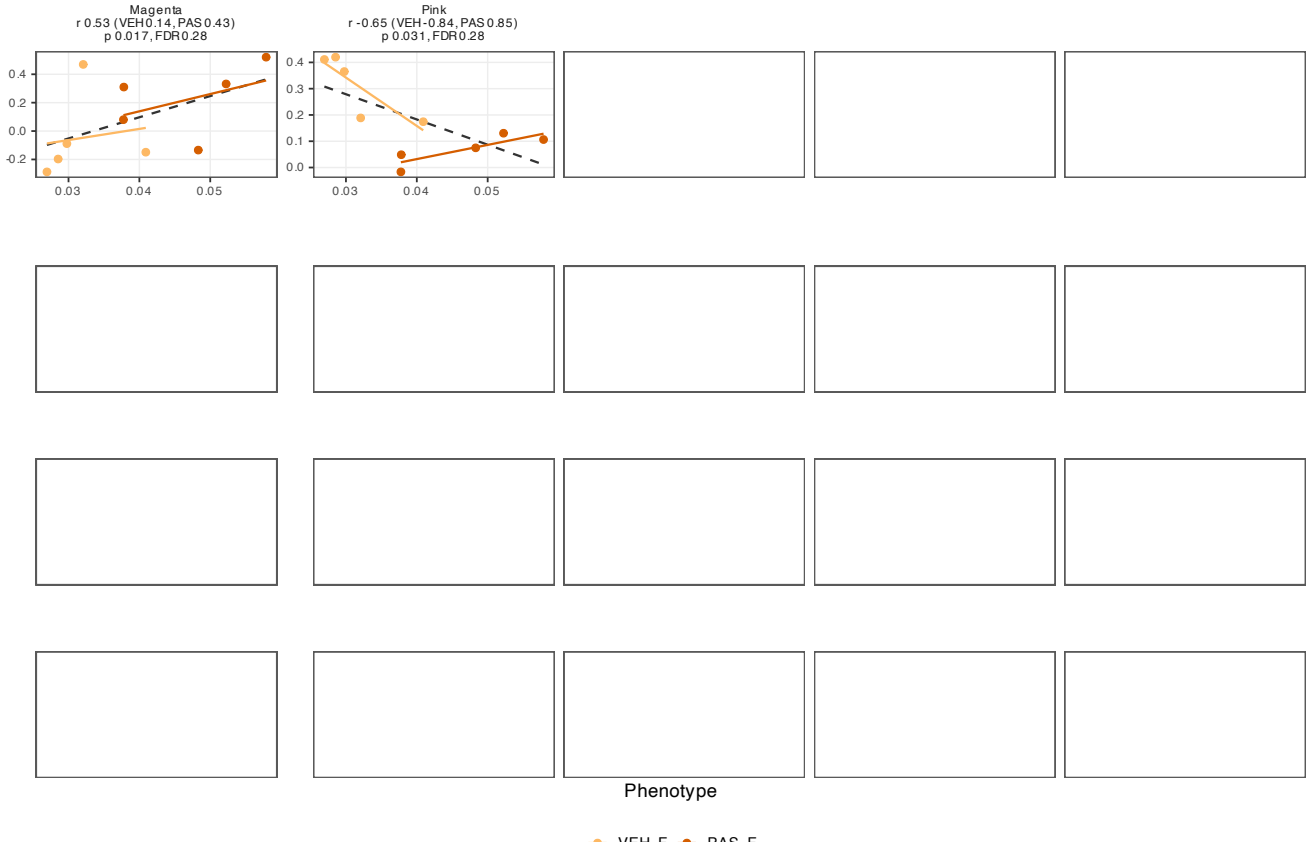
Female: Modules associated with CSA, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



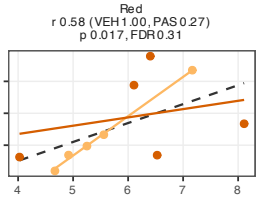
Female: Modules associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 2)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



Female: Modules associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



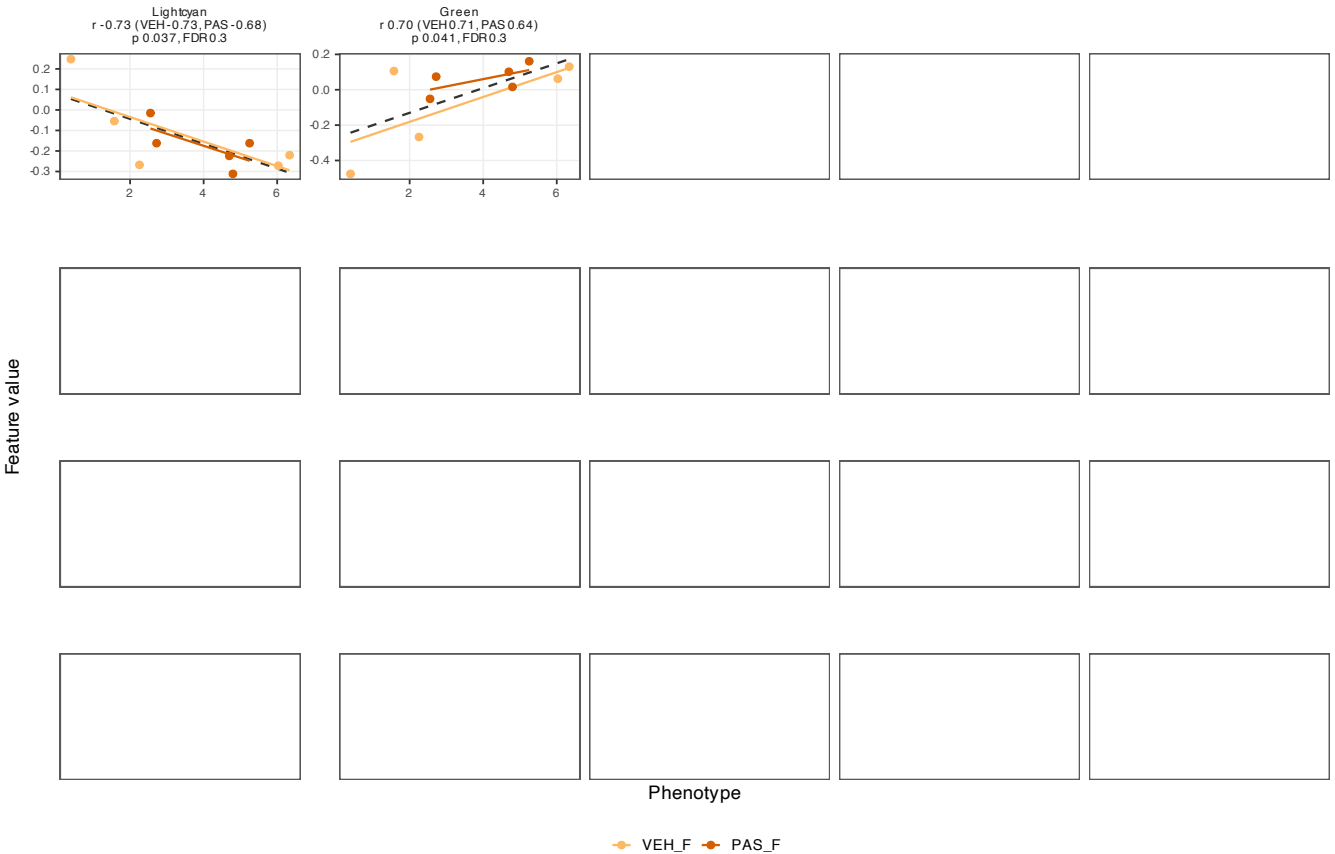
Feature value

Phenotype

VEH_F PAS_F

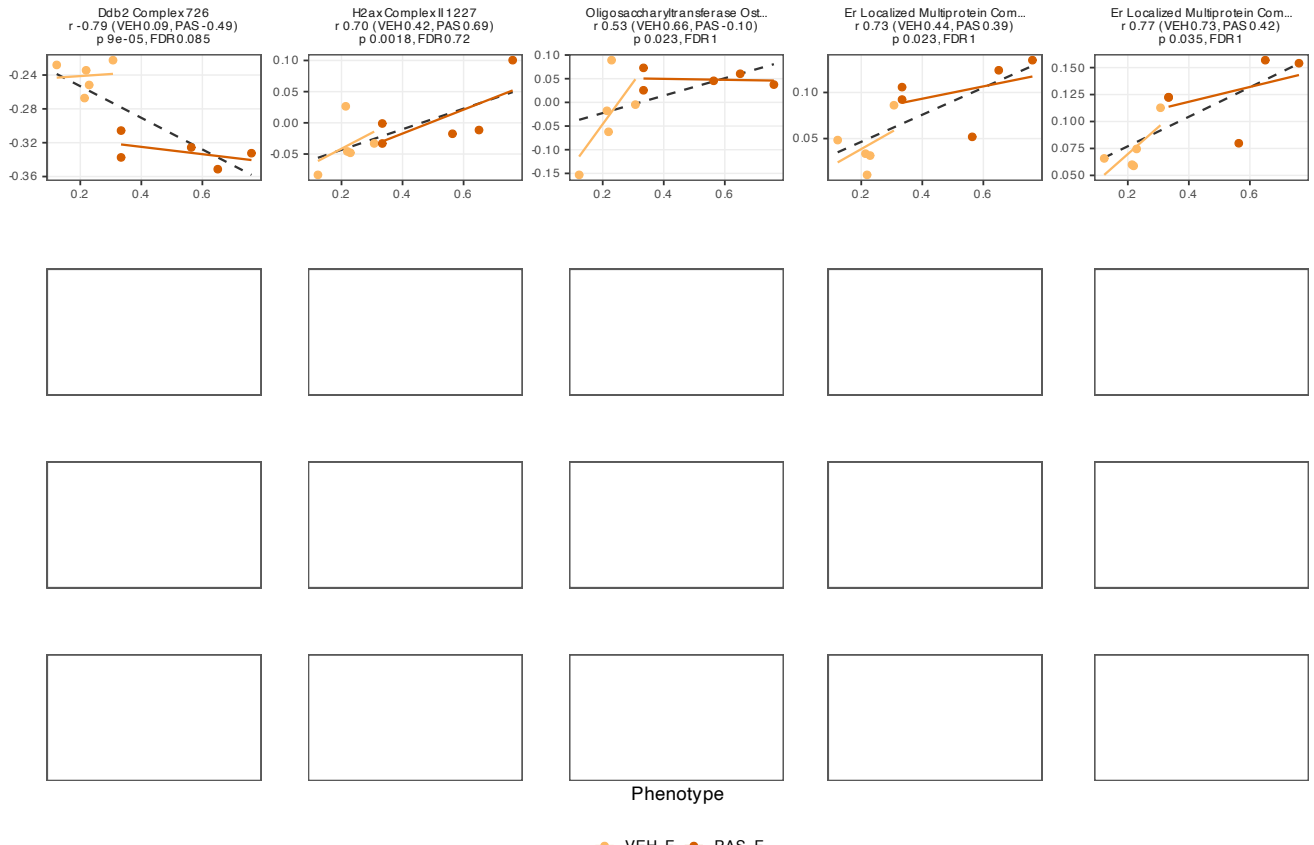
Female: Modules associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 2)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



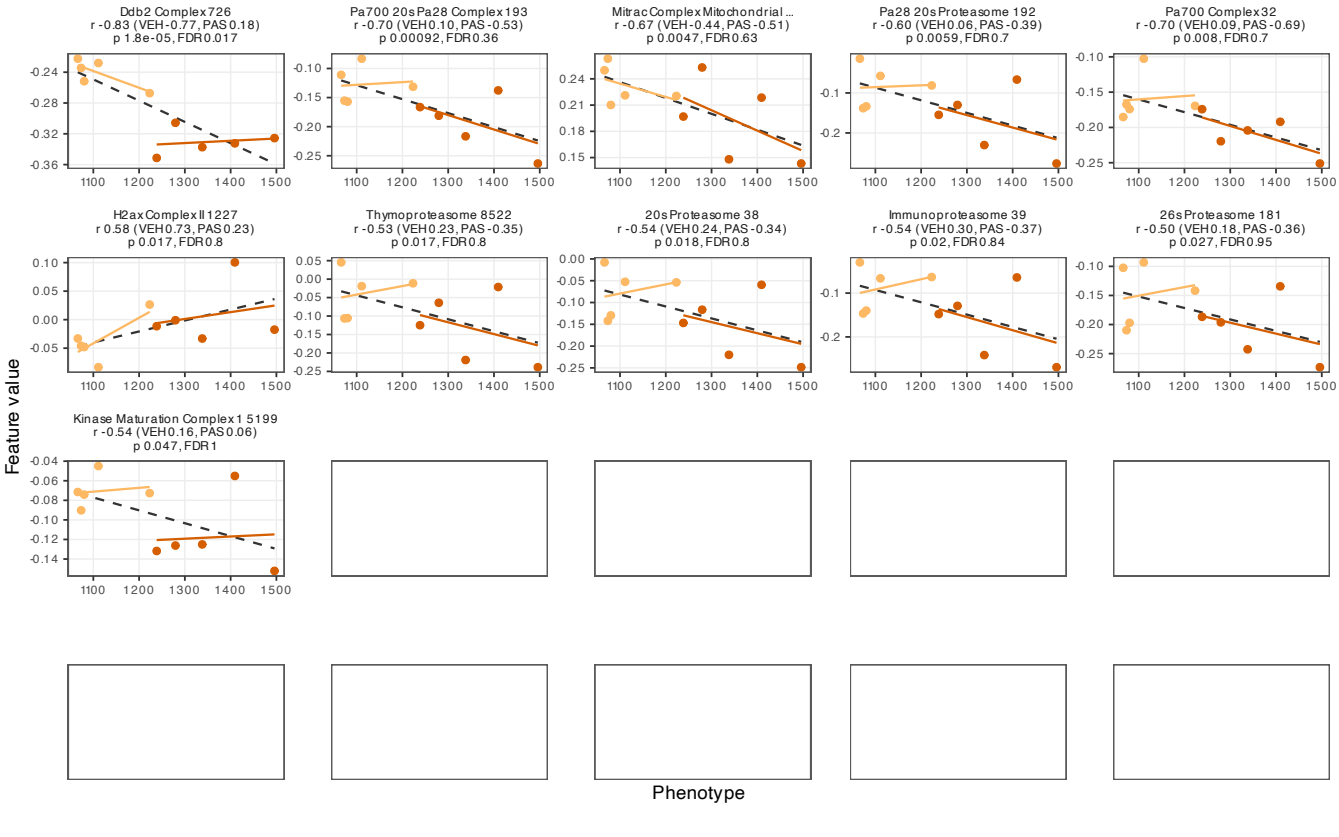
Female: Complexes associated with Capillary density, every association at p ≤ 0.05 (n = 5)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



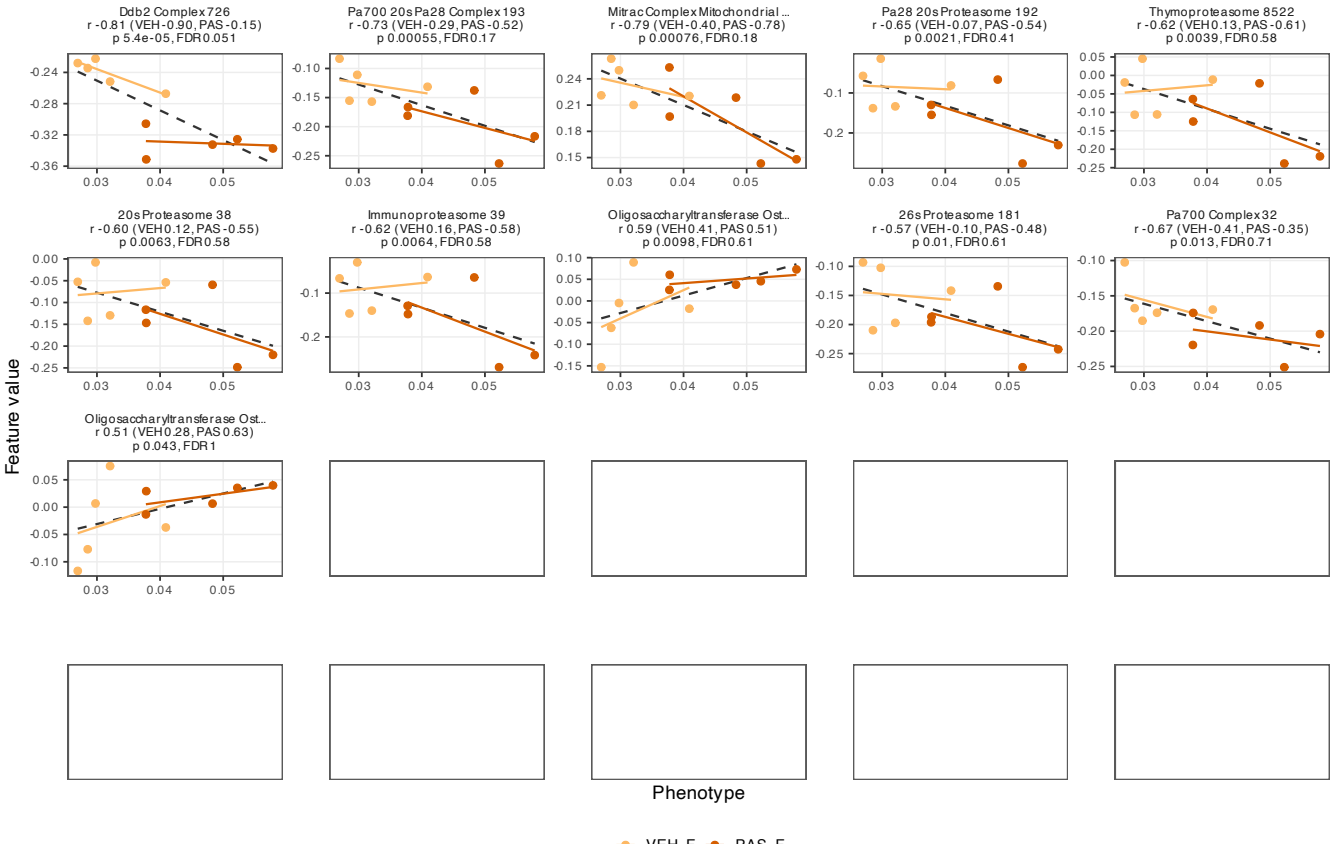
Female: Complexes associated with CSA, every association at p ≤ 0.05 (n = 11)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



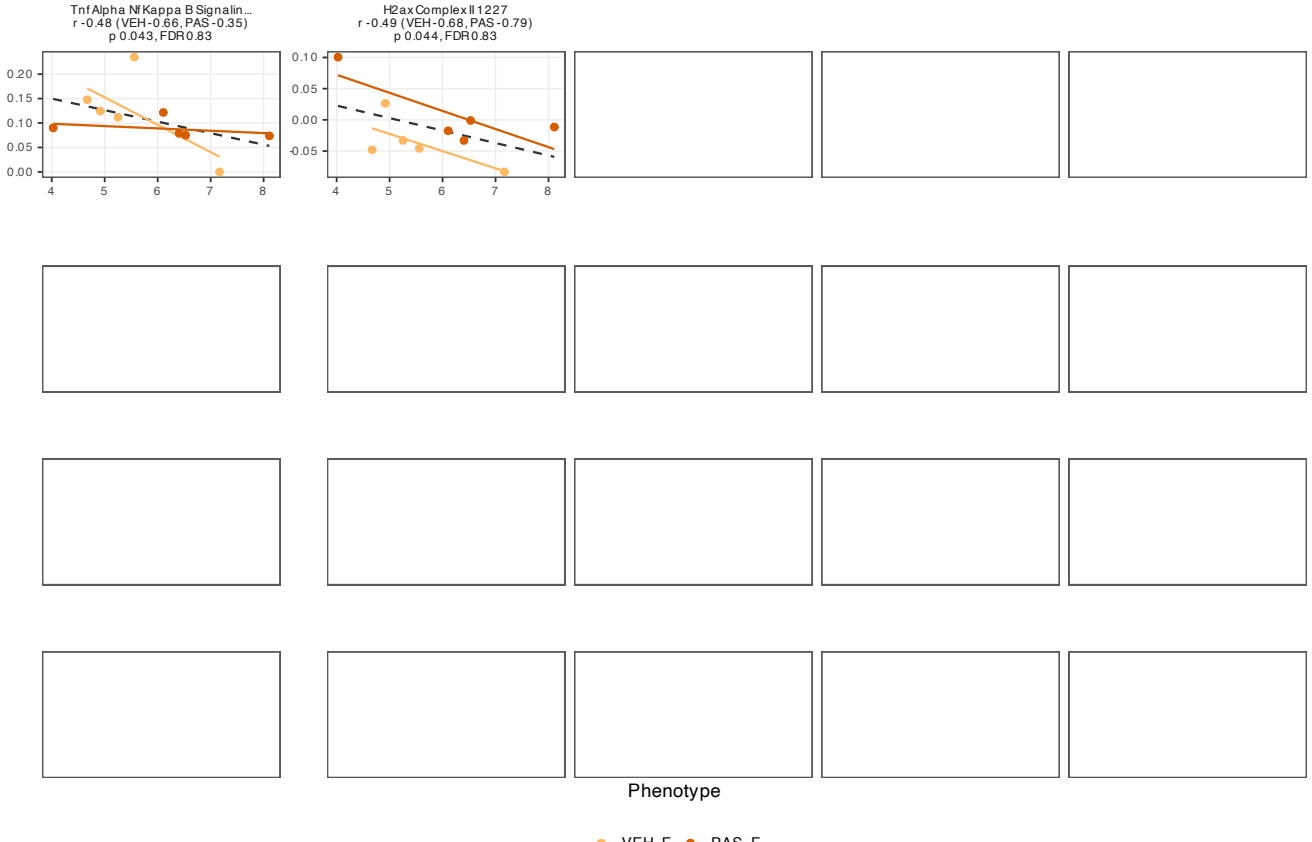
Female: Complexes associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 11)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



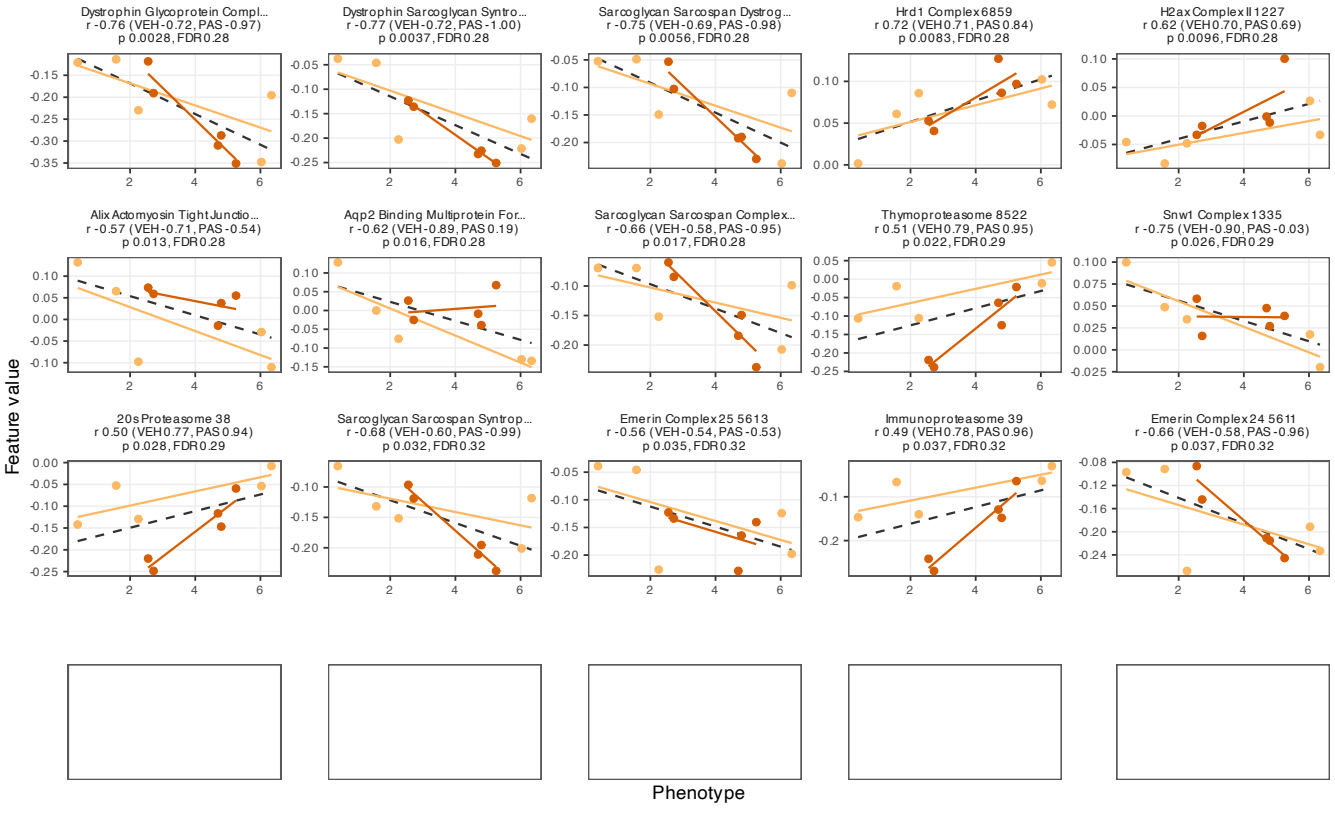
Female: Complexes associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 2)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response. Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



Female: Complexes associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 15)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



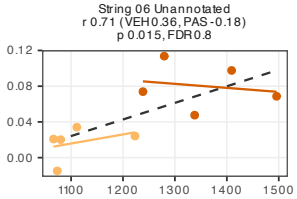
Female: Clusters associated with Capillary density, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +138% (p 0.0052).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



Female: Clusters associated with CSA, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +22% (p 0.002).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



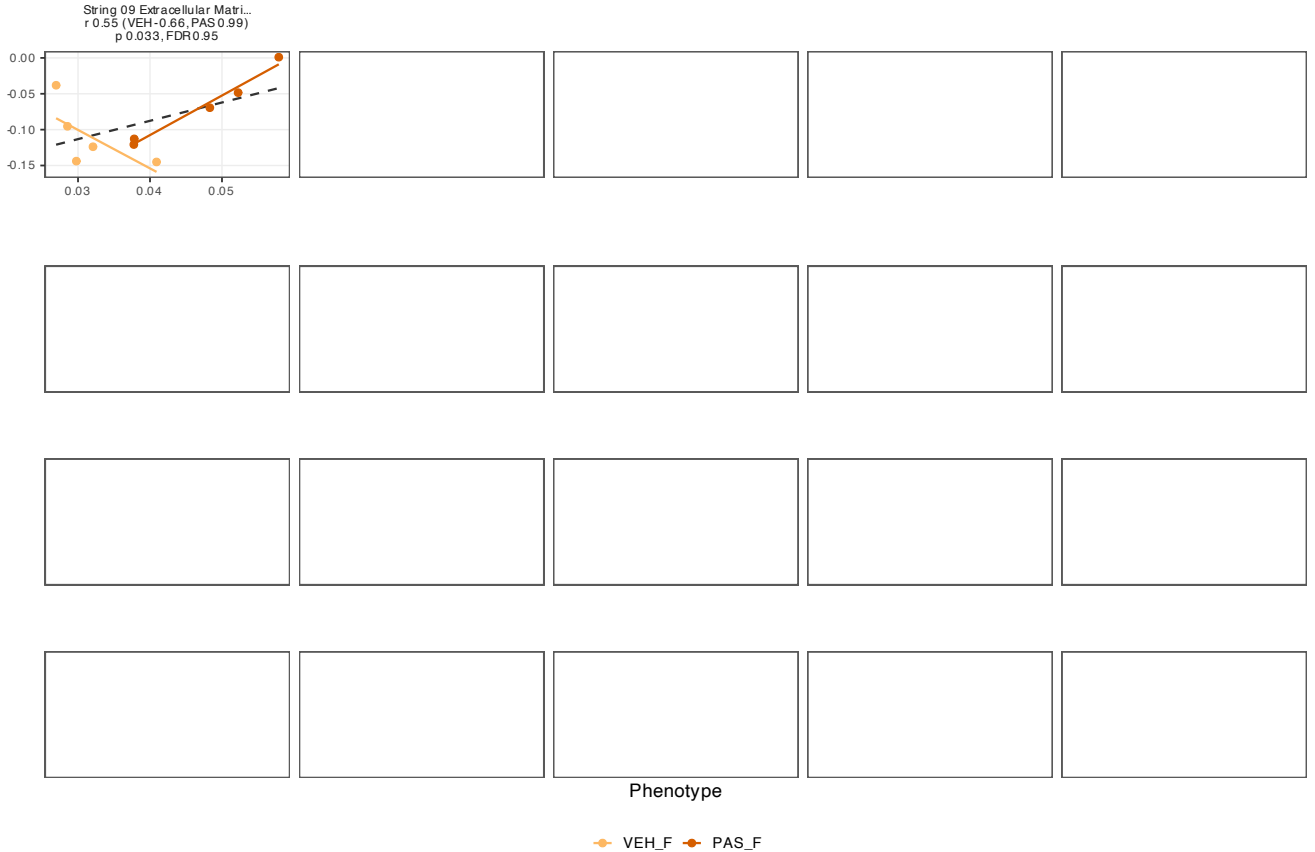
Feature value

Phenotype

VEH_F PAS_F

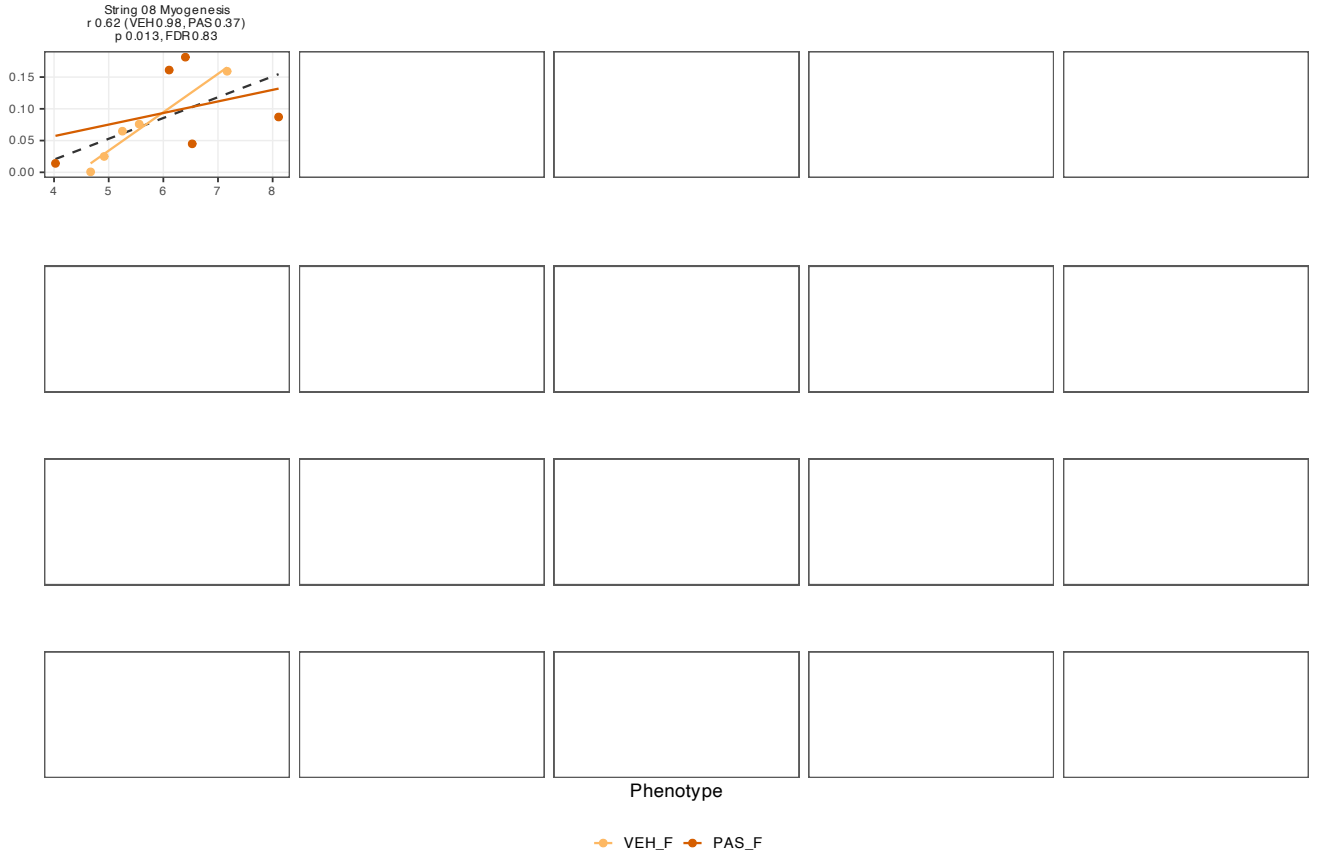
Female: Clusters associated with Satellite cells / fibre, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +47% (p 0.0093).
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



Female: Clusters associated with Soleus pyruvate, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +12% (p 0.45), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association



Female: Clusters associated with Soleus octanoyl, every association at p ≤ 0.05 (n = 1)

Phenotype effect of PAS in this sex: +73% (p 0.35), so these associations follow variation between mice, not a PAS response.
Lines within vehicle and within PAS (5 mice each); dashed: all 10 mice, the tested association

